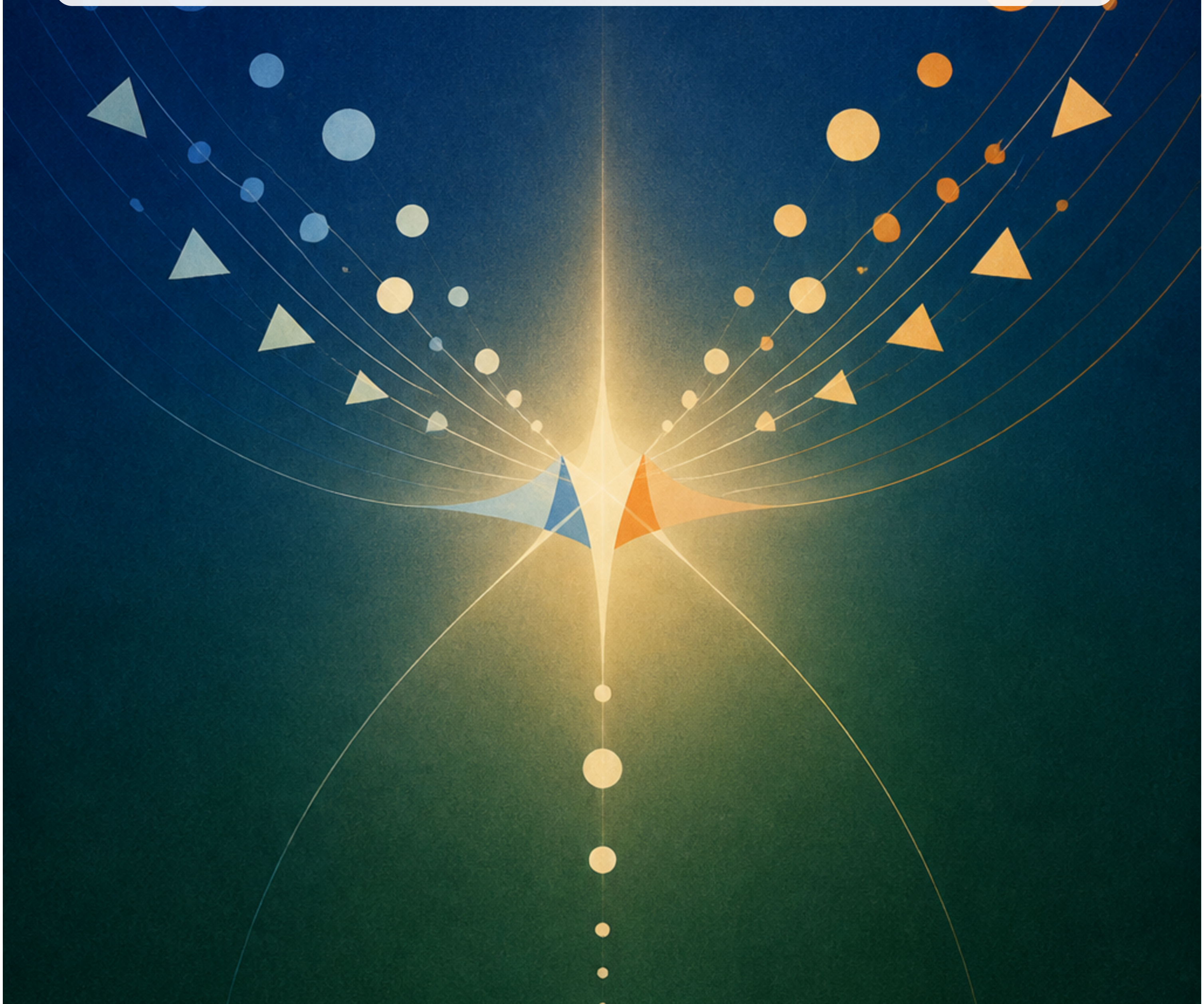
Particle Physics 3: Supersymmetry and Grand Unification

Supplementary track, 2010 Spring



Lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book

About These Notes

These independently edited companion notes follow lectures by Leonard Susskind. They were reconstructed from machine transcripts, subtitles, and selected blackboard frames, then checked against the available source material. They are not an original manuscript by Professor Susskind and have not been reviewed or endorsed by him.

The edition was prepared by LazyingArt LLC using the open-source Video2Book workflow. Machine transcripts remain available separately and may contain recognition errors. Editorial clarifications and nontrivial reconstructions are identified in footnotes.

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Chapter 1

Renormalization, Naturalness, and the Hierarchy Problem

Overview. The central ideas of this course grew out of puzzles about renormalization in the Standard Model. We begin, therefore, with the physical meaning of renormalization: eliminate degrees of freedom that are too small and too fast for the question at hand, then encode their effects in an effective description. A molecule supplies the prototype. Dimensional analysis then carries the same reasoning into scalar quantum field theory, where it exposes the quadratic sensitivity of a scalar mass, the hierarchy problem of the Higgs field, the protection enjoyed by fermions and gauge bosons, and the still more severe problem of vacuum energy.

1.1 Physics at the Scale of the Question

Most of the ideas to be developed this quarter originated in questions about renormalization—sometimes puzzles, sometimes useful observations—in the Standard Model. Renormalization combines two habits of thought. The first is to remove from a description structures at distances so short that they are irrelevant to the question being asked. The second is dimensional analysis, which often determines the short-distance answer before any difficult integral is evaluated.

This elimination of detail happens throughout physics. If we wish to understand a nucleus, we may begin with quarks and quantum chromodynamics. After calculating the masses and spins of the proton and neutron, together with their effective nuclear forces, we can forget their quark substructure for many purposes. Nuclear motion is slow enough that a nonrelativistic quantum mechanics of protons and neutrons is often the more useful description.

At the next level, atomic physics does not ordinarily need the internal constitution of a nucleus. Nuclear physics tells us which nuclei and isotopes exist and supplies such properties as mass and electric charge. We then begin again with a new set of effective elementary objects: nuclei and electrons. Quarks, protons, and neutrons have not ceased to exist; their effects have been compressed into the parameters carried by the nuclei.

The amount retained depends on the next question. An atomic calculation may need the nuclear mass because it enters the reduced mass, and it certainly needs the charge because that determines the Coulomb field. It usually does not need to know which quark carries which fraction of the nuclear momentum. There are roughly ninety elemental charges, with more nuclear species once isotopes are counted. That catalogue can be taken as input. Electrons, on the other hand, cannot yet be eliminated: their motion is exactly what makes atomic structure.

After solving the atomic problem, we may similarly regard atoms as the building blocks of molecules. Susskind's whimsical list of cookie atoms, coffee atoms, chocolate atoms, and ketchup atoms makes the serious point that a new description begins at every level. Once molecular properties and forces are known, molecules can be joined into larger Tinkertoy-like assemblies without reopening the electron problem at every step.

Each passage is a coarse-graining. The new account may be less exact in an absolute sense, but it is far more useful at its intended scale. Quantum field theory makes this issue unavoidable because a field has modes at every wavelength, including arbitrarily short ones. We therefore sum the effects of wavelengths too short to resolve and replace them by new effective parameters. That operation, together with dimensional analysis, is the physical content of renormalization.

1.2 A Molecule as the Prototype

Consider two atoms and ask for a simple description of their relative motion. Each atom contains a heavy nucleus and a cloud of electrons. Even the proton is roughly two thousand times heavier than an electron, and heavier nuclei increase the separation of scales. A useful picture is a pair of bowling balls surrounded by rapidly moving flies: the nuclei respond slowly while the electronic state rearranges quickly.

Let the nuclear positions and momenta be R_a and P_a, with common mass M, and let the electron coordinates and momenta be r_i and q_i, with mass m_e. Suppressing conventional factors such as $4\pi\epsilon_0$, the Hamiltonian has the schematic form

$$\begin{aligned} H = \frac{P_1^2}{2M} + \frac{P_2^2}{2M} + \frac{e^2}{R_{12}} + \sum_i \frac{q_i^2}{2m_e} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}} \\ - \sum_i \left(\frac{e^2}{|R_1 - r_i|} + \frac{e^2}{|R_2 - r_i|} \right). \end{aligned} \tag{1.1}$$

The first line contains the nuclear kinetic energies, their mutual Coulomb repulsion, and the electronic kinetic energy. The second line includes electron-electron repulsion and electron-nucleus attraction. For nuclei of general charges, the corresponding factors of Z_a are restored; the lecture keeps the charges schematic because the separation of time scales is the issue.

There are two characteristic motions. The nuclei are heavy and slow; the electrons are light and fast enough to look like a blur. In the first approximation, fix the nuclear positions and group all electron-dependent terms into an electronic Hamiltonian,

$$H_{\mathrm{el}}(R_1, R_2). \tag{1.2}$$

For each fixed nuclear configuration, solve

$$H_{\mathrm{el}}(R_1, R_2)\psi_0(r_1, \ldots, r_N; R_1, R_2) = E_{\mathrm{el}}(R_1, R_2)\psi_0. \tag{1.3}$$

We retain the lowest electronic eigenvalue. The electrons have not been ignored; all their rapid motion has been summarized by the function $E_{\mathrm{el}}(R_1, R_2)$.

The logic is adiabatic. During one rapid electronic adjustment, the nuclei have moved so little that their coordinates can be treated as parameters. After the electronic eigenproblem is solved, those same coordinates are allowed to move again. At each slowly changing configuration the electron cloud follows its instantaneous ground state. A more accurate treatment could retain excited electronic surfaces and transitions between them, but the leading surface already exhibits the renormalization idea.

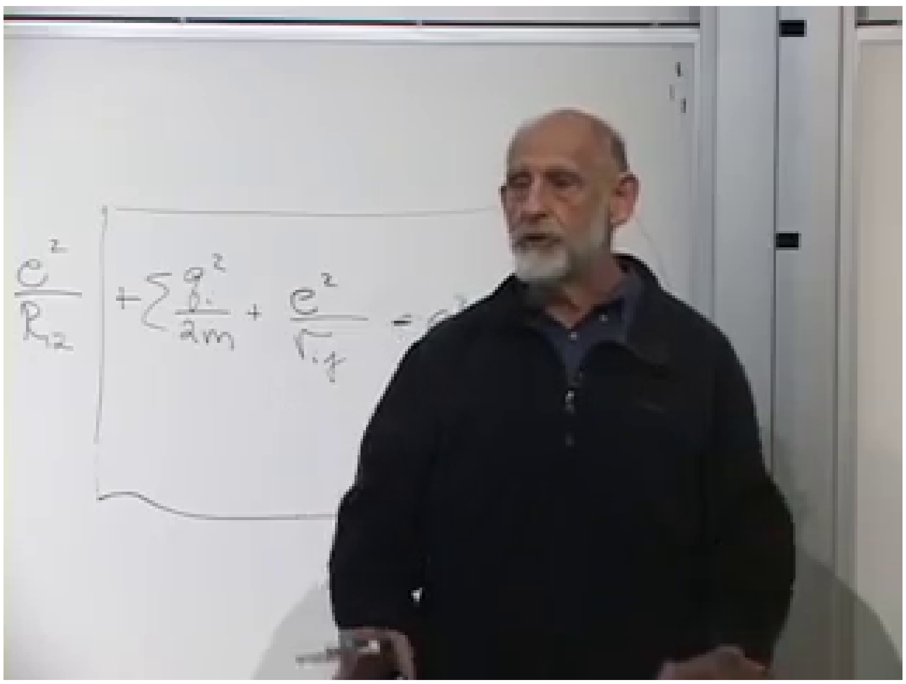

Figure 1.1: The molecular Hamiltonian remains on the board as the electronic ground-state problem is introduced. The typed form in Eq. (1.1) records the terms obscured in this frame.

Lecture frame: 00:16:19.

Question from the audience. Are the nuclear momenta simply being set to zero?[a]

Response. Only temporarily. In the leading approximation the nuclei are treated as nailed down because they move much more slowly than the electrons. Their momenta will be restored when we solve the effective nuclear problem, so this approximation is not a violation of the uncertainty principle.

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[a]Lecture timestamp: 00:15:39.

The electronic ground-state energy is now a contribution to the potential felt by the nuclei:

$$V_{\text{eff}}(R_1, R_2) = \frac{e^2}{R_{12}} + E_{\text{el}}(R_1, R_2). \tag{1.4}$$

Rotational and translational invariance reduce it, for two isolated atoms, to a function of their separation R_{12}. At very short distance the nuclear Coulomb repulsion dominates. At very large distance there is little interaction. Between these limits the electronic cloud can produce an attractive region and a minimum. The nuclear wavefunction then sits near that minimum and describes a bound molecule.

Notice the order of operations. We did not guess an empirical attraction and discard the electrons. We first solved their quantum problem in the fixed nuclear background. Only then did we erase their explicit coordinates and retain the resulting eigenvalue. The short-distance physics survives in the shape, depth, and curvature of V_{eff}. The location of its minimum sets the equilibrium separation, its curvature controls small vibrations, and its asymptotic form controls whether a bound state exists. One effective function carries all of that electronic information into the slower problem.

Thus a complicated many-electron Hamiltonian has become a two-body nuclear Hamiltonian with a new potential. We eliminated the fast degrees of freedom, but retained their complete low-energy effect. The potential has been renormalized. This familiar Born–Oppenheimer reasoning is not usually introduced under that name, yet it contains precisely the idea we need:

$$\begin{gathered}\text{fast microscopic dynamics}\\ \Downarrow\\ \text{effective parameters for slow dynamics}\end{gathered} \tag{1.5}$$

1.3 Natural Units and the Dimension of a Field

In quantum field theory, the modes to be eliminated may be described equivalently as short-distance, short-wavelength, high-frequency, or high-energy modes. Before drawing diagrams, let us fix the dimensional bookkeeping.

Ordinary mechanics uses independent units of length, time, and mass. In particle physics we set

$$\hbar = c = 1. \tag{1.6}$$

Then energy and mass are equivalent by $E = mc^2$, length and time are equivalent because $c = 1$, and $\hbar = 1$ makes momentum the inverse of length. Only one dimension remains:

$$[M] = [E] = [p] = L^{-1}, \qquad [x] = [t] = L. \tag{1.7}$$

This does not erase dimensions. It chooses conversion factors so that a time can be quoted as a length and a mass as an inverse length. Once one unit is specified, every other one follows. This is why a cutoff may be described either as a short distance δ or as a high energy $\Lambda = 1/\delta$: they are the same scale in two languages.

Take one real scalar field ϕ, with a schematically normalized Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi)(\partial^\mu \phi) - V(\phi), \qquad S = \int d^4x\, \mathcal{L}. \tag{1.8}$$

The Lagrangian density is local: it describes one spacetime point. Classical field equations follow by making the action stationary. In quantum theory the same action appears in the path-integral phase $\exp(iS)$. Either viewpoint makes the dimension of S decisive. With $\hbar$ restored, the phase is $\exp(iS/\hbar)$; after setting $\hbar = 1$, S itself must be dimensionless.

Question from the audience. Does the action have units of $\hbar$?[a]

Response. Yes. Since $\hbar$ has been set equal to one, the action is dimensionless in these units. The four-dimensional measure has dimension L^4, so the Lagrangian density must have dimension L^{-4}.

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[a] Lecture timestamp: 00:26:13.

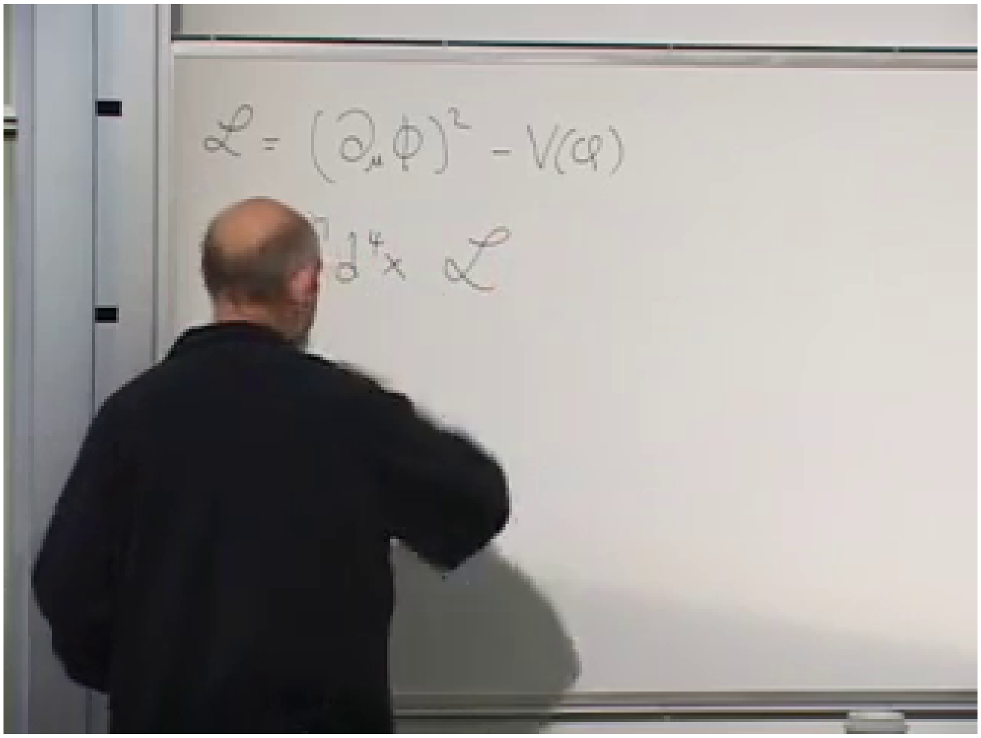

Figure 1.2: The local scalar Lagrangian density and its spacetime integral into the action.
Lecture frame: 00:26:08.

The kinetic term now fixes the dimension of the field:

$$[S] = 1, \qquad [\mathcal{L}] = L^{-4}, \tag{1.9}$$

$$[\partial_\mu] = L^{-1}, \qquad [(\partial\phi)^2] = L^{-2}[\phi]^2 = L^{-4},$$
$$[\phi] = L^{-1}. \tag{1.10}$$

This small calculation will determine the ultraviolet size of every diagram that follows.

Expand the potential as

$$V(\phi) = \frac{m^2}{2}\phi^2 + g\phi^3 + \lambda\phi^4 + \cdots . \tag{1.11}$$

Every term must have dimension L^{-4}. Since $[\phi^n] = L^{-n}$,

$$[m^2] = L^{-2}, \qquad [g] = L^{-1}, \qquad [\lambda] = 1. \tag{1.12}$$

The quartic coupling is dimensionless. Such couplings are especially important in renormalization because their scale dependence cannot be read off from a simple power of length.

Question from the audience. Does g have units of mass?[a]

Response. Yes. The factor ϕ^3 has dimension L^{-3}, while the potential density must have dimension L^{-4}. Therefore $[g] = L^{-1}$, which is a mass in natural units.

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[a] Lecture timestamp: 00:29:52.

1.4 Vertices and the Short-Distance Propagator

The Feynman rules are built from vertices and propagators. The term $g\phi^3$ gives a three-line vertex carrying a factor proportional to g; $\lambda\phi^4$ gives a four-line vertex carrying a factor proportional to λ. The quadratic term gives a local two-point insertion. It may be pictured as a particle entering, being absorbed, and being re-emitted, with a coefficient proportional to $m^2/2$. Numerical normalizations, signs, factors of i, and symmetry factors are deliberately suppressed because the lecture is extracting scale dependence rather than computing a cross section.

These drawings have an operational reading. A three-point vertex can absorb two quanta and emit one, or represent any crossing-related assignment of its three lines. A four-point vertex can describe two particles entering and two leaving. The quadratic cross is still a legitimate two-line vertex even though it is normally incorporated into the free propagator. We display it because every unresolved two-point process will shortly be recognized by comparison with it.

A propagator joins two spacetime points. Its line is not a claim that a quantum particle followed a straight classical path. It represents the amplitude to create a field excitation at x and remove or detect it at y:

$$\Delta(x-y) = \langle 0|\phi(y)\phi(x)|0\rangle. \tag{1.13}$$

Read the expression from the right. Acting on the vacuum with $\phi(x)$ creates a one-particle component at x. Acting with $\phi(y)$ on the bra removes the component at y. Their vacuum matrix element is the overlap amplitude between creation at the first point and detection at the second. The symbols x and y denote whole four-vectors, not merely two Cartesian coordinates.

States carry no physical dimension, while each field contributes L^{-1}. Therefore

$$[\Delta] = L^{-2}. \tag{1.14}$$

For a massless field there is no intrinsic length scale, so dimensional analysis forces the short-distance behavior

$$\Delta(x-y) \sim \frac{1}{(x-y)^2}. \tag{1.15}$$

The denominator is the Lorentz-invariant squared separation, with the usual prescription needed at its singularities.[1]

As y approaches x, the propagator diverges. This singularity of propagation over a vanishing separation is the source from which ultraviolet divergences in more elaborate diagrams are built.

1.5 How a Scalar Mass Is Renormalized

For a scalar field the dynamical parameter is m^2, as the relativistic dispersion relation already suggests:

$$E = \sqrt{\mathbf{p}^2 + m^2}. \tag{1.16}$$

Let δ be the shortest distance our effective description resolves. We shall not distinguish processes whose vertices are separated by less than δ. Any such unresolved process with one incoming and

[1] Editorial clarification: For power counting it is convenient to Wick rotate to Euclidean separation, where the magnitude is unambiguously proportional to $1/|x-y|^2$. This does not change the ultraviolet powers used below.

one outgoing scalar has the same observable two-point shape as a mass insertion and must therefore contribute to the effective mass.

Think of an accelerator as a microscope. If its finest resolution is δ, an absorption at one point followed by an emission a distance much smaller than δ away cannot be distinguished from a truly local event. We deliberately blur the vertex over that scale. Renormalization is the rule that transfers all such sub-resolution alternatives into the coefficient of the corresponding local operator.

1.5.1 The first quartic correction

Begin with a single ϕ^4 vertex. Two legs are the external particle; the other two join into a short loop emitted and reabsorbed at nearly the same point. Smearing the coincident propagator over the cutoff distance gives

$$\Delta(0) \;\longrightarrow\; \Delta(\delta) \sim \frac{1}{\delta^2}, \tag{1.17}$$

and hence

$$\delta m^2_{(1)} \sim \frac{\lambda}{\delta^2}. \tag{1.18}$$

If m_0^2 denotes the input or bare parameter, the low-resolution two-point coefficient has the schematic form

$$\frac{m^2_{\text{eff}}}{2} \sim \frac{m_0^2}{2} + c_1 \frac{\lambda}{\delta^2}, \tag{1.19}$$

where c_1 contains the ignored numerical and convention-dependent factors. The laboratory mass measured in experiments that do not resolve distances below δ is this renormalized quantity, not m_0 by itself.

There are therefore two indistinguishable ways for the observed two-point event to occur. The original quadratic vertex can absorb and re-emit the particle, or the quartic interaction can do so while briefly emitting and reabsorbing a virtual scalar. Quantum mechanics adds the amplitudes. Once the loop lies below the resolution scale, its separate history is gone from the effective description, but its contribution to the measured coefficient is not.

1.5.2 The next quartic correction

Now connect two quartic vertices so that one external particle enters at x, one leaves from $x+\Delta$, and three internal propagators join the vertices. There are two powers of λ, while the propagators give Δ^{-6}. Quantum amplitudes must be summed over every unresolved place from which the outgoing particle can emerge, so the relative four-vector is integrated:

$$\delta m^2_{(2)} \sim \lambda^2 \int_{|\Delta| \gtrsim \delta} d^4\Delta \, \frac{1}{|\Delta|^6}. \tag{1.20}$$

Question from the audience. Does the small mark at the vertex mean there are only three particles there?[a]

Response. No. We are still using the quartic interaction. Four lines meet at each vertex; in this two-point diagram one line is external and three run between the two vertices. The second vertex again has four incident lines.

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[a] Lecture timestamp: 00:46:35.

The measure supplies four powers of length and the three propagators supply minus six. Since the ultraviolet cutoff is the only scale in the estimate,

$$\int_{|\Delta|\gtrsim\delta} \frac{d^4\Delta}{|\Delta|^6} \sim \frac{1}{\delta^2}, \qquad \delta m_{(2)}^2 \sim \frac{\lambda^2}{\delta^2}. \tag{1.21}$$

Here Δ^μ is a four-component integration variable; one integrates over its time and three spatial components. The lowercase δ is the fixed cutoff. The lecture initially uses the same spoken name for both and later corrects the notation.

The same result is visible by writing the four-dimensional measure radially. Up to an angular constant,

$$\int_\delta \frac{d^4\Delta}{|\Delta|^6} \sim \int_\delta \frac{R^3\,dR}{R^6} = \int_\delta \frac{dR}{R^3} \sim \frac{1}{\delta^2}. \tag{1.22}$$

This is why the large-distance upper limit is unimportant for the ultraviolet estimate: the integral is dominated by its lower endpoint.

Higher diagrams repeat the pattern. A diagram with more quartic vertices adds a higher power of λ, but dimensional analysis still permits a quadratically divergent mass contribution:

$$\begin{aligned} m_{\text{eff}}^2 = m_0^2 + \frac{1}{\delta^2}\left(c_1\lambda + c_2\lambda^2 + c_3\lambda^3 + \cdots\right) \\ + \text{ weaker terms.} \end{aligned} \tag{1.23}$$

1.5.3 The cubic interaction and a logarithm

Two cubic vertices contain two propagators between nearby points. Their contribution scales as

$$\delta m_g^2 \sim g^2 \int_{|\Delta|\gtrsim\delta} \frac{d^4\Delta}{|\Delta|^4}. \tag{1.24}$$

Now the measure and denominator carry equal powers. The remaining dependence is logarithmic rather than a power:

$$\delta m_g^2 \sim c_g g^2 \log\left(\frac{1}{\mu\delta}\right). \tag{1.25}$$

The reference scale μ makes the logarithm dimensionless. This is the fully dimensional version of the lecture's schematic $g^2 \log \delta$. A logarithm varies much more slowly than any inverse power, so $1/\delta^2$ dominates as the cutoff becomes short.

Again the radial form makes the distinction transparent:

$$\int_\delta^{R_{\text{IR}}} \frac{d^4\Delta}{|\Delta|^4} \sim \int_\delta^{R_{\text{IR}}} \frac{R^3\,dR}{R^4} = \log\left(\frac{R_{\text{IR}}}{\delta}\right). \tag{1.26}$$

Equal powers in measure and denominator produce a logarithm. More powers in the denominator produce an inverse power of the cutoff. The lecture is not asking us to evaluate angular factors; it is teaching us to recognize the ultraviolet answer on sight.

Question from the audience. To which base is the logarithm taken?[a]

Response. Use the natural logarithm. Changing the base only multiplies it by a numerical constant, and that constant is absorbed into the coefficient whose factors of 2, π, and symmetry numbers have not been calculated here.

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[a] Lecture timestamp: 00:57:42.

Question from the audience. How can one take the logarithm of a length, and where do the units of the mass correction come from?[a]

Response. A logarithm always compares two scales; properly it is the logarithm of a dimensionless ratio such as $1/(\mu\delta)$. The coupling g has one power of mass, so g^2 already has exactly the mass-squared dimension required for the correction.

[a]Lecture timestamp: 00:58:32.

Question from the audience. A Feynman diagram gives an amplitude. Is its squared magnitude a probability?[a]

Response. Yes, when the amplitude is placed in a physical transition calculation with the required state and phase-space factors. Here its more immediate role is different: an unresolved two-point amplitude has the same local form as the mass term and therefore shifts the effective mass parameter.

[a]Lecture timestamp: 00:59:41.

1.6 Every Parameter Becomes Effective

The infinite family of mass diagrams does not disappear merely because λ is small. Perturbation theory orders them by powers of the coupling, so successive terms may become smaller, but there are still infinitely many. The same short-distance elimination also renormalizes the interactions themselves.

Question from the audience. Even for mass renormalization, are there not infinitely many diagrams?[a]

Response. There are. If λ is sufficiently small, the perturbative series may be useful because higher powers are successively suppressed. Nevertheless, the renormalized parameter is defined by the whole series, not by the first drawing alone.

[a]Lecture timestamp: 01:00:35.

Operationally, a local $\lambda\phi^4$ interaction is a two-particle-in, two-particle-out amplitude. If a more complicated four-point process occurs entirely inside an unresolved region, it is indistinguishable at low resolution from the original local vertex. Such diagrams are therefore lumped into an effective λ. Their leading scale dependence is typically logarithmic in four-dimensional ϕ^4 theory. The lecture does not calculate that beta function; it uses the example to make the general statement:

$$\begin{gathered}\text{parameters measured at low energy}\\ \neq \\ \text{parameters entered at the microscopic cutoff}\end{gathered} \tag{1.27}$$

The four-point example matters because it removes a possible misunderstanding. Renormalization is not a special repair applied only to a mass divergence. Every operator allowed in the Lagrangian can be reproduced by unresolved diagrams with the same external legs and symmetries. A two-point process renormalizes a quadratic coefficient, a four-point process renormalizes a quartic coefficient, and vacuum-to-vacuum processes will renormalize the constant term. The low-energy Lagrangian is the compact ledger in which all of those short-distance processes are recorded.

Question from the audience. Since Δ has units of length, is a logarithm of it not dimensionally strange? And did the board interchange uppercase and lowercase delta?[a]

Response. The logarithm really means a ratio to another scale and is therefore dimensionless. Yes, the notation was briefly interchanged: Δ^μ is the integration variable, while δ is the lower distance cutoff that remains in the answer.

[a]Lecture timestamp: 01:01:34.

This is the field-theory counterpart of the molecule. We solved for the unseen fast electrons and put their result into a potential. Here we sum unseen short-distance processes and put their result into masses, couplings, and every other allowed local coefficient. Everything is shifted.

Question from the audience. Could you repeat whether the measured quantity includes scales smaller than the experiment can see?[a]

Response. It does. Their individual fluctuations are not resolved, but their summed effect is present. All diagrams involving shorter scales are collected into the single effective coefficient that a low-energy experiment calls, for example, the physical mass squared.

[a]Lecture timestamp: 01:05:13.

Question from the audience. Must the input term m_0^2 then cancel the other contributions?[a]

Response. If the measured scalar mass is far smaller than the individual ultraviolet contributions, yes. The required cancellation is the fine-tuning problem to which we now turn.

[a]Lecture timestamp: 01:06:00.

1.7 The Higgs Hierarchy Problem

Imagine that the cutoff is the Planck length, so that the effective field theory is being asked to summarize fluctuations from laboratory scales down to the scale where quantum gravity must intervene. In particle-physics units,

$$\delta \sim M_{\mathrm{Pl}}^{-1} \sim 10^{-19}\,\mathrm{GeV}^{-1}, \qquad \frac{1}{\delta^2} \sim 10^{38}\,\mathrm{GeV}^2. \tag{1.28}$$

The precise use of the reduced or unreduced Planck scale is irrelevant to the enormous comparison.

The distance hierarchy is equally striking. The Planck length is about nineteen orders of magnitude below a proton radius and roughly sixteen or seventeen orders below the distances directly probed by the accelerators being discussed. Squaring that separation is what turns a large hierarchy of lengths into the roughly thirty-four-place cancellation in a scalar mass-squared parameter.

For the Higgs field, a scalar, the desired weak scale is of order a few hundred GeV, not 10^{19} GeV. Yet the terms in Eq. (1.23) can naturally be of Planck-scale size. Their signs have not been tracked: some may be positive and some negative, and the bare mass squared may also have either sign. But to leave an electroweak-scale remainder, those huge pieces must cancel to roughly thirty-four decimal places. Susskind first says seventeen digits, then corrects himself because the comparison is between squared masses. It is the squared ratio that measures this version of the tuning.

Question from the audience. Is λ smaller or larger than one?[a]

Response. It is assumed to be modestly smaller than one. Perturbation theory would become intractable if successive powers grew, and a pure scalar theory also cannot be extrapolated consistently with an arbitrarily large quartic coupling. But λ is not remotely as small as 10^{-38}; the lecture takes a percent-scale number for orientation.

[a] Lecture timestamp: 01:09:50.

Question from the audience. Even if $\lambda < 1$, do not many terms remain important because each carries the large cutoff factor?[a]

Response. Yes. Higher powers eventually suppress the series, but a large collection of terms can contribute before that happens. They and the bare term would all have to conspire to leave the extraordinarily small physical answer.

[a] Lecture timestamp: 01:10:43.

Recall why the Higgs mass parameter is known to be weak-scale. In the lecture's convention, write the symmetry-breaking potential as

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4. \tag{1.29}$$

The negative quadratic coefficient makes the origin unstable, while the positive quartic term stabilizes the potential. Differentiating,

$$\frac{dV}{d\phi} = -\mu^2\phi + \lambda\phi^3 = \phi(-\mu^2 + \lambda\phi^2), \tag{1.30}$$

gives the nonzero minimum

$$\phi_{\text{min}}^2 = f^2 = \frac{\mu^2}{\lambda}. \tag{1.31}$$

The board writes the quartic schematically as $\lambda\phi^4$ but quotes Eq. (1.31); inserting the conventional factor $1/4$ makes those two lines consistent without changing the argument.

The sign of $-\mu^2$ is important but does not rescue naturalness. It is the negative curvature at the origin that drives spontaneous symmetry breaking. Renormalization shifts that curvature by large positive and negative pieces. The observed broken vacuum requires their sum not merely to be small, but to have the particular small negative value that places the minimum at the weak scale.

The W- and Z-boson masses determine f to be of order 200 GeV in the rough normalization used in the lecture. If λ is neither enormous nor fantastically tiny, μ must be of the same general order. The observed weak-scale coefficient is therefore the residue of contributions that may each be tens of orders of magnitude larger.

This is called the Higgs fine-tuning problem, the hierarchy problem, or the gauge-hierarchy problem. The hierarchy is the immense ratio between the ordinary weak scale and a possible unification or Planck scale. A small number by itself need not be paradoxical. What is disturbing here is a small number assembled from large, apparently unrelated pieces that must cancel with extreme precision. The problem was recognized in the late 1970s and early 1980s, and the search for a mechanism behind that cancellation motivates supersymmetry and several competing ideas.

The infinite perturbative series sharpens rather than removes the puzzle. A percent-scale λ makes sufficiently high powers small, but many terms can occur before that suppression wins. It is not enough for one loop to cancel the bare mass accidentally. The effective weak scale must remain stable after the rest of the short-distance processes are included. A structural relation among the terms would be an explanation; an unrelated digit-by-digit cancellation is only a tuning.

1.8 Why a Fermion Mass Behaves Better

The other Standard Model particles do not require independent tunings of the same kind. For a Dirac fermion, the mass term is

$$\mathcal{L}_m = -m\bar{\psi}\psi = -m\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right). \tag{1.32}$$

It contains the mass, not its square, and it couples left and right chiralities. A local mass insertion therefore changes L into R, or R into L.

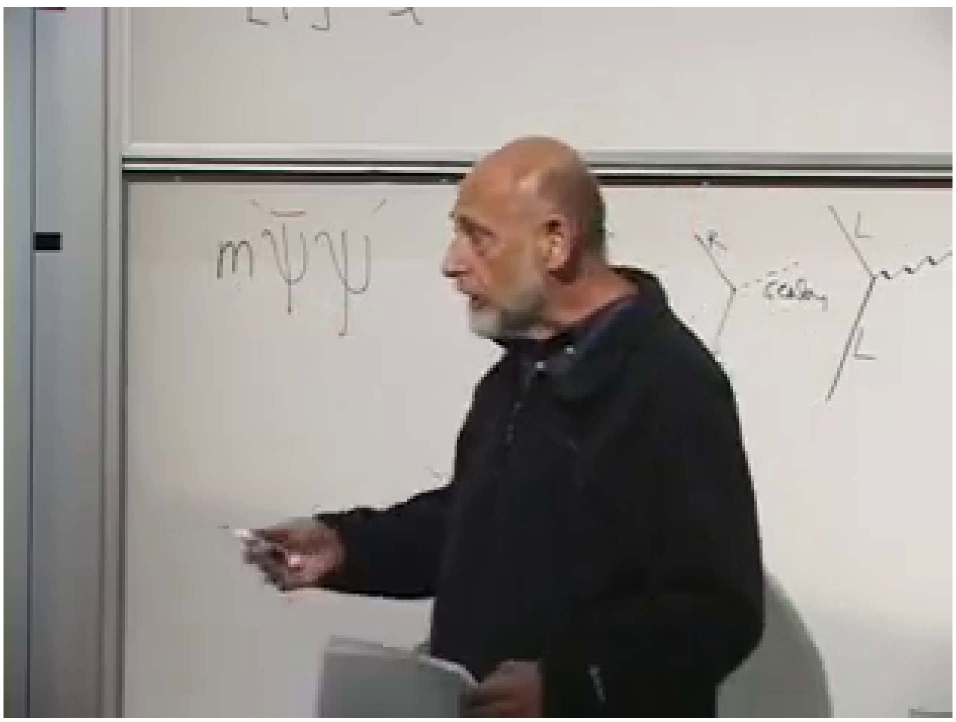

Figure 1.3: The fermion mass term and the board's chirality-changing line sketches. The diagrams are reconstructed algebraically below because part of the chalk work is occluded.

Lecture frame: 01:23:36.

The spoken lecture often says handedness or helicity. The exact protection is most cleanly stated in terms of chirality. For an ultrarelativistic particle the helicity picture supplies the intended intuition, but chirality is the quantity carried by the left- and right-handed fields in Eq. (1.32).

Gauge interactions preserve chirality:

$$L \xrightarrow{\gamma} L, \qquad R \xrightarrow{\gamma} R. \tag{1.33}$$

A Yukawa interaction with a scalar connects opposite chiralities:

$$L \xleftrightarrow{\phi} R. \tag{1.34}$$

Suppose first that the electron starts with $m_0 = 0$. Emitting and reabsorbing any number of photons cannot produce a mass, because every gauge vertex preserves chirality. A sequence that starts left remains left. Thus pure quantum electrodynamics consistently permits a massless electron: its electromagnetic self-energy cannot create an additive mass from zero.

In the line picture, start with a left-chiral electron. At the first photon vertex it is still left-chiral; after propagation and reabsorption of the photon it is still left-chiral. Repeating the construction only inserts more chirality-preserving vertices. The external two-point function can be renormalized, but it cannot acquire the left-right structure that defines a mass.

Could scalar emission do it? One scalar vertex changes L to R, but an emitted scalar must be reabsorbed. A two-point self-energy graph therefore contains an even number of scalar vertices. The second flip reverses the first, so the complete graph again preserves the external chirality. No matter how complicated the internal graph becomes, emission and reabsorption pair the vertices and prevent an additive L-to-R term from appearing when the starting mass is zero.

This is a parity argument on the number of flips. One scalar vertex flips chirality once; closing every internal scalar line requires a second endpoint and hence a second flip. Four scalar vertices flip four times, and so on. An even number returns the fermion to its original chirality. The conclusion is not restricted to the simplest loop drawn on the board.

A nonzero mass correction becomes possible only after the diagram already contains a mass insertion. For example, photon emission and reabsorption can surround one $m_0\bar{\psi}\psi$ insertion. The gauge vertices contribute two powers of the electric charge, but the sole chirality flip contributes the original mass:

$$\begin{aligned} m_e &= m_0 + c_1 e^2 m_0 + c_2 e^4 m_0 + \cdots \\ &= m_0 \left(1 + c_1 e^2 + c_2 e^4 + \cdots\right). \end{aligned} \tag{1.35}$$

If $m_0 = 0$, the result remains zero. If m_0 is small, every correction is proportional to that small number. We may still ask why the electron is light relative to a fundamental scale, but we are no longer subtracting enormous unrelated numbers to obtain it. In this sense a small fermion mass is technically natural.

The mass insertion makes the allowed sequence explicit. Begin with L. Photon emission preserves L; the insertion m_0 changes L to R; photon reabsorption preserves R. The diagram now has the external chirality of a mass term, but its value necessarily contains the factor m_0. This is the crucial contrast with Eq. (1.18), whose scalar correction remains nonzero even when the starting scalar mass is set to zero.

Gauge bosons have an analogous protection: gauge symmetry forbids an independent mass term, so a gauge boson that begins massless remains massless to all perturbative orders. In the Standard Model, the masses of fermions and the weak gauge bosons arise in response to the Higgs expectation value. Control the Higgs scale, and the remaining light masses do not require separate copies of the same catastrophic tuning.

This identifies the question supersymmetry is meant to address: can a theory prevent the Higgs scalar from receiving an enormous additive mass correction? The answer is yes in suitably supersymmetric settings, because bosonic and fermionic contributions are related rather than independent. The detailed mechanism begins in the next lecture.

1.9 The Other Fine-Tuning: Vacuum Energy

There is one still more severe renormalization problem, but it becomes observable only when gravity is included. In nongravitational physics, adding a constant to every energy changes no experiment. The electron mass, for example, is not an absolute energy; it is the extra energy required to add an electron to the vacuum:

$$m_e = E(\text{vacuum} + \text{one electron}) - E(\text{vacuum}) \tag{1.36}$$

in the electron rest frame. If the vacuum and every excited state shift together, their differences remain unchanged.

Nevertheless, vacuum-to-vacuum diagrams do shift the vacuum energy. In $\lambda\phi^4$ theory, join the four legs of one quartic vertex into two short loops. Each coincident propagator supplies $1/\delta^2$, so

$$\delta\rho_{\text{vac}} \sim c_{\text{vac}} \frac{\lambda}{\delta^4}. \tag{1.37}$$

More complicated vacuum diagrams carry different coupling factors and coefficients, but dimensional analysis typically leaves the same quartic cutoff sensitivity for an energy density.

The absence of external lines is exactly what identifies the correction. Two-point graphs compare a one-particle state with the vacuum and alter a mass; four-point graphs alter scattering; a graph with no incoming or outgoing particles changes the vacuum-to-vacuum amplitude itself. One quartic vertex supplies λ, and its two short loops supply δ^{-2} apiece. Dimensional analysis then requires the δ^{-4} energy-density scale.

Without gravity one may choose the additive zero of energy and ask who cares. General relativity removes that freedom: stress-energy sources spacetime curvature, so a uniform vacuum energy makes empty space gravitate. It appears as a cosmological constant and is connected observationally with dark energy. A tiny measured vacuum energy must then survive contributions of the form δ^{-4}.

This is why the question changes when gravity is turned on. In ordinary quantum mechanics, shifting every level by the same constant leaves all transition frequencies untouched. In gravity, the constant is not inert: the vacuum stress tensor is proportional to the metric and curves spacetime. What had been an arbitrary zero of energy becomes a source in Einstein's equation.

Thus there are at least two extreme naturalness puzzles. The lecture quotes roughly 123 decimal places for the vacuum-energy tuning and roughly 30–35 for the Higgs mass tuning. The exact counts depend on the ultraviolet scale and conventions, but the contrast is not in doubt. Supersymmetry is a potential response to the Higgs problem discussed here; once supersymmetry is broken, it does not by itself explain the observed cosmological constant.[2]

[2] Editorial clarification: This is also why an exactly supersymmetric boson–fermion cancellation is not a phenomenological solution to the observed vacuum energy: supersymmetry must be broken in our world.

1.10 Closing Classroom Questions

Question from the audience. The Higgs correction depends on the cutoff δ, but the cutoff seems arbitrary. Could it be the Planck length, ten times the Planck length, or something else? Does that make the fine-tuning arbitrary?[a]

Response. The numerical contribution depends on the shortest scale to which the effective theory is trusted. That dependence is physical bookkeeping, not an escape. Modes down to any scale where the Standard Model still makes sense are modes whose effects must be included. Choose δ at the distance where new physics is expected—perhaps the Planck scale, perhaps a unification scale near 10^{16} GeV. The contribution accumulated down to that scale is already a lower bound on the problem. Extending the same theory to still shorter distances only makes the quadratic sensitivity larger. We are obligated to explain what cancels the contribution from every scale between the laboratory and that ultraviolet boundary.

[a]Lecture timestamp: 01:33:50.

Question from the audience. What defines a scalar particle, and what is an example?[a]

Response. The Higgs is the intended example. A scalar does not transform into different components under a spatial rotation; equivalently, its particle quanta have spin zero. A photon is described by a vector field, and the gravitational field by a tensor. At the time of this 2010 lecture, no elementary spin-zero particle had yet been detected.[b]

[a]Lecture timestamp: 01:37:26.

[b]Editorial clarification: The Higgs boson was discovered at CERN in 2012 and its measured properties are consistent with spin zero.

Question from the audience. Is the Z particle also a gauge boson?[a]

Response. Yes. The Z, W^+, and W^- are spin-one gauge bosons, as is the photon. Their field equations and interactions are non-Abelian or electroweak generalizations of the Maxwell-like gauge description.

[a]Lecture timestamp: 01:38:31.

Question from the audience. Why does the bosonic ground-state contribution not cancel the fermionic one in the vacuum energy?[a]

Response. Fermion loops and corresponding boson loops can indeed enter with opposite signs. Cancellation is not automatic, however: the particle masses, multiplicities, and couplings must match. Even a small mismatch leaves a large cutoff-sensitive residue, and the infinite families of diagrams must match order by order. Exact supersymmetry enforces just such boson–fermion relations. Generic nonsupersymmetric particle content does not, and broken supersymmetry leaves a residual vacuum-energy problem.

[a]Lecture timestamp: 01:39:10.

1.11 Perspective

The route through the lecture is deliberate. A molecular calculation first shows that eliminating fast variables is neither neglect nor magic: their ground-state energy becomes a potential for the slow variables. Quantum field theory repeats that operation across wavelengths. Dimensional analysis then reveals which effective parameters are sensitive to the cutoff. Scalar masses receive additive terms proportional to δ^{-2}; a small Higgs scale therefore appears unnaturally delicate. Fermion masses are multiplicatively renormalized because chirality protects the zero-mass limit, and gauge symmetry similarly protects massless gauge bosons. Vacuum energy is more sensitive still, scaling as δ^{-4}, and gravity makes its absolute value observable.

Supersymmetry enters the course at exactly this point, not as ornamental new mathematics but as an attempt to make the scalar sector behave more like the protected sectors. The next task is to understand what symmetry can relate bosons and fermions strongly enough to organize those cancellations.

Chapter 2

Fermion Signs and Supersymmetric Cancellations

Overview. Supersymmetry relates bosons to fermions, but the reason for wanting such a relation is easier to see before writing its algebra. We begin with the topology of rotations, make the sign of a fermion wavefunction observable through interference, derive the minus sign of a closed fermion loop from exchange antisymmetry, and then use the opposite bosonic and fermionic loop signs to motivate cancellations of vacuum energy and scalar self-energy. The second half asks how badly supersymmetry may be broken, how one might detect superpartners, and why exact supersymmetry would erase ordinary atomic shell structure.

2.1 Before Supersymmetry: A Peculiarity of Rotations

Supersymmetry is a symmetry between fermions and bosons. Before trying to construct it, however, it is worth recalling how differently these two kinds of particles respond to rotations. That question itself begins with a topological fact that is easy to demonstrate and surprisingly difficult to forget once one has seen it.

Hold one hand palm upward and rotate the wrist through 2π. The hand returns to its original orientation, but the arm plainly has not returned to an equivalent configuration: it is twisted. Continue through another 2π, moving the arm around as necessary, and the twist can be removed. Thus a 2π rotation is not continuously deformable to no rotation while the surroundings are held fixed, whereas a 4π rotation is. The hand demonstration is a proof by hand-waving in the most literal sense.

The same point can be made without relying on anatomy. Put a ball in a fixed box and attach the ball to the walls with any number of flexible, stretchable, but unbreakable strings. Rotate the ball once through 2π. The strings become tangled, and no motion of the strings alone restores the original configuration. Rotate the ball through a total angle 4π, and the strings can be carried around it until the tangle disappears. The walls stand for spatial infinity: the comparison is meaningful only because the boundary is not rotated along with the ball.

Question from the audience. Could the tangle be removed by rotating the box? [a]

Response. That changes the problem. The box represents spatial infinity and is held fixed. We are comparing a rotation of the object relative to its surroundings, not a common rotation of object and surroundings.

[a]Lecture timestamp: 00:02:53.

Question from the audience. Why not simply rotate the ball by another 2π? [a]

Response. That is precisely what distinguishes the two cases. Without rotating the ball further, the tangle left by one 2π turn cannot be removed. After the second turn, giving a total of 4π, it can.

[a]Lecture timestamp: 00:03:20.

In modern group language, the rotation group has two topological classes of closed paths:

$$\pi_1(SO(3)) \cong \mathbb{Z}_2. \tag{2.1}$$

A single traversal is nontrivial, but traversing twice gives the trivial class. Equivalently, $SU(2)$ is the double cover of $SO(3)$. This is the geometric prehistory of the fermion sign.
[1]

2.2 What a 2π Rotation Does to a Quantum State

Now nail down the position of a particle and retain only its spin state, denoted by $|\psi\rangle$. Integer-spin particles are bosons and half-integer-spin particles are fermions. A physical rotation can be implemented by putting a charged spin in a strong magnetic field and rotating the field, or by allowing the spin to precess through the desired angle in a fixed field.

Question from the audience. How can an electron spin actually be rotated in the laboratory? [a]

Response. A magnetic field supplies the handle. One may rotate a sufficiently strong field so that the spin follows it, or let the spin precess around a fixed field. Grabbing the electron by hand is not a competitive experimental method.

[a]Lecture timestamp: 00:05:26.

Suppose a full 2π rotation multiplies the state by a phase:

$$R(2\pi)\,|\psi\rangle = \xi\,|\psi\rangle\,. \tag{2.2}$$

Performing the operation twice gives

$$R(4\pi)\,|\psi\rangle = \xi^2\,|\psi\rangle\,. \tag{2.3}$$

Since the 4π path is topologically trivial, it must act as the identity in this comparison. Therefore

$$\xi^2 = 1, \qquad \xi = +1 \text{ or } -1. \tag{2.4}$$

[1]Editorial clarification: The language of $SO(3)$, $SU(2)$, and the fundamental group makes the topology explicit. The arm demonstration and the string demonstration establish the same two-valued behavior geometrically.

Bosons realize the plus sign and fermions the minus sign:

$$R(2\pi)\,|\psi_B\rangle = +\,|\psi_B\rangle\,, \tag{2.5}$$
$$R(2\pi)\,|\psi_F\rangle = -\,|\psi_F\rangle\,. \tag{2.6}$$

For a state of definite spin j, the compact statement is

$$R(2\pi) = (-1)^{2j}. \tag{2.7}$$

The transcript briefly says that the second possibility is multiplication by 2π; the surrounding derivation and the board argument make clear that the intended factor is -1.

At first this seems physically empty. If an entire wavefunction changes sign,

$$\Psi \longmapsto -\Psi, \tag{2.8}$$

then probabilities are unchanged because

$$(-\Psi)^*(-\Psi) = \Psi^*\Psi. \tag{2.9}$$

Take two ensembles of identically prepared electrons, rotate every member of one ensemble by 2π, and leave the other ensemble alone. No measurement on an isolated member can reveal which ensemble it came from. An overall phase is not an observable.

2.3 Making the Minus Sign Relative

The sign becomes observable when it is imposed on one coherent branch of a wavefunction but not the other. Send electrons one at a time through a two-path apparatus. Do not determine which path is taken, because which-path information destroys interference. Let the upper and lower amplitudes be ψ_1 and ψ_2, so that the recombined state is

$$\Psi = \psi_1 + \psi_2. \tag{2.10}$$

Imagine temporarily trapping each branch in a magnetic region. Rotate only the field associated with the second branch through 2π. If the particle is a fermion, the state becomes

$$\Psi' = \psi_1 - \psi_2. \tag{2.11}$$

This is no longer a global sign. The relative phase shifts interference maxima into minima. If the two returning packets are mirror images about the middle of the detector, the antisymmetric combination vanishes at the center:

$$\Psi'(0) = \psi_1(0) - \psi_2(0) = 0. \tag{2.12}$$

The electron is therefore never detected at that symmetry point. Rotating both branches would instead produce $-\psi_1 - \psi_2 = -\Psi$, which has no observable effect.

The lecture emphasizes the relational nature of the experiment. Rotating one magnetic region relative to the other is analogous to rotating the ball while holding the box fixed. Rotating both regions together is analogous to rotating ball and box together. A relative 4π rotation removes the effect because the fermion state then returns with a plus sign. The experiment was proposed by Yakir Aharonov and Leonard Susskind in 1967 and was later realized in interferometric form, as the lecture notes in passing. The same principle can be applied to electrons or charged ions: a shifted fringe pattern identifies a half-integer-spin state, while an unshifted pattern identifies an integer-spin state.

Question from the audience. If the relative field is rotated through 4π, does the shifted pattern return to the original one? [a]

Response. Yes. Two sign changes multiply to $+1$, so the relative phase responsible for exchanging maxima and minima disappears.

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[a]Lecture timestamp: 00:12:21.

Question from the audience. How literally should the strings in the box be associated with a fermion? [a]

Response. They are a topological mnemonic, not physical strings attached to an electron. A boson behaves as though a 2π turn leaves no memory; a fermion retains precisely the sign memory represented by the nontrivial tangle.

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[a]Lecture timestamp: 00:13:43.

Question from the audience. Are there two electrons in the two magnetic regions, one rotated and one not? [a]

Response. No. There is one electron in a coherent spatial superposition. It has an amplitude to occupy either branch, and the apparatus must not reveal which branch it occupied. Rotating one region changes one amplitude relative to the other; rotating both changes only the common phase.

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[a]Lecture timestamp: 00:14:29.

The angle 2π is special because the spin orientation returns to itself while the spinor acquires a sign. A generic relative rotation changes the spin orientation as well. If the two path states return with spin states $|\chi_1\rangle$ and $|\chi_2\rangle$, the interference term is proportional to

$$\langle \chi_1 \,|\, \chi_2 \rangle \,. \tag{2.13}$$

Its magnitude controls fringe visibility.[2] Orthogonal spin states carry perfect which-path information and do not interfere.

Question from the audience. What happens if one measures which magnetic region contained the electron? [a]

Response. The interference pattern is destroyed. Any record that distinguishes the two paths removes the coherent addition of their amplitudes.

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[a]Lecture timestamp: 00:18:03.

[2]Editorial clarification: The spin-overlap formula is a standard completion of the lecture's qualitative discussion of partial and complete loss of interference.

Question from the audience. If all spins begin in the same direction and the final spin is measured, does the 2π operation itself change that direction? [a]

Response. No. After 2π the physical spin direction is the same; only the fermionic branch has acquired a minus sign. That is why this operation can change interference without leaving a different spin orientation as a path marker.

[a]Lecture timestamp: 00:18:17.

Question from the audience. What if the field is turned only through π, rather than 2π? [a]

Response. Then the spin direction changes. An up-spin branch and a down-spin branch are distinguishable and cannot interfere. Intermediate angles reduce the visibility continuously; a 180° relative turn destroys it completely.

[a]Lecture timestamp: 00:18:43.

2.4 Exchange Symmetry and the Sign of a Closed Loop

The rotation discussion has isolated one fermionic minus sign. A second, closely related sign appears when identical particles are exchanged. For a two-particle wavefunction,

$$P_{12}\Psi_B(1,2) = +\Psi_B(1,2) = +\Psi_B(2,1), \tag{2.14}$$
$$P_{12}\Psi_F(1,2) = -\Psi_F(1,2) = -\Psi_F(2,1). \tag{2.15}$$

Bosonic states are symmetric and fermionic states antisymmetric.

We can now determine the relative sign of a closed particle loop without evaluating its integral. Consider a vacuum Feynman diagram with no external legs. It may be pictured as a particle tracing a closed path in spacetime, or as a particle-antiparticle pair being created and later annihilated. Such a vacuum diagram contributes to the vacuum amplitude and hence to the vacuum energy.

Start with two disconnected copies of the loop. Disconnected diagrams multiply, so if one loop has sign σ, the pair has sign

$$\sigma^2 = +1. \tag{2.16}$$

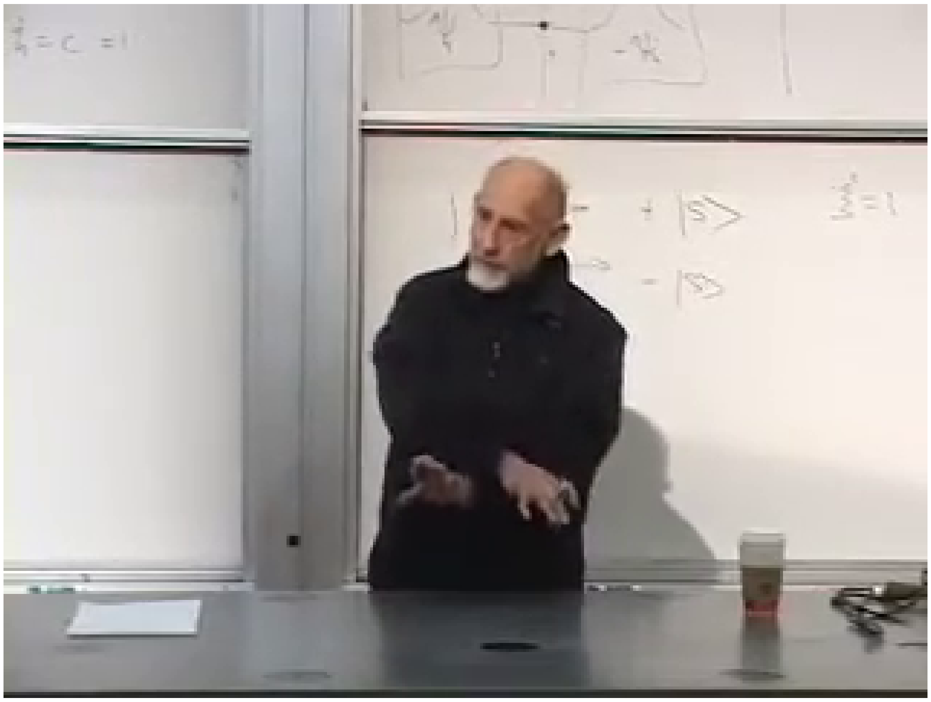
Cut both loops at a common time, say $t = 0$. At that slice there are two identical particles. Reconnect them without exchange and the two separate loops return. Exchange the two particles before reconnecting and the topology becomes one loop. The exchange contributes $+1$ for bosons and -1 for fermions. Since the starting two-loop sign was positive, the resulting one-loop signs are

$$\begin{aligned} \text{sign(closed boson loop)} &= +1, \\ \text{sign(closed fermion loop)} &= -1. \end{aligned} \tag{2.17}$$

More generally, a diagram with L_F independent closed fermion loops carries the factor

$$(-1)^{L_F}. \tag{2.18}$$

The lecture deliberately derives this rule from exchange rather than invoking Grassmann integration. The elementary argument makes the physical origin of the sign visible.



Lecture frame: 00:24:27.

Figure 2.1: The exchange argument in state notation. Only the ket label is securely legible in the frame; the antisymmetry equations are typed in the text rather than attributed to obscured chalk.

2.5 Opposite Loop Signs and Vacuum Energy

A bosonic vacuum loop raises the vacuum contribution with one sign, while a fermionic loop enters with the opposite sign. This is immediately suggestive because the observed dark-energy density is extremely small compared with naive cutoff estimates in quantum field theory. The lecture quotes the notorious discrepancy as roughly 123 orders of magnitude. The precise exponent depends on which ultraviolet scale is used; the durable point is the fantastic mismatch.

The value of a loop depends on more than statistics. It depends on mass, charge, multiplicity, and interactions. Yet if a boson and fermion match in all relevant properties, their equal-magnitude contributions can cancel:

$$\begin{gathered} m_B = m_F, \qquad q_B = q_F, \qquad g_B = g_F, \\ \Delta\rho_{\text{vac}}^{(B)} + \Delta\rho_{\text{vac}}^{(F)} = 0. \end{gathered} \tag{2.19}$$

This observation does not solve the cosmological-constant problem. It merely supplies something that ordinary bosonic loops alone could not supply: contributions of the opposite sign. In an exactly unbroken globally supersymmetric theory, vacuum-energy cancellations can indeed be exact under the appropriate assumptions; after supersymmetry is broken, the observed tiny nonzero value remains unexplained.[3]

Exact matching is already known not to describe nature. If the electron had a spin-zero partner with the same mass and electric charge, electron-positron annihilation through a virtual photon

[3] Editorial clarification: This qualification separates the lecture's motivation from the much harder cosmological-constant problem.

could produce a pair of them:

$$e^- e^+ \longrightarrow \gamma^* \longrightarrow \widetilde{e}^- \widetilde{e}^+. \tag{2.20}$$

A degenerate charged scalar would not have escaped ordinary collider experiments. Thus the boson-fermion reflection cannot be an exact symmetry of the observed spectrum. This is a reflection in particle type, not a reflection in physical space.

2.6 The More Immediate Target: Scalar Self-Energy

Vacuum energy is the first illustration, but the lecture's main concern is the radiative instability of scalar masses. A scalar that begins with zero bare mass can receive a quadratically cutoff-sensitive correction. Schematically,

$$\Delta m_H^2 \sim \frac{g_B^2}{16\pi^2}\Lambda^2 \tag{2.21}$$

from a bosonic loop. Fermion masses have a different protection: when a massless fermion has an exact chiral symmetry, radiative corrections cannot generate an additive mass term. That was the left-right argument of the preceding lecture.

Now imagine a symmetry rigid enough to place a scalar and a fermion in one multiplet and to match their interactions. A fermion loop then appears with the opposite sign:

$$\Delta m_H^2 \sim \frac{g_B^2 - g_F^2}{16\pi^2}\Lambda^2 + \cdots . \tag{2.22}$$

If supersymmetry enforces $g_B = g_F$ and equal masses, the quadratic terms cancel. The fermion's protection is thereby transmitted to its bosonic partner. This is not merely a numerical coincidence between two diagrams; the symmetry must enforce the cancellation through every order and every allowed interaction.

The broad blackboard comparison shows the lecture moving from isolated loops to paired loop corrections of propagating particles.

After the classroom break, the argument restarts with what is called the zeroth-order picture. This is not an order in perturbation theory. It means the roughest useful account:

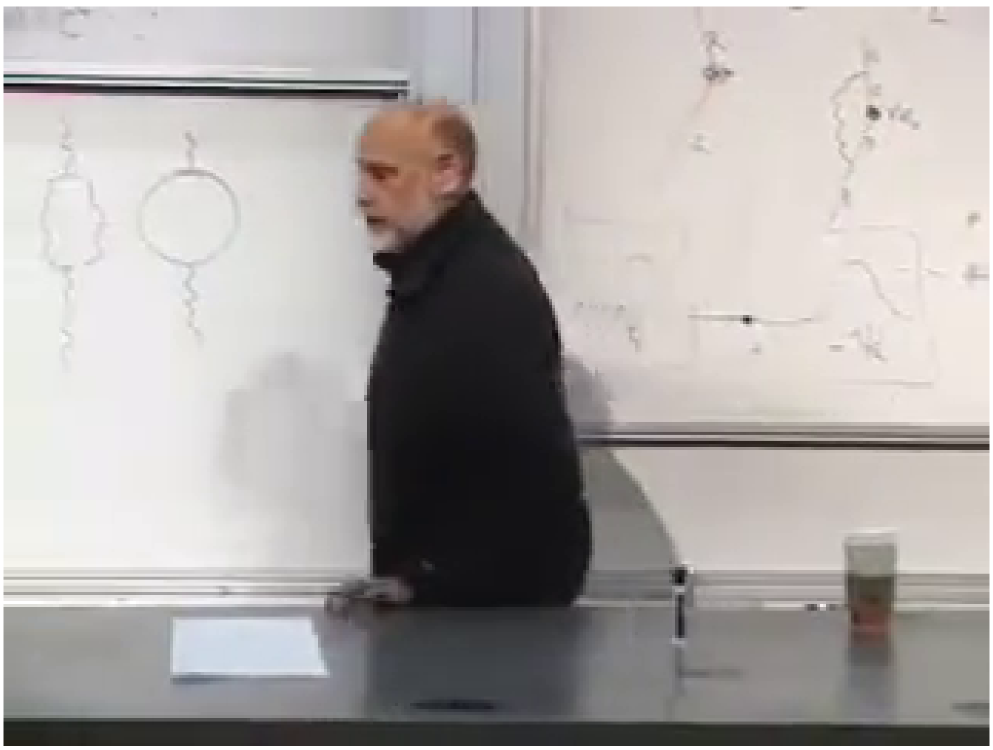
> For every particle there is a superpartner of the opposite statistics. Every fermion has a bosonic partner, and every boson has a fermionic partner.

In an exact supersymmetric theory the partners would also be degenerate, meaning equal in mass. Nature is not degenerate in this way. Still, if the masses are unequal but not too far apart, the cancellation may remain good enough to prevent an explosively large scalar correction. With soft supersymmetry breaking, the dangerous scale is replaced schematically by the superpartner splitting:

$$\Delta m_H^2 \sim \frac{g^2}{16\pi^2}\,\Delta m_{\mathrm{SUSY}}^2 \ln\frac{\Lambda^2}{\Delta m_{\mathrm{SUSY}}^2} \tag{2.23}$$

up to model-dependent coefficients.[4]

[4] Editorial clarification: This formula expresses the lecture's qualitative statement that imperfect mass matching leaves a residual correction controlled by the splitting; it is not copied from the board.



Lecture frame: 00:40:23.

Figure 2.2: A board-wide comparison of ordinary and partner loop topologies. The typed sketch preserves only the visible topological contrast; it does not assign obscured particle labels.

2.7 Broken Supersymmetry and the 2010 Experimental Bet

No superpartner had been observed when this lecture was recorded in 2010. The immediate inference was that the partners, if real, must be heavier than the known particles and heavier than the reach of earlier machines. Supersymmetric models also distinguish particles whose masses arise only after electroweak symmetry breaking from states allowed to carry larger symmetry-preserving masses. The lecture offers this as a rationale for why the visible half of a supersymmetric spectrum might be light while the unseen half is heavier, while candidly calling the arrangement a stretch. Susskind states his own expectation that superpartners would be detected, then immediately marks it as a guess rather than a result.

The Large Hadron Collider was therefore central to the lecture's outlook. It was expected to search both for the Higgs boson and for superpartners. These statements must be read in their 2010 setting: the lecture is explaining the experimental bet before the Higgs discovery and before later LHC search results.

Why should a naturalness-motivated superpartner not sit at the Planck scale? Because it would then fail at the task assigned to it. If the boson and fermion masses differ enormously, their loop integrals cease to cancel above the lighter mass. The residual scalar correction is set roughly by the heavy splitting rather than by the weak scale. Supersymmetry at an arbitrarily remote scale is still a mathematical symmetry, but it no longer gives a persuasive explanation for a light Higgs sector.

The lecture translates this into its era's expectation that the relevant splittings should lie within roughly a few hundred GeV and hence within LHC reach. That numerical estimate is historical and model-dependent; the logical chain is more important:

$$\begin{gathered}\text{Higgs protection}\\ \Downarrow\\ \text{effective boson–fermion cancellation}\\ \Downarrow\\ \Delta m_{\text{SUSY}}\ \text{not too large.}\end{gathered} \tag{2.24}$$

Question from the audience. Could all fermionic contributions merely cancel all bosonic contributions in the aggregate, without one-to-one partners? [a]

Response. They could cancel numerically, but that would be an accidental cancellation as finely tuned as the original problem. The supersymmetric proposal is stronger: reasonable postulates enforce matched contributions, while broken supersymmetry explains why the masses are only approximately matched. Vacuum energy was the illustration; the immediate concern here is the Higgs self-energy.

[a]Lecture timestamp: 00:45:01.

Question from the audience. Is calling an aggregate cancellation unnatural already assuming a reductionist way of organizing the theory? [a]

Response. That is a fair challenge. Nobody has a final definition of naturalness, and the search is partly a matter of groping for a compelling explanation. Supersymmetry and spontaneous supersymmetry breaking provide known mathematical frameworks that can enforce the desired pattern. The unresolved questions are whether nature uses that framework and whether it is unique.

[a]Lecture timestamp: 00:47:51.

2.8 Direct Production, Virtual Effects, and Decay Chains

A heavy particle need not be produced on shell to influence an experiment. It can occur virtually inside a loop and slightly modify an ordinary electron-positron scattering amplitude. Precision electroweak measurements therefore constrain superpartners even when the center-of-mass energy is below the direct pair-production threshold

$$\sqrt{s} < 2m_{\widetilde{p}}. \tag{2.25}$$

The lecture's 2010 assessment is that precision data were already squeezing supersymmetric parameter space but had not ruled it out.

Question from the audience. Could superpartners be inferred indirectly, as earlier heavy particles were constrained through precise production rates before direct observation? [a]

Response. Yes. A virtual superpartner loop changes an ordinary amplitude slightly, so the problem becomes high-precision electroweak phenomenology rather than direct production. A clean, sufficiently energetic electron-positron collider would be especially useful. Proton collisions are messier because a proton is a composite state of quarks and gluons; one is really colliding fluctuating constituents, not two elementary particles with sharply known initial conditions.

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[a] Lecture timestamp: 00:49:01.

One student asks about running the reaction in reverse, beginning with abundant superparticles. The obstacle is not time-reversal symmetry but availability. Heavy superpartners must first be created, and most are expected to decay rapidly into lighter states.

Question from the audience. If a universe contained many superparticles, could two of them collide and turn into ordinary particles? [a]

Response. In principle the relevant reactions have reverse channels, but heavy superpartners are not expected to remain freely abundant. They decay toward lower-energy states, so an experiment must first supply enough collision energy to create them.

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[a] Lecture timestamp: 00:53:43.

This raises the dark-matter question. In many supersymmetric models a discrete selection rule prevents the final superpartner in a decay chain from disappearing entirely. Then every heavier superpartner cascades toward the lightest superpartner, or LSP. If the LSP is electrically neutral, it can be stable and difficult to detect, making it a possible dark-matter candidate. The transcript once says "dark energy" in this exchange, but the audience's question and the particle description unambiguously concern dark matter.

Question from the audience. Does rapid superpartner decay conflict with the proposal that a superpartner could be dark matter? [a]

Response. No. Most superpartners may decay while the lightest one remains stable in models that require a superpartner at the end of each decay chain. Electric charge generally raises additional phenomenological difficulties, so the viable LSP is likely neutral. It is not undetectable in principle, but it would not leave the obvious ionizing track of a charged selectron.

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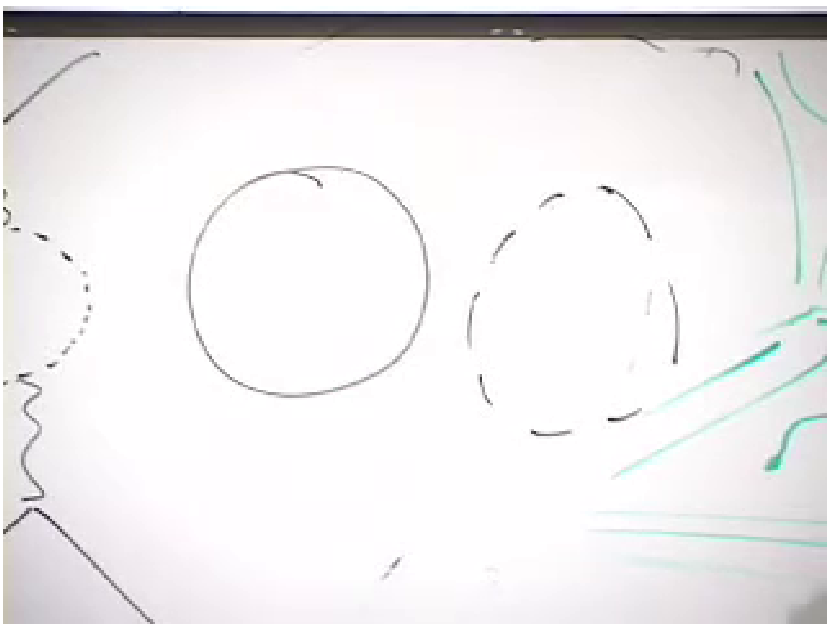
[a] Lecture timestamp: 00:54:53.

2.9 Matching Means Matching the Couplings

Mass matching alone is insufficient. Compare an electron vacuum loop with a selectron loop. Their opposite statistics supply opposite signs, but equal magnitudes require equal masses. Now attach two external photons. The coefficients also contain the electric couplings, so cancellation requires equal charges:

$$\begin{gathered} q_{\tilde{e}} = q_e, \\ \mathcal{A}^{(e)}_{\text{loop}} + \mathcal{A}^{(\tilde{e})}_{\text{loop}} = 0, \\ \text{provided the full couplings match.} \end{gathered} \tag{2.26}$$

A mismatch at any level leaves an uncancelled divergent term. Supersymmetry is valuable because it relates interactions as well as names and spins.



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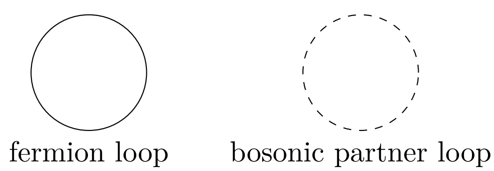

Figure 2.3: Solid and dashed loops compared during the charge-matching argument. The cropped external marks are not reconstructed because the frame alone does not determine every attachment.

Question from the audience. Could superpartners have escaped detection simply because their coupling constants are different? [a]

Response. Not if the same diagrams are to cancel. Adding photon interactions tests the electric charge of each line. Unequal couplings give unequal numerical loop factors and restore the large remainder. The symmetry must match the appropriate charges and interactions.

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[a] Lecture timestamp: 00:56:23.

Question from the audience. Should the photon's superpartner also be massless? [a]

Response. It would be massless in an exact supersymmetric theory because it would be degenerate with the photon. The observed world is not exactly supersymmetric, so a physical photino-like state need not share the photon's mass.

[a]Lecture timestamp: 00:58:01.

2.10 The Superpartner Dictionary

The naming rules are inelegant but useful. A boson receives a fermionic partner whose name often ends in *-ino*: photon and photino, the W and wino, the Z and zino, gluon and gluino, Higgs and Higgsino, graviton and gravitino. An ordinary fermion receives a bosonic partner whose name usually begins with *s-*: electron and selectron, quark and squark.

ordinary state	superpartner and spin change
scalar Higgs, $j = 0$	Higgsino, $j = \frac{1}{2}$; $\Delta j = +\frac{1}{2}$
gauge boson, $j = 1$	gaugino, $j = \frac{1}{2}$; $\Delta j = -\frac{1}{2}$
matter fermion, $j = \frac{1}{2}$	sfermion, $j = 0$; $\Delta j = -\frac{1}{2}$
graviton, $j = 2$	gravitino, $j = \frac{3}{2}$; $\Delta j = -\frac{1}{2}$

The general relation is that members of a supermultiplet differ by half a unit of spin:

$$\Delta j = \frac{1}{2}. \tag{2.27}$$

The rough lecture examples pair a spin-$\frac{1}{2}$ matter fermion with a scalar and a spin-one gauge boson with a spin-$\frac{1}{2}$ gaugino. A complete interacting model can mix states after symmetry breaking, so the physical mass eigenstates need not retain these simplest names.

Question from the audience. Could all the new scalar partners undergo spontaneous symmetry breaking of their own? [a]

Response. Scalar fields can acquire expectation values, but arbitrary extra symmetry breaking would be phenomenologically disastrous. A viable supersymmetric model must arrange its scalar potential so that color, electric charge, and other required symmetries are not broken in the wrong way.

[a]Lecture timestamp: 01:01:27.

Question from the audience. Does the Higgsino have special interactions corresponding to the Higgs quartic term? [a]

Response. Supersymmetry matches interactions as well as particles. The relation is not obtained by naively replacing every Higgs line with a Higgsino line, but the scalar quartic and the relevant fermionic and auxiliary interactions belong to one constrained structure. Those details have not yet been developed in the lecture.

[a]Lecture timestamp: 01:02:19.

Question from the audience. Does the 2π-versus-4π rotation theorem have analogues in other dimensions? [a]

Response. Yes, but dimensionality changes the possibilities. In two spatial dimensions exchange paths can wind around one another, allowing anyons whose exchange phases are more general than the bosonic $+1$ and fermionic -1. Such excitations occur in condensed-matter systems, but they are outside the present course's route.

[a] Lecture timestamp: 01:03:07.

2.11 Why Exact Supersymmetry Would Erase Chemistry

We finally ask what an exactly supersymmetric world would be like. Exactness here means equal masses and matched interactions for every particle-superpartner pair. The electron would have a bosonic selectron with the same mass and electric charge, and the massless photon would have a massless fermionic photino:

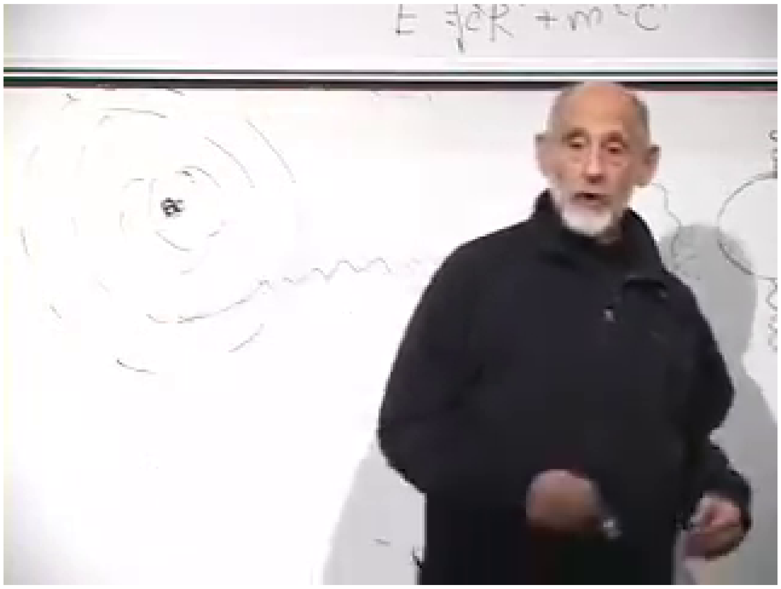
$$m_{\widetilde{e}} = m_e, \qquad q_{\widetilde{e}} = q_e, \qquad m_{\widetilde{\gamma}} = m_\gamma = 0. \tag{2.28}$$

The familiar periodic structure of chemistry depends on Fermi statistics. Hydrogen has one electron in its lowest orbital. Helium accommodates two there because the spin labels differ. Lithium's third electron must occupy a higher orbital: the Pauli exclusion principle prevents three identical electrons from occupying the same one-particle state. That outer electron has more energy and would prefer to fall inward, ordinarily emitting a photon, but the filled lower states block the transition.

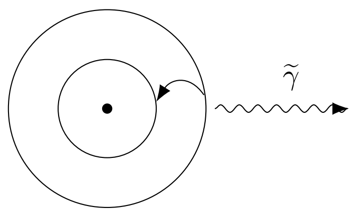
Exact supersymmetry opens another channel:

$$e^-_{\text{outer}} \longrightarrow \widetilde{e}^-_{\text{inner}} + \widetilde{\gamma}. \tag{2.29}$$

The initial electron and emitted photino are fermions, while the remaining selectron is a boson. Because selectrons obey Bose rather than Fermi statistics, the final selectron is not excluded from an already occupied low orbital. The valence electron can therefore convert and fall inward.



Lecture frame: 01:09:18.



selectron drops to a lower orbital

Figure 2.4: The atomic transition used to show why exact supersymmetry would change chemistry. The outgoing wavy line is visible; its identification as a photino comes from the transcript.

The point is not confined to a lithium-like atom. In a large atom, valence electrons are what create chemical variety. If they can repeatedly convert into bosonic selectrons, they can crowd into the lowest energetically available region rather than filling shells. The result would resemble a concentrated bosonic population near the nucleus, not the familiar architecture of valence orbitals. Molecules built from directional bonds and partially filled shells would not survive in recognizable form. The lecture calls the hypothetical lithium descendant *slithium*, but the joke carries a serious conclusion: exact supersymmetry would produce matter unlike ordinary chemistry and therefore unlike ordinary biology.

Question from the audience. Could the electron-to-selectron transition be so weakly coupled that ordinary atoms would survive? [a]

Response. Not in the exact theory being imagined. Supersymmetry relates the transition to the electromagnetic interaction, so its coupling is fixed by the electric charge rather than freely chosen to be tiny. The lecture gives a rough lifetime of order a microsecond for a lithium-like atom, not as a precision calculation but to emphasize that the decay would be rapid.

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[a] Lecture timestamp: 01:10:23.

We therefore know from everyday matter, quite apart from collider searches, that supersymmetry cannot be exact at low energy. The separation between particle and superpartner masses is enormous compared with atomic or proton scales, even if it would be tiny compared with a unification or Planck scale.

Question from the audience. Why, then, is supersymmetry broken? [a]

Response. The lecture offers a minimal selection statement: if it were not broken, observers built from ordinary chemistry would not be present to ask the question. That does not replace a dynamical theory of supersymmetry breaking. It explains only why an exactly supersymmetric low-energy world could not be ours.

[a]Lecture timestamp: 01:11:19.

Breaking supersymmetry is mathematically possible and in many constructions rather generic, though arranging realistic breaking is far from automatic. Conversely, exactly supersymmetric quantum field theories possess enough special structure that some can be analyzed much more deeply than generic theories. They serve both as candidates for organizing particle physics and as unusually powerful mathematical laboratories. The next lecture turns from this physical motivation toward more of that structure.

2.12 The Thread of the Argument

The route to supersymmetry has been deliberately indirect:

1. A 2π spatial rotation belongs to a nontrivial topological class, while a 4π rotation is deformable to the identity.
2. A fermion state acquires a minus sign under 2π, and a two-path experiment makes that sign observable when it is relative rather than global.
3. Fermionic antisymmetry under exchange gives every closed fermion loop an extra minus sign relative to a boson loop.
4. Opposite loop signs permit symmetry-enforced cancellations of vacuum and scalar self-energy contributions when masses and couplings match.
5. Exact matching is absent in nature. If supersymmetry protects the Higgs sector, its breaking cannot make the relevant partners arbitrarily heavy.
6. Exact low-energy supersymmetry would allow electrons to convert into bosonic selectrons and collapse the shell structure on which ordinary chemistry depends.

Supersymmetry is therefore attractive and impossible in its naive exact form for the same reason: it ties bosons and fermions together tightly. That tight relation can tame dangerous quantum corrections, but if it survived unbroken into atomic physics, the world would cease to resemble the one around us.

Chapter 3

Ultraviolet Divergences and the Supersymmetric Cure

Overview. Before introducing the formal mathematics of supersymmetry, we should be precise about the problem it is meant to solve. We shall describe ultraviolet divergences in both position space and momentum space, derive the short-distance behavior of scalar and fermion propagators by dimensional analysis, and estimate the leading loop corrections to a scalar mass. A scalar loop and a fermion loop have the same quadratic cutoff dependence but opposite signs. Supersymmetry becomes compelling when we ask what principle could make their couplings, multiplicities, and masses match well enough for that cancellation to be structural rather than accidental.

3.1 Why Return to Divergences?

The intention was to begin the machinery of supersymmetry immediately. It is better, however, to delay the abstraction for one lecture and rebuild the motivation. The earlier account of divergences and mass renormalization went by quickly, and supersymmetry will seem like an arbitrary mathematical invention unless we first see exactly what needs to be repaired.

Feynman diagrams enter particle physics in two closely related ways. Most familiarly, they organize scattering amplitudes. We specify incoming and outgoing particles and sum the amplitudes represented by all diagrams of the required order. They also organize the construction of an *effective Lagrangian*. Here the word "effective" is not a compliment meaning unusually successful. It means that the description is approximate: inaccessible short-distance details have been eliminated or integrated out, and their influence is encoded in the parameters and interactions that remain.

Every such description has a range of validity. We may specify that range by a shortest trustworthy distance, denoted here by δ, or by a largest trustworthy momentum,

$$\Lambda \sim \delta^{-1}. \tag{3.1}$$

The two cutoffs are not separate assumptions. They are Fourier-dual descriptions of the same ultraviolet limit.

3.2 One Propagator, Two Descriptions

The propagator is the amplitude for creating or inserting a particle at one spacetime point and removing or detecting it at another. Let the two points be x^μ and y^μ, and write their separation as

$$\Delta^\mu = y^\mu - x^\mu. \tag{3.2}$$

We reserve the lower-case symbol δ for the cutoff, so that the separation and the regulator cannot be confused.

Translation invariance implies that a free propagator depends on x and y only through Δ. The same object can be Fourier transformed:

$$G(\Delta) = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot\Delta}\, \widetilde{G}(k). \tag{3.3}$$

When the question "What are the Fourier conjugates to position?" is put to the room, the answer is momentum. Consequently, the position-space description of a loop as a propagator returning to its starting point is equivalent to the momentum-space description as an integral over every momentum that can circulate around the loop. Momentum-space rules require one to impose momentum conservation at every vertex; position space keeps the present power-counting argument more transparent.

This equivalence identifies the ultraviolet problem before we calculate anything. In position space, a loop probes the coincident-point limit $\Delta \to 0$. In momentum space, it probes the limit $|k| \to \infty$. Large momentum means short wavelength, so a short-distance singularity and a high-momentum divergence are the same phenomenon in different coordinates.

The lecture treats the simplest spin classes: scalar fields, fermion fields, and, in the background, gauge fields. Scalars have spin zero and are represented by a one-component field $\phi(x)$. When the lecture was recorded in 2010, the Higgs had not yet been discovered and was described as the principal scalar expected in particle physics. That historical setting explains the playful phrase "higgly-wiggly"; the field was physically urgent but not yet experimentally established.

For a real scalar field, the propagator may be written schematically as

$$G_\phi(x, y) = \langle 0|T\{\phi(x)\phi(y)\}\,|0\rangle = G_\phi(\Delta). \tag{3.4}$$

The time-ordering convention and normalization will not matter for our power counting. The important fact is that the two field insertions describe creation at one point and annihilation at the other.

Question from the audience. Are x and y positions in space or in spacetime? [a]

Response. They are spacetime points. Each is a four-vector. The propagator therefore depends on their four-dimensional separation and, in a Lorentz-invariant vacuum, on the corresponding invariant interval.

[a]Lecture timestamp: 00:09:31.

A spin-one-half particle is described by a Dirac field $\psi_i(x)$. The index i is a spinor index, not a spatial component, and the propagator is therefore a matrix:

$$S_{ij}(x, y) = \left\langle 0|T\left\{\psi_i(x)\bar{\psi}_j(y)\right\}|0\right\rangle = S_{ij}(\Delta). \tag{3.5}$$

A question to the room supplies the component count: a four-dimensional Dirac field has four complex components. A massless Weyl field may be described with two complex components, so left- and right-handed fields can be treated separately. For the present dimensional argument, only the existence of the spinor matrix structure matters.

Because S_{ij} is a 4×4 matrix, it can be expanded in the basis of Dirac matrices. We shall suppress those indices at first and restore them when they become necessary. This is not a claim that the matrix structure is absent; it is a deliberate simplification while we establish the ultraviolet power law.

Question from the audience. When does the conjugate or daggered field have to appear in a propagator? [a]

Response. For a real neutral scalar in this schematic discussion, the same field contains creation and annihilation operators, so no distinct daggered field is required. For a charged field, particle and antiparticle operators must be distinguished. Thus ψ, $\psi^\dagger$, and covariantly $\bar{\psi} = \psi^\dagger\gamma^0$, play different roles for an electron.

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[a] Lecture timestamp: 00:14:06.

The physical Higgs particle is neutral, although the underlying Standard Model Higgs field is an electroweak doublet. The lecture is using the real neutral scalar as its local model; that simplification is sufficient for the ultraviolet estimate.[1]

3.3 Natural Units and the Dimension of a Scalar Field

Set $\hbar = c = 1$. Length and time then have the same units, while energy, mass, and momentum have units of inverse length:

$$[m] = [E] = [p] = L^{-1}. \tag{3.6}$$

The action appears in the phase e^{iS}, so it is dimensionless. A representative scalar action in four spacetime dimensions is

$$S_\phi = \int d^4x \left[\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 + \cdots\right]. \tag{3.7}$$

The conventional factors of $1/2$ and $1/4!$ do not affect dimensional analysis.

Begin with the kinetic term. Since $[d^4x] = L^4$ and $[\partial_\mu] = L^{-1}$, dimensionlessness gives

$$1 = L^4\left(L^{-1}[\phi]\right)^2, \tag{3.8}$$

$$[\phi] = L^{-1}. \tag{3.9}$$

The mass term provides a check:

$$L^4[m^2][\phi^2] = L^4L^{-2}L^{-2} = 1. \tag{3.10}$$

The next question to the room is the dimension of the quartic coupling. It is dimensionless in four spacetime dimensions:

$$L^4[\lambda][\phi]^4 = L^4[\lambda]L^{-4} = 1 \quad\Longrightarrow\quad [\lambda] = L^0. \tag{3.11}$$

[1] Editorial clarification: This distinction prevents the schematic real-scalar example from being mistaken for a complete statement about the electroweak Higgs multiplet.

This makes the ϕ^4 interaction special in four dimensions. The dimensions of ϕ^6 and higher interactions follow by the same rule, but their couplings carry negative powers of mass and therefore announce a more restricted effective-theory role.

Thus the field dimensions already tell us which interactions can appear with dimensionless coefficients. The same method will determine both propagators and the Yukawa coupling without evaluating a detailed Feynman integral.

3.4 The Scalar Propagator at Short Distance

Consider first a massless scalar. There is no intrinsic length scale, so the only available scale is the separation Δ. Translation invariance leaves only Δ, Lorentz invariance reduces the dependence to its invariant magnitude, and the product of two scalar fields has dimension L^{-2}. Up to a numerical constant,

$$G_0(\Delta) \sim \frac{1}{\Delta^2}. \tag{3.12}$$

Factors such as $4\pi^2$ cannot be fixed by dimensional reasoning alone.

A mass introduces the Compton length m^{-1}. Its effect is easiest to understand from the dispersion relation

$$E = \sqrt{\boldsymbol{p}^2 + m^2}, \qquad E = |\boldsymbol{p}| \quad (m = 0). \tag{3.13}$$

At $|\boldsymbol{p}| \gg m$, the mass is negligible. Since high momentum probes short distance, the massive propagator must approach the massless $1/\Delta^2$ behavior as $\Delta \to 0$. At distances much larger than m^{-1}, propagation is suppressed. The lecture represents that fact schematically by

$$G_m(\Delta) \sim \frac{e^{-m|\Delta|}}{\Delta^2}. \tag{3.14}$$

This is deliberately not an exact formula. In four-dimensional Euclidean space the exact free propagator is proportional to $mK_1(mr)/r$, with $r = |\Delta|$; it has the same $1/r^2$ ultraviolet singularity and an exponential large-distance decay accompanied by a power-law prefactor.[2]

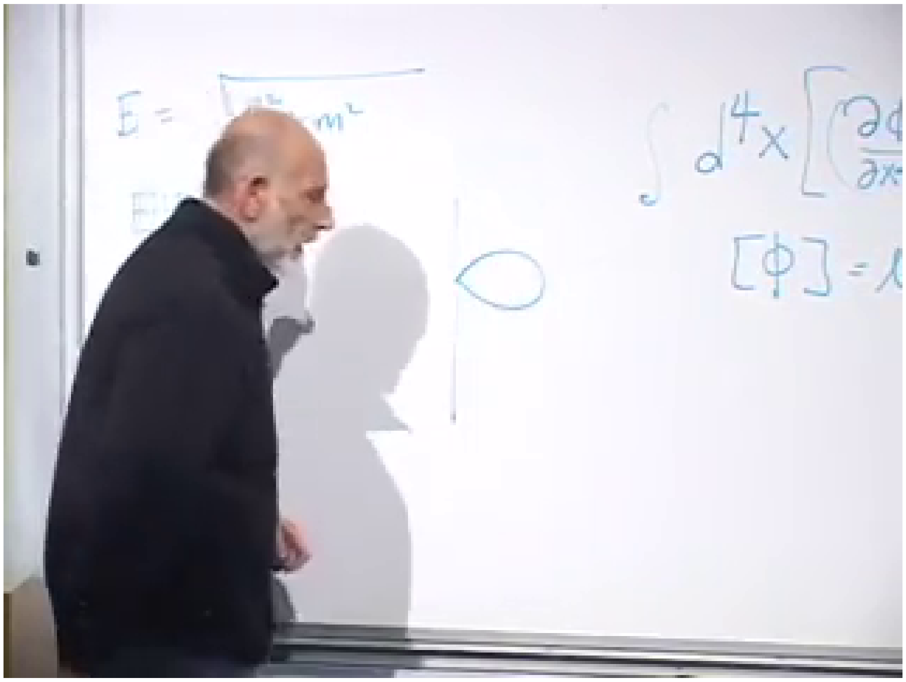
The crucial lesson is simple: mass softens long-distance propagation but does not repair the coincident-point singularity. Near enough to a point, every finite mass becomes irrelevant.

3.5 The Scalar Tadpole and Fine Tuning

The quartic interaction $\lambda\phi^4$ produces a four-line vertex. Join two of its legs to one another and leave two external scalar legs. The resulting tadpole diagram corrects the coefficient of ϕ^2, hence the scalar mass squared.

The preserved frame catches the exact transition from dimensional analysis to this loop correction. The dispersion relation remains on the left, the scalar action and field dimension remain partly visible on the right, and the small tadpole has just appeared in the center.

[2] Editorial clarification: The Bessel-function form supplies the standard precise completion of the lecture's schematic exponential and is not required for its power-counting argument.



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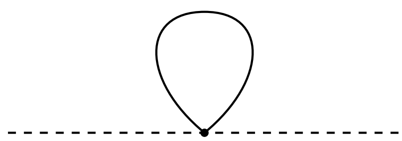

Figure 3.1: The board at the scalar self-energy transition, followed by a conservative schematic redraw of the tadpole topology. The screenshot securely shows the loop, the relativistic and massless energy formulas, and a partial action; obscured coefficients and labels are not claimed as visible.

At the vertex the scalar may be absorbed and re-emitted while the other two legs propagate from the point back to itself. Apart from numerical and combinatorial factors, the correction is

$$\delta m_\phi^2 \sim \lambda G_\phi(0). \tag{3.15}$$

The coincident propagator is infinite in the continuum approximation. An effective theory does not claim validity at arbitrarily short distance, so impose $|\Delta| \geq \delta$. Then

$$G_\phi(0) \longrightarrow G_\phi(\delta) sim \frac{1}{\delta^2}, \qquad \delta m_\phi^2 \sim \frac{\lambda}{\delta^2} sim \lambda\Lambda^2. \tag{3.16}$$

The divergence is called quadratic because it contains two powers of the momentum cutoff. Momentum-space integration gives the same conclusion: excessive contribution comes from arbitrarily large loop momentum. Whether the shortest scale is eventually set by grand unification, quantum gravity, or something not yet imagined does not change the effective-theory diagnosis.

This is a fine-tuning problem. If δ is extremely small and λ is not extraordinarily tiny, the radiative correction is much larger than the scalar mass squared observed at low energy. The bare coefficient and the loop correction must then cancel to implausibly many digits:

$$m^2_{\text{physical}} = m^2_{\text{bare}} + \delta m_\phi^2. \tag{3.17}$$

The massive propagator does not save the situation. At the distances responsible for the divergence, $e^{-m|\Delta|} \simeq 1$. The ultraviolet loop has forgotten the mass long before it reaches the cutoff.

3.6 Fermion Dimensions and Their Propagator

We now repeat the same dimensional analysis for a Dirac fermion. Its free action is

$$S_\psi = \int d^4x\, \bar{\psi}\,(i\gamma^\mu \partial_\mu - m)\,\psi. \tag{3.18}$$

The gamma matrices are numerical and dimensionless. The kinetic term contains only one derivative, so

$$1 = L^4[\bar{\psi}]L^{-1}[\psi], \tag{3.19}$$
$$[\psi] = [\bar{\psi}] = L^{-3/2}. \tag{3.20}$$

Half-integer powers should not be alarming; fermions bring halves into many places.

Question from the audience. Would it not be easier to derive $[\psi]$ from the mass term? [a]

Response. Yes. The mass term gives $L^4L^{-1}[\bar{\psi}\psi] = 1$, hence $[\bar{\psi}\psi] = L^{-3}$ and $[\psi] = L^{-3/2}$. The kinetic derivation was used first because it remains meaningful in the massless theory and displays why the fermion mass appears as m, not m^2.

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[a] Lecture timestamp: 00:36:02.

Two fermion fields have dimension L^{-3}, so the lecture's compact dimensional shorthand for the massless propagator is

$$S(\Delta) \sim \frac{1}{|\Delta|^3}. \tag{3.21}$$

Restoring its matrix structure gives the more precise numerator required by Lorentz covariance:

$$S_{ij}(\Delta) \propto \frac{(\gamma_\mu \Delta^\mu)_{ij}}{(\Delta^2)^2}. \tag{3.22}$$

The numerator adds one power of length, and the denominator therefore contains four powers of the interval; the net dimension remains L^{-3}. As with the scalar, a fermion mass suppresses the propagator at large separation but leaves this short-distance scaling unchanged.

3.7 Why a Closed Fermion Loop Is Negative

We can now ask whether a fermionic diagram might cancel the bad bosonic self-energy. The scalar tadpole has a positive sign, up to positive numerical factors such as powers of π and combinatorial coefficients. A closed fermion loop carries the opposite sign.

Question from the audience. Is the minus sign of the fermion loop related to the earlier discussion of one full rotation? [a]

Response. It belongs to the same fermionic sign structure, but the useful proof here proceeds through exchange antisymmetry rather than the geometric orientation of a loop. Cut two loops, exchange the identical particles, and reconnect them. The exchange turns two loops into one and contributes a minus sign only for fermions.

[a] Lecture timestamp: 00:40:21.

The argument is worth giving in full. Suppose a one-loop diagram has sign σ, either plus or minus. Two disconnected identical loops have sign

$$\sigma^2 = +1. \tag{3.23}$$

Cut both loops at the same time. We now have two identical particles propagating between the cuts. Their ordinary two-particle wavefunction is symmetric for bosons and antisymmetric for fermions:

$$\Psi_B(x_1, x_2) = +\Psi_B(x_2, x_1), \tag{3.24}$$
$$\Psi_F(x_1, x_2) = -\Psi_F(x_2, x_1). \tag{3.25}$$

The antisymmetry also explains the Pauli principle: if both fermions are put in the same state, interchange changes nothing physically but must reverse the wavefunction, forcing it to vanish.

Reconnect the cut lines after exchanging the two particles. Topologically, the two loops have become one. The unexchanged two-loop amplitude was positive. The exchange leaves it positive for bosons and reverses it for fermions. Therefore

$$\begin{aligned} \text{sign(closed boson loop)} &= +1, \\ \text{sign(closed fermion loop)} &= -1. \end{aligned} \tag{3.26}$$

This conclusion is general: each independent closed fermion loop supplies a factor of -1.

3.8 The Yukawa Loop

Return to the scalar mass. A boson cannot turn into a single fermion because fermion number parity would change. The simplest fermionic correction has the scalar split into a fermion-antifermion pair, which propagates around a loop and recombines. The interaction is a Yukawa coupling,

$$\mathcal{L}_Y = -g\phi\bar{\psi}\psi. \tag{3.27}$$

For the Higgs, the fermions may be quarks or leptons. There are two interaction points and hence two factors of g.

The scalar tadpole involved one point, whose overall position could be fixed because translation invariance merely supplies the spacetime volume. The Yukawa loop involves two points. We may fix one point, or equivalently fix their average position, but we must integrate over their relative separation Δ. Each fermion propagator contributes the short-distance factor $1/|\Delta|^3$, so

$$\delta m^2_{\phi,F} \sim -g^2 \int_{|\Delta|\geq\delta} d^4\Delta \, \frac{1}{|\Delta|^6}. \tag{3.28}$$

After the two propagators, the separation integral, and the two couplings have been written, one last factor is requested from the room. The answer is the minus sign. The two propagators close into a fermion loop, so the entire otherwise positive short-distance integral is multiplied by -1.

Dimensional analysis already determines the divergence. Four powers of length come from $d^4\Delta$, six occur in the denominator, and the only remaining length scale is the cutoff:

$$\int_{|\Delta|\geq\delta} \frac{d^4\Delta}{|\Delta|^6} \sim \frac{1}{\delta^2}. \tag{3.29}$$

In four-dimensional Euclidean radial coordinates this can be made explicit:

$$\int_{\delta}^{R} d^4\Delta\, \frac{1}{|\Delta|^6} = 2\pi^2 \int_{\delta}^{R} dr\, \frac{r^3}{r^6} = \pi^2\left(\frac{1}{\delta^2} - \frac{1}{R^2}\right). \tag{3.30}$$

Spinor traces and propagator normalizations alter the coefficient, not the quadratic power of the cutoff.[3]

Question from the audience. Could reversing or choosing the integration limits change the sign? [a]

Response. No. This is not a one-dimensional oriented integral whose limits generate the physical sign. It is an integral over four components, or more cleanly over a positive radial measure. Its ultraviolet magnitude is positive. The negative contribution comes solely from the closed fermion loop.

[a]Lecture timestamp: 00:51:06.

Question from the audience. What is the dimension of the Yukawa coupling g? [a]

Response. It is dimensionless in four dimensions. Since $[\phi] = L^{-1}$ and $[\bar{\psi}\psi] = L^{-3}$, the product $\phi\bar{\psi}\psi$ has dimension L^{-4}, exactly canceled by d^4x.

[a]Lecture timestamp: 00:51:51.

The leading fermionic correction is therefore

$$\delta m^2_{\phi,F} \sim -C_F\frac{g^2}{\delta^2} = -C_F g^2\Lambda^2, \tag{3.31}$$

where $C_F > 0$ contains numerical, spin, and multiplicity factors. The scalar loop has the corresponding form

$$\delta m^2_{\phi,B} \sim +C_B\frac{\lambda}{\delta^2} = +C_B\lambda\Lambda^2. \tag{3.32}$$

We have not yet solved the fine-tuning problem, but we have found something indispensable: a contribution with the same ultraviolet degree and the opposite sign.

3.9 What the Cancellation Requires

It would be an extraordinary accident if unrelated couplings happened to obey

$$C_B\lambda = C_F g^2 \tag{3.33}$$

[3]Editorial clarification: The radial integral is a standard explicit version of the dimensional estimate made in the lecture. It is written in Euclidean distance, where the positivity and ultraviolet power counting are transparent.

to precisely the accuracy required by a large hierarchy of scales. The lecture abbreviates this as $g^2 = \lambda$, suppressing convention-dependent and multiplicity factors. A genuine solution needs a mathematical structure that forces the relation rather than merely permits it.

That is the supersymmetric motive. Supersymmetry relates bosons and fermions and restricts their couplings to combinations that make many otherwise dangerous divergent terms cancel. The cancellation is then repeated across diagrams because it follows from a symmetry, not because parameters have been adjusted one loop at a time.

Mass matching matters as well. If boson and fermion masses are identical, their full propagators can be paired most closely. If the masses differ slightly, the complete finite answers do not cancel exactly, because the propagators differ at distances comparable to their Compton wavelengths. Their common short-distance asymptotics can nevertheless cancel the divergent part:

$$\mathcal{I}_B(\Delta) - \mathcal{I}_F(\Delta) \quad \text{may be finite as} \quad \Delta \to 0, \tag{3.34}$$

where $\mathcal{I}_B$ and $\mathcal{I}_F$ denote the complete bosonic and fermionic loop integrands after vertices, numerators, and multiplicities are included. The propagators themselves have different dimensions and are not being subtracted directly. Even when the ultraviolet terms cancel, a finite residual may remain. This is the opening for approximate or broken supersymmetry. Exact equality gives the cleanest cancellation; sufficiently controlled breaking may still remove the ultraviolet catastrophe while leaving observable mass splittings.

We have therefore reached supersymmetry without yet writing its algebra. The chain of reasoning is complete:

$$\begin{gathered} G_\phi(\Delta) \sim \Delta^{-2}, \qquad \delta m^2_{\phi,B} \sim +C_B \lambda \Lambda^2, \\ S_\psi(\Delta) \sim \Delta^{-3}, \qquad \delta m^2_{\phi,F} \sim -C_F g^2 \Lambda^2, \\ \text{symmetry-enforced matching} \\ \Longrightarrow \quad \text{quadratic divergences cancel.} \end{gathered} \tag{3.35}$$

The next task is to understand the abstract symmetry principle that enforces those relations and to see concrete examples of how it acts.

Chapter 4

From Ordinary Symmetry to Grassmann Algebra

Overview. Supersymmetry is a symmetry, so we begin by recalling what that word means in quantum mechanics. A physical transformation is represented on Hilbert space by a unitary operator; continuous transformations are generated infinitesimally, and their commutators encode how the transformations compose. A genuine symmetry preserves energy, which is equivalent to commuting with the Hamiltonian. Supersymmetry then changes one assumption shared by all familiar examples: its generators turn bosons into fermions and fermions into bosons. Constructing such generators leads naturally to creation and annihilation operators, anticommutators, and the finite Grassmann algebra in which odd quantities square to zero.

4.1 Symmetry Relates Different Configurations

A symmetry is not merely a decorative regularity. It is an operation that relates one physical configuration to another. Left-right reflection, for example, relates the two hands. If the person and the surrounding laws possessed exact left-right symmetry, corresponding properties of the two hands would agree. More generally, objects or states connected by an exact symmetry must share the physical quantities that the symmetry preserves. In particle physics that frequently means equal masses; in atomic physics it means equal energies for states that lie in the same rotational multiplet.

Our aim is supersymmetry, but it is useful first to recall continuous symmetries such as rotations and translations. The question of whether supersymmetry is "continuous" in the ordinary sense is already awkward. Its infinitesimal parameters are not ordinary commuting real numbers. It is best regarded as a graded extension of continuous symmetry, and the Grassmann arithmetic introduced below makes that statement precise enough for our present purposes. [1]

[1] Editorial clarification: Modern terminology describes supersymmetry as a continuous graded symmetry. The qualification explains the lecture's warning that the usual continuous-versus-discrete question does not fit supersymmetry cleanly.

4.2 Unitary Operations on Hilbert Space

A quantum system is represented by a state vector $|\psi\rangle$. A physical symmetry operation is represented by an operator U acting on that state:

$$|\psi\rangle \longmapsto U\,|\psi\rangle\,. \tag{4.1}$$

For a spatial rotation, the axis and angle determine U. If a state contains particles at several positions, the transformed ket represents all of them at their rotated positions. A translation works similarly: it replaces the state by one in which the entire configuration has been displaced.

Question from the audience. The state vector belongs to an often infinite-dimensional Hilbert space. Are we rotating Hilbert space or ordinary three-dimensional space? [a]

Response. Both descriptions refer to the same operation at different levels. The unitary transformation acts on the Hilbert-space vector and represents the physical rotation of the configuration in ordinary space. It is not literally a three-dimensional rotation of the abstract Hilbert space; it is the representation there of a spatial rotation.

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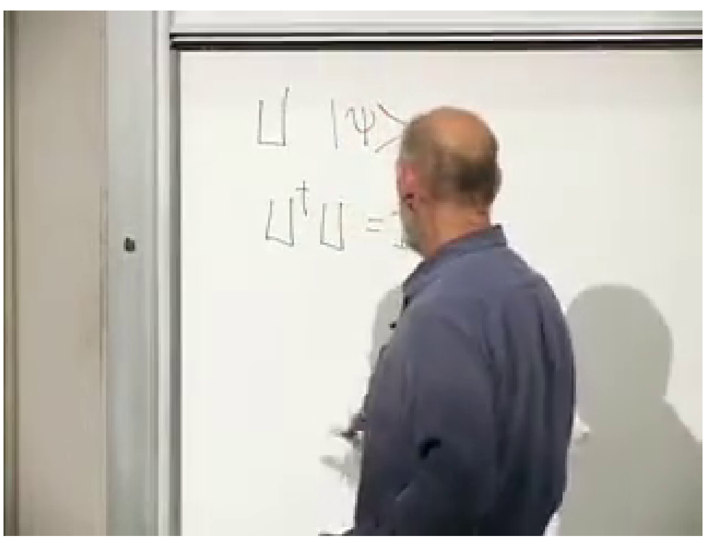
[a]Lecture timestamp: 00:04:20.

The essential condition is unitarity:

$$U^\dagger U = UU^\dagger = \mathbf{1}, \qquad U^{-1} = U^\dagger. \tag{4.2}$$

This preserves inner products and therefore probabilities:

$$\langle U\psi \,|\, U\psi\rangle = \langle\psi|\, U^\dagger U\,|\psi\rangle = \langle\psi \,|\, \psi\rangle\,. \tag{4.3}$$



Lecture frame: 00:05:56.420.

Figure 4.1: The early board transition from a transformed state to the unitarity condition. The ket and daggered product are visible, but the lecturer obscures their right-hand ends; the complete equations appear in the derivation below.

Strictly speaking, Wigner's theorem also permits antiunitary symmetries such as time reversal. The transformations connected continuously to the identity, which are the subject here, are represented unitarily. [2]

4.3 Infinitesimal Transformations and Generators

Doing nothing is represented by $U = \mathbf{1}$, not $U = 0$. A sufficiently small transformation differs from the identity by a term linear in its small parameter. For a rotation through a small angle ϵ,

$$U(\epsilon) = \mathbf{1} + i\epsilon L + O(\epsilon^2). \tag{4.4}$$

Unitarity to first order requires

$$U^\dagger U = \left(\mathbf{1} - i\epsilon L^\dagger\right)(\mathbf{1} + i\epsilon L) + O(\epsilon^2) \tag{4.5}$$

$$= \mathbf{1} + i\epsilon(L - L^\dagger) + O(\epsilon^2), \tag{4.6}$$

and hence

$$L^\dagger = L. \tag{4.7}$$

The operator L is the infinitesimal generator.

Rotations require three generators, one for each independent axis:

$$L_x, \qquad L_y, \qquad L_z. \tag{4.8}$$

A rotation about an arbitrary axis $\hat{\boldsymbol{n}}$ is generated by

$$L_{\hat{\boldsymbol{n}}} = \hat{\boldsymbol{n}} \cdot \boldsymbol{L}. \tag{4.9}$$

Question from the audience. Can the generators for the x, y, and z axes be added to make a rotation about another axis? [a]

Response. Yes. Their linear combinations generate rotations about intermediate axes. Three independent components are enough to build every infinitesimal spatial rotation.

---

[a] Lecture timestamp: 00:07:50.

For rotations, these generators are the components of angular momentum. The notation is useful more generally because every continuous symmetry is understood locally through its generators.

4.4 The Commutator as a Multiplication Table

The rotation generators obey

$$[L_x, L_y] = iL_z, \qquad [L_y, L_z] = iL_x, \qquad [L_z, L_x] = iL_y, \tag{4.10}$$

[2] Editorial clarification: The antiunitary possibility is a standard qualification; it does not alter the lecture's discussion of rotations, translations, or supersymmetry generators.

or, compactly,

$$[L_i, L_j] = i\epsilon_{ijk}L_k. \tag{4.11}$$

The Levi-Civita symbol vanishes if two indices agree and is $+1$ or -1 for even or odd permutations of $1, 2, 3$. The important structural fact is closure: the commutator of two generators is another generator.

Why should a commutator describe the geometry of rotations? Perform a small rotation about x, then one about y, undo the x-rotation, and finally undo the y-rotation. The operations do not cancel because rotations about different axes do not commute.

Question from the audience. Can this noncommutation be pictured through an airplane's roll, pitch, and yaw? [a]

Response. Yes. Successive rotations about body axes provide an immediate geometric demonstration that changing the order changes the final orientation. The algebra below is the infinitesimal version of that familiar fact.

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[a]Lecture timestamp: 00:12:36.

Let

$$U_x(\epsilon) = \mathbf{1} + i\epsilon L_x, \qquad U_y(\delta) = \mathbf{1} + i\delta L_y. \tag{4.12}$$

Because operators act on kets from right to left, the stated sequence is represented by

$$\mathcal{C} = U_y^{-1}(\delta)U_x^{-1}(\epsilon)U_y(\delta)U_x(\epsilon). \tag{4.13}$$

Question from the audience. Why may the undoing operations be written as inverses rather than daggers? [a]

Response. Because the transformations are unitary: $U^\dagger = U^{-1}$. For an infinitesimal transformation, taking the inverse simply reverses the sign of the first-order term, $U_x^{-1} = \mathbf{1} - i\epsilon L_x + O(\epsilon^2)$.

---

[a]Lecture timestamp: 00:13:25.

Multiplying the four factors on the board became a lively exercise: all terms linear in ϵ or δ cancel, the room tracked the mixed terms, and a temporary factor-of-two confusion exposed how easy it is to lose operator order. The clean second-order result is

$$\mathcal{C} = \mathbf{1} + \epsilon\delta[L_x, L_y] + O(\epsilon^2\delta, \epsilon\delta^2) \tag{4.14}$$

$$= \mathbf{1} + i\epsilon\delta L_z + O(\epsilon^2\delta, \epsilon\delta^2). \tag{4.15}$$

Thus the closed sequence leaves a small rotation about z through an angle proportional to $\epsilon\delta$. [3]

Question from the audience. Would changing the order of the inverse operations make the product the identity? [a]

Response. If each transformation is placed immediately beside its own inverse, the pairs cancel. The nontrivial commutator sequence deliberately separates those pairs; its failure to collapse measures the order dependence.

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[a]Lecture timestamp: 00:25:23.

[3]Editorial clarification: The displayed multiplication fixes the order and coefficient after the exploratory classroom calculation. Reversing the convention for the sequence reverses the sign but not the conclusion that the residual transformation is governed by the commutator.

Longer products do become more complicated, but they contain no new principle.

Question from the audience. What happens when there are more than three generators or many small transformations? [a]

Response. There are more commutators and more bookkeeping, but knowing the closed algebra lets one determine how arbitrarily many infinitesimal transformations combine. Finite transformations can be assembled from many small ones.

[a] Lecture timestamp: 00:25:34.

In this sense the commutator algebra is the multiplication table of a continuous group written near the identity.

4.5 Translations Complete the Spatial Example

Spatial translations are generated by momentum. With the sign convention used in the lecture,

$$\begin{aligned} T_x(a) &= \mathbf{1} + iaP_x + O(a^2), \\ T_y(b) &= \mathbf{1} + ibP_y + O(b^2), \end{aligned} \tag{4.16}$$

where a and b are distances rather than angles.

Two translations commute. Displace a point in x, then in y, undo the x-translation, and undo the y-translation; the point returns to its origin:

$$[P_i, P_j] = 0. \tag{4.17}$$

Question from the audience. Does translating or rotating a physical object require forces to start and stop the motion? [a]

Response. That is not the operation being defined. A symmetry transformation compares the original state with the mathematical state in which the whole configuration has been displaced or rotated. It does not describe a dynamical trajectory, acceleration, or work done while moving the apparatus.

[a] Lecture timestamp: 00:28:45.

A rotation and a translation do not commute. Translate a point, rotate it about the origin, translate back along the original axis, and undo the rotation; the point does not return to where it began. The mixed algebra is

$$[L_i, P_j] = i\epsilon_{ijk}P_k. \tag{4.18}$$

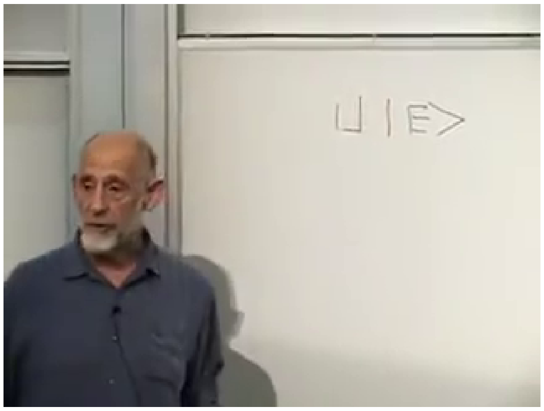
This relation says that momentum itself transforms as a vector under rotations. The exact index exercise is less important here than the general lesson: transformations of Euclidean space are encoded in the commutators among their rotation and translation generators.

4.6 A Symmetry Must Preserve Energy

We now pass from the composition of transformations to dynamics. If U is genuinely a symmetry of a system, applying it cannot change the energy. Let $|E\rangle$ be an energy eigenstate:

$$H\,|E\rangle = E\,|E\rangle\,. \tag{4.19}$$

The symbol E on the right is a number; $|E\rangle$ is a vector labeled by that eigenvalue.



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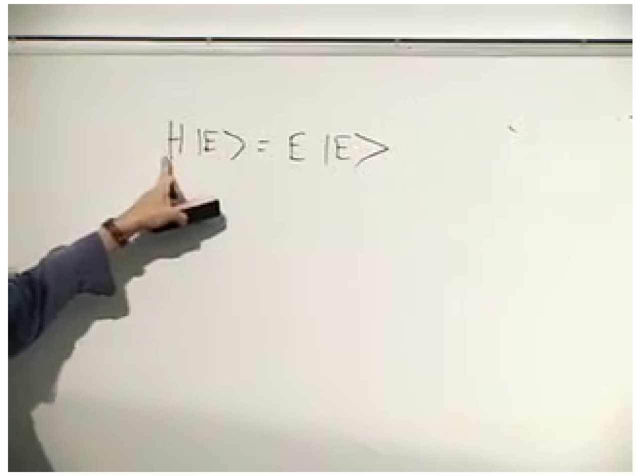

Lecture frame: 00:36:22.376.

Figure 4.2: The transformed ket $U|E\rangle$ and the defining eigenvalue equation $H|E\rangle = E|E\rangle$, isolated in successive board frames.

If U is a symmetry, then $U|E\rangle$ has the same eigenvalue:

$$HU|E\rangle = EU|E\rangle\,. \tag{4.20}$$

Since E is a scalar, it commutes with U, and the original eigenvalue equation may be used:

$$\begin{aligned} HU|E\rangle &= EU|E\rangle && (4.21)\\ &= UE|E\rangle && (4.22)\\ &= UH|E\rangle\,. && (4.23)\end{aligned}$$

Therefore

$$(HU - UH)|E\rangle = 0. \tag{4.24}$$

Question from the audience. Could the derivation be repeated from its first relation? [a]

Response. Start with $H|E\rangle = E|E\rangle$. Symmetry says $U|E\rangle$ has the same energy, so $HU|E\rangle = EU|E\rangle$. Move the scalar E through U, replace $E|E\rangle$ by $H|E\rangle$, and subtract. The result is $(HU - UH)|E\rangle = 0$.

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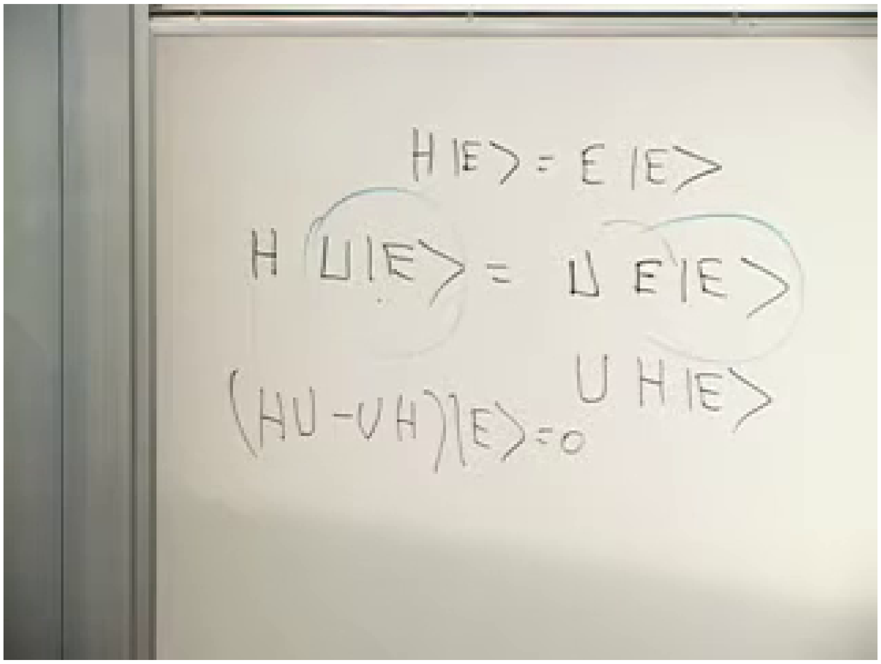
[a] Lecture timestamp: 00:35:55.



Lecture frame: 00:38:51.066.

Figure 4.3: The full board derivation. Curved guide marks connect the equal-energy statement to the substitution $E\,|E\rangle = H\,|E\rangle$; the last line is $(HU - UH)\,|E\rangle = 0$.

The argument holds for every energy eigenstate. Assuming those states form a complete basis, an operator that annihilates every one of them is the zero operator:

$$[H, U] = 0. \tag{4.25}$$

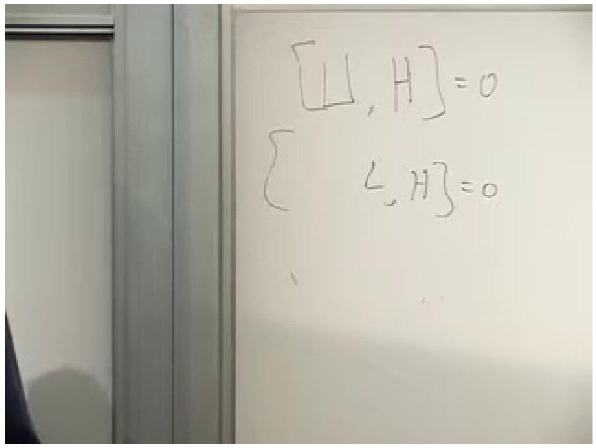
This is the criterion for a time-independent unitary transformation to be a symmetry of the Hamiltonian.

For an infinitesimal transformation,

$$U(\epsilon) = \mathbf{1} + i\epsilon L_i + O(\epsilon^2), \tag{4.26}$$

the identity already commutes with H, so each generator must satisfy

$$[L_i, H] = 0. \tag{4.27}$$



Lecture frame: 00:42:14.296.

Figure 4.4: The symmetry criterion condensed on the board: $[U, H] = 0$, followed by its generator form, standardized from the partly soft lower label as $[L_i, H] = 0$.

If L_i has no explicit time dependence, the Heisenberg equation gives

$$\frac{dL_i}{dt} = i[H, L_i] = 0 \qquad (\hbar = 1). \tag{4.28}$$

Thus symmetry generators are conserved charges. Spatial translation symmetry gives momentum conservation; rotation symmetry gives angular-momentum conservation. [4]

4.7 Transform the Whole Physical System

The phrase "rotation is a symmetry" can be misused if only part of the apparatus is transformed. A free spin can be rotated without changing its energy. Place it in a fixed magnetic field and rotate only the spin, however, and its Zeeman energy changes. That does not disprove rotational symmetry; it means the field was held fixed while the spin was rotated relative to it. Rotate the spin and field together, or use an external field that is itself invariant, and the energy is unchanged.

The same issue appears in a hydrogen atom. Translating only the electron relative to the proton changes their separation and therefore the Coulomb energy. Translation symmetry acts on the complete electron-proton system, moving both by the same amount.

Question from the audience. Must an external field itself possess the symmetry? [a]

Response. Either the fixed external field must already be invariant under the operation, or it must be transformed along with the rest of the system. A relative rotation between a spin and an unrotated field is a different physical configuration, not the symmetry transform of the whole setup.

---

[a] Lecture timestamp: 00:43:09.

Rotational degeneracy in an atom gives a clean positive example. For a fixed angular momentum l, rotations mix states with magnetic quantum numbers

$$m = -l, -l+1, \ldots, l. \tag{4.29}$$

[4] Editorial clarification: The explicit-time-derivative qualification is standard and is implicit in the lecture's conservation-law statement.

In the absence of an external magnetic or electric field, all members of the multiplet have the same energy:

$$E_{l,m} = E_l. \tag{4.30}$$

The symmetry relates distinct states and forces their degeneracy.

4.8 What Supersymmetry Changes

Every familiar symmetry discussed so far preserves boson-versus-fermion character. A rotation takes an electron to another electron, not to a photon. Color rotations mix the three colors of a quark but do not turn the quark into a gluon. Ordinary internal and spacetime symmetries may mix components or quantum labels, yet their generators are bosonic and do not change fermion parity.

Supersymmetry is different:

$$Q\,|B\rangle \sim |F\rangle\,, \qquad Q\,|F\rangle \sim |B\rangle\,. \tag{4.31}$$

Its multiplets cross the boson-fermion divide. In an exact supersymmetric theory, partners in one multiplet have the same mass. This is precisely the pairing required by the previous lectures: matched bosonic and fermionic loops can cancel the large ultraviolet corrections to scalar masses. The construction is difficult to visualize because no ordinary spatial motion turns an integer-spin state into a half-integer-spin state. We therefore proceed algebraically.

4.9 A Prototype Supercharge

Let $A, A^\dagger$ annihilate and create a boson, and let $C, C^\dagger$ annihilate and create a fermion. The simplest Hermitian operator that interchanges their one-particle states is

$$Q = C^\dagger A + A^\dagger C. \tag{4.32}$$

The first term removes a boson and creates a fermion; the second removes a fermion and creates a boson. If Q acts on a one-boson state, the $A^\dagger C$ term vanishes because there is no fermion to annihilate, while $C^\dagger A$ performs the conversion. On a one-fermion state the roles reverse.

Question from the audience. Does this construction create a single particle that is a superposition of a boson and a fermion? [a]

Response. Q is an operator, not a state. Acting on a pure bosonic state it produces a fermionic state, and acting on a pure fermionic state it produces a bosonic state. Acting linearly on an already general superposition produces another superposition, just as any linear operator does.

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[a] Lecture timestamp: 00:52:37.

The room next tests the vacuum. Since both annihilation operators kill it,

$$A\,|0\rangle = 0, \qquad C\,|0\rangle = 0, \tag{4.33}$$

we have

$$Q\,|0\rangle = 0. \tag{4.34}$$

The result is the zero vector, not a second copy of the vacuum.

The prototype is intentionally schematic. A spin-zero boson can be mapped to either component of a spin-one-half fermion, so the generator itself needs a spinor index:

$$Q_\alpha \left|B\right\rangle \sim \left|F_\alpha\right\rangle . \tag{4.35}$$

In the lecture's simplified two-component picture,

$$Q_\alpha \left|F_\beta\right\rangle \sim \delta_{\alpha\beta} \left|B\right\rangle . \tag{4.36}$$

Question from the audience. How can the state-space dimensions match when a spin-zero boson has one spin component and a spin-half fermion has two? [a]

Response. A single unindexed Q is insufficient. There are spinor components Q_α, one capable of producing each fermion spin component. The supercharge itself carries half a unit of spin because it changes spin by half a unit.

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[a] Lecture timestamp: 00:55:02.

The Q_α are fermionic, or more precisely Grassmann-odd, operators. Every term in them contains an odd number of fermion operators, which is why they reverse fermion parity. Ordinary measurable observables are even. One should distinguish an odd supercharge from its odd transformation parameter: a finite supersymmetry transformation is built from an even combination such as $\epsilon^\alpha Q_\alpha + \bar{\epsilon}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}}$, where both ϵ and Q are odd. [5]

4.10 Bosonic and Fermionic Operator Algebras

For bosonic modes labeled by i, j,

$$[A_i, A_j] = 0, \qquad [A_i^\dagger, A_j^\dagger] = 0, \qquad [A_i, A_j^\dagger] = \delta_{ij}. \tag{4.37}$$

The labels may denote momentum, position, spin, or any complete one-particle basis. Because creation operators commute,

$$A_i^\dagger A_j^\dagger \left|0\right\rangle = A_j^\dagger A_i^\dagger \left|0\right\rangle . \tag{4.38}$$

The two-boson state is symmetric under interchange.

Fermionic operators obey anticommutation relations:

$$\{C_i, C_j\} = 0, \qquad \{C_i^\dagger, C_j^\dagger\} = 0, \qquad \{C_i, C_j^\dagger\} = \delta_{ij}, \tag{4.39}$$

where

$$\{X, Y\} = XY + YX. \tag{4.40}$$

For a single mode,

$$\{C^\dagger, C^\dagger\} = 2(C^\dagger)^2 = 0, \tag{4.41}$$

so

$$(C^\dagger)^2 = 0. \tag{4.42}$$

[5] Editorial clarification: The lecture informally calls the generators Grassmann numbers. More precisely, the supercharges are odd operators, while the anticommuting parameters are Grassmann numbers; their product is even and can appear in a finite transformation.

The operator is not zero; its square is. It can create one fermion in the mode, but a second application gives zero. This is Pauli exclusion in operator language.

For distinct labels,

$$C_i^\dagger C_j^\dagger |0\rangle = -C_j^\dagger C_i^\dagger |0\rangle\,, \tag{4.43}$$

which is the antisymmetry of the two-fermion state. The algebraic replacement

$$\begin{gathered}\text{bosons: commutators}\\ \Updownarrow\\ \text{fermions: anticommutators}\end{gathered} \tag{4.44}$$

encodes the statistical distinction.

Many bosons may occupy one mode. A large photon occupation can therefore behave as a classical electromagnetic field with an ordinary numerical amplitude. Fermions cannot accumulate in one mode, and an odd fermionic field does not acquire an ordinary commuting classical value. This motivates a different arithmetic for fermionic bookkeeping.

4.11 Grassmann Numbers and Parity

Ordinary numbers commute:

$$\alpha_i\alpha_j = \alpha_j\alpha_i. \tag{4.45}$$

Grassmann generators anticommute:

$$\theta_i\theta_j = -\theta_j\theta_i. \tag{4.46}$$

Setting $i = j$, over a number system in which $2 \neq 0$, gives

$$\theta_i^2 = 0. \tag{4.47}$$

No ordinary numerical value such as six or seven can be assigned to θ_i. These symbols form an algebra designed to keep track of fermionic signs.

Ordinary coefficients commute with Grassmann generators:

$$\alpha\theta_i = \theta_i\alpha. \tag{4.48}$$

Question from the audience. What kind of object results when an ordinary number multiplies a Grassmann number? [a]

Response. It remains Grassmann-odd. Multiplication by an ordinary coefficient changes its scale, not its parity. The span of the generators is a vector space over the ordinary numbers.

[a] Lecture timestamp: 01:12:32.

The algebra has a $\mathbb{Z}_2$ parity. Ordinary coefficients and products of an even number of generators are even; a generator or a product of an odd number is odd:

$$\begin{aligned}\text{even}\cdot\text{even} &= \text{even},\\ \text{even}\cdot\text{odd} &= \text{odd},\\ \text{odd}\cdot\text{odd} &= \text{even}.\end{aligned} \tag{4.49}$$

Question from the audience. Is this analogous to the multiplication rules for real and imaginary numbers, or a generalization like the introduction of i? [a]

Response. It is a generalization of arithmetic, but the analogy should not be pushed far. Imaginary numbers still commute and satisfy $i^2 = -1$; Grassmann-odd quantities anticommute and square to zero. The shared even-odd multiplication pattern does not make the algebras the same.

[a] Lecture timestamp: 01:14:25.

A product such as $\theta_1\theta_2$ is even and commutes with every homogeneous element of the Grassmann algebra, but it is not an ordinary number. It is nonzero while

$$(\theta_1\theta_2)^2 = 0. \tag{4.50}$$

Question from the audience. If $\theta_1\theta_2$ is even, why is it not simply an ordinary number? [a]

Response. Even parity means that it commutes in the graded algebra; it does not mean that it is a real or complex scalar. Its nilpotence distinguishes it immediately from a nonzero ordinary number.

[a] Lecture timestamp: 01:17:36.

4.12 Functions of Grassmann Variables

With one generator, every polynomial terminates after first order:

$$f(\theta) = \alpha + \beta\theta. \tag{4.51}$$

All powers θ^n with $n \geq 2$ vanish. With two generators the most general element is

$$f(\theta_1, \theta_2) = \alpha + \beta_1\theta_1 + \beta_2\theta_2 + \gamma\theta_1\theta_2. \tag{4.52}$$

Writing the last monomial in reversed order changes its sign:

$$\theta_2\theta_1 = -\theta_1\theta_2. \tag{4.53}$$

Question from the audience. Does multiplying θ_1 by a coefficient β create a third independent Grassmann variable? [a]

Response. No. It creates another vector in the span of θ_1, not a new basis generator. With two independent generators, every odd linear term is $\beta_1\theta_1 + \beta_2\theta_2$.

[a] Lecture timestamp: 01:18:39.

For three generators, the ordered basis is

$$1, \quad \theta_1, \theta_2, \theta_3, \quad \theta_1\theta_2, \theta_1\theta_3, \theta_2\theta_3, \quad \theta_1\theta_2\theta_3. \tag{4.54}$$

In general, n generators produce 2^n ordered basis monomials. Any analytic function of finitely many Grassmann generators truncates to such a polynomial because repeated generators vanish. [6]

[6] Editorial clarification: The 2^n-dimensional basis states explicitly the pattern the lecture constructs for one, two, and three generators.

4.13 Grassmann Differentiation

For one variable define

$$\frac{\partial}{\partial\theta}\alpha = 0, \qquad \frac{\partial}{\partial\theta}\theta = 1, \tag{4.55}$$

so that

$$\frac{\partial}{\partial\theta}(\alpha + \beta\theta) = \beta. \tag{4.56}$$

For two variables, choose left derivatives and keep the canonical order $\theta_1\theta_2$:

$$\frac{\partial^L f}{\partial\theta_1} = \beta_1 + \gamma\theta_2, \tag{4.57}$$

$$\frac{\partial^L f}{\partial\theta_2} = \beta_2 - \gamma\theta_1. \tag{4.58}$$

The minus sign in the second line arises because the odd derivative must pass through θ_1 before differentiating θ_2.

This is expressed by the graded product rule. If A has parity $|A| \in \{0, 1\}$, then

$$\frac{\partial^L}{\partial\theta}(AB) = \left(\frac{\partial^L A}{\partial\theta}\right) B + (-1)^{|A|} A \left(\frac{\partial^L B}{\partial\theta}\right). \tag{4.59}$$

The rule is ordinary in spirit but contains the sign required by anticommutation. Grassmann integration can also be defined and, because the function space is finite, reduces to a short table. The lecture postpones that table.

4.14 The Exponential Puzzle, Resolved

With a single Grassmann generator and ordinary coefficient a,

$$e^{a\theta} = 1 + a\theta, \tag{4.60}$$

because every higher power vanishes. Two exponentials involving the same generator obey

$$\begin{aligned} e^{a\theta}e^{b\theta} &= (1 + a\theta)(1 + b\theta) && (4.61)\\ &= 1 + (a + b)\theta && (4.62)\\ &= e^{(a+b)\theta}. && (4.63) \end{aligned}$$

The room then tries distinct generators. Direct multiplication gives

$$e^{\theta_1}e^{\theta_2} = (1 + \theta_1)(1 + \theta_2) = 1 + \theta_1 + \theta_2 + \theta_1\theta_2. \tag{4.64}$$

At first the board expansion suggests that the square of $\theta_1 + \theta_2$ might supply the last term. An audience member catches the sign:

$$(\theta_1 + \theta_2)^2 = \theta_1^2 + \theta_1\theta_2 + \theta_2\theta_1 + \theta_2^2 = 0. \tag{4.65}$$

Question from the audience. Does the square term cancel because it contains $\theta_1\theta_2 + \theta_2\theta_1$? [a]

Response. Yes. That correction changes the conclusion: $e^{\theta_1+\theta_2} = 1 + \theta_1 + \theta_2$, while the ordered product $e^{\theta_1}e^{\theta_2}$ also contains $\theta_1\theta_2$. The naive exponential addition law does not apply to distinct anticommuting exponents.

[a] Lecture timestamp: 01:24:00.

There is no contradiction. Even ordinary matrices satisfy $e^A e^B = e^{A+B}$ only when A and B commute. Here

$$[\theta_1, \theta_2] = \theta_1\theta_2 - \theta_2\theta_1 = 2\theta_1\theta_2. \tag{4.66}$$

This commutator is even and commutes with both generators, so the Baker-Campbell-Hausdorff series terminates:

$$e^{\theta_1}e^{\theta_2} = \exp\left(\theta_1 + \theta_2 + \frac{1}{2}[\theta_1, \theta_2]\right) = e^{\theta_1+\theta_2+\theta_1\theta_2}. \tag{4.67}$$

Since the exponent on the right is nilpotent, it expands to exactly $1 + \theta_1 + \theta_2 + \theta_1\theta_2$. [7]

Question from the audience. What kind of number is $1 + \theta_1$, and are ordinary and Grassmann parts allowed to be added? [a]

Response. It is an inhomogeneous element of the Grassmann algebra, with an even part 1 and an odd part θ_1. Such sums are allowed, but they are neither ordinary scalars nor homogeneous odd elements.

[a] Lecture timestamp: 01:26:16.

Near the end, another apparent paradox is tested:

$$(\theta_1 + \theta_2)^2 = 0. \tag{4.68}$$

This does not imply $\theta_1\theta_2 = 0$, because the two cross terms cancel rather than vanish separately.

Question from the audience. If every odd Grassmann element squares to zero, does that mean the product of any two Grassmann generators is zero? [a]

Response. No. The equation $(\theta_1 + \theta_2)^2 = 0$ says $\theta_1\theta_2 + \theta_2\theta_1 = 0$. It proves anticommutation, not that either ordered product vanishes. Distinct generators can have a nonzero product.

[a] Lecture timestamp: 01:32:19.

Question from the audience. Are we assuming the ordinary power-series definition when we write these exponentials? [a]

Response. Yes, and that definition is consistent because every series terminates. What fails is not the exponential itself but the unjustified identity $e^A e^B = e^{A+B}$ for noncommuting A and B. The commutator correction restores the proper composition law.

[a] Lecture timestamp: 01:34:42.

This exploratory ending is appropriate. Ordinary symmetries led us from unitary transformations to generators and commutators; supersymmetry led us from odd generators to anticommuting variables.

[7] Editorial clarification: This BCH calculation supplies the resolution that the live lecture leaves as a question. The lecture correctly discovers the failure of the naive law but ends before identifying the missing commutator term.

In both cases, the multiplication law is the central object. For rotations it appears in $[L_i, L_j]$; for Grassmann exponentials it appears in the commutator term that prevents naive addition. The algebra is unfamiliar, but it is coherent, finite, and exactly suited to the fermionic transformations that supersymmetry requires.

Chapter 5

Supercharges, Closure, and Superspace

Overview. Supersymmetry is difficult not because its rules are numerous, but because the rules join ideas that ordinary physics keeps separate. We begin with Grassmann variables, whose odd elements anticommute and whose functions terminate after finitely many terms. Their derivative algebra then provides the model for odd symmetry generators. A bosonic oscillator and a fermionic oscillator give the simplest supercharges: their anticommutator is the Hamiltonian, so exchanging bosons and fermions is inseparable from time translation. The same algebra can be represented geometrically by translations in Grassmann directions of an enlarged spacetime called superspace.

5.1 Why We Must Back Up Before Going Forward

Supersymmetry is among the most abstract constructions in physics. It resembles rotations and translations because it is a symmetry, but it does not fit inside ordinary spacetime without enlarging what we mean by a coordinate. Its natural parameters are Grassmann quantities: objects that can be added and multiplied, yet anticommute rather than commute.

That is why a direct assault on the final algebra would be unhelpful. We shall first make the odd arithmetic usable, then recall how ordinary continuous symmetries are organized, and only then combine the two ideas. The route is longer, but each step prepares the next.

5.2 Grassmann Algebra in Working Form

5.2.1 Odd and even elements

Let ϵ_i denote independent Grassmann generators. They obey

$$\epsilon_i \epsilon_j = -\epsilon_j \epsilon_i, \qquad \{\epsilon_i, \epsilon_j\} = 0. \tag{5.1}$$

Setting $i = j$ gives

$$2\epsilon_i^2 = 0, \qquad \epsilon_i^2 = 0, \tag{5.2}$$

over the real or complex numbers. The generator is not zero; only its square is zero.

Each homogeneous element has a parity. A product containing an odd number of Grassmann generators is odd, while a product containing an even number is even:

$$\text{odd} \times \text{odd} = \text{even}, \tag{5.3}$$
$$\text{odd} \times \text{even} = \text{odd}, \tag{5.4}$$
$$\text{even} \times \text{even} = \text{even}. \tag{5.5}$$

Even elements commute with homogeneous elements. Two odd elements anticommute. Thus

$$n\epsilon_i = \epsilon_i n, \qquad \epsilon_i\epsilon_j = -\epsilon_j\epsilon_i, \tag{5.6}$$

when n is even.

An even element need not be an ordinary real or complex number. The product $\epsilon_1\epsilon_2$, for example, is even and nilpotent. Ordinary numbers form the scalar part of the algebra, while even products provide further commuting elements.

Question from the audience. Is there a distinction between the Grassmann algebra and the Grassmann numbers, or are they the same thing? [a]

Response. The elements are the Grassmann numbers; the algebra is the collection of those elements together with their addition and multiplication rules. One may therefore speak loosely of the same object, but it is useful to distinguish the things being manipulated from the rules by which they are manipulated.

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[a] Lecture timestamp: 00:06:45.

5.2.2 Functions terminate

A function of one Grassmann variable has only two possible powers:

$$f(\epsilon) = a + b\epsilon. \tag{5.7}$$

There is no ϵ^2 term. With two independent generators,

$$f(\epsilon_1, \epsilon_2) = a + b_1\epsilon_1 + b_2\epsilon_2 + c\epsilon_1\epsilon_2. \tag{5.8}$$

The expansion stops after both generators have appeared. For n independent generators, the ordered monomials form a 2^n-dimensional basis:

$$1, \quad \epsilon_i, \quad \epsilon_i\epsilon_j, \quad \ldots, \quad \epsilon_1\epsilon_2\cdots\epsilon_n. \tag{5.9}$$

[1]

When two real generators are combined into a complex pair $\theta, \bar{\theta}$, a general function can be written, with a chosen ordering, as

$$f(\theta, \bar{\theta}) = a + \bar{b}\,\theta + \bar{\theta}\,b + c\,\bar{\theta}\theta. \tag{5.10}$$

The bar indicates the complex-pair notation used here. Reversing the final order changes a sign:

$$\bar{\theta}\theta = -\theta\bar{\theta}. \tag{5.11}$$

[1] Editorial clarification: The finite 2^n-dimensional statement is the general pattern behind the one- and two-generator examples. An infinite Grassmann algebra can also be defined, but every polynomial used here involves only finitely many generators.

It is usually convenient to assign the whole function a definite parity. If f is even, then a and c are even while b and $\bar{b}$ are odd. If f is odd, those assignments reverse. The coefficients may themselves belong to a larger Grassmann algebra, provided they do not depend on the particular variables with respect to which we are expanding.

Question from the audience. Can a Grassmann variable be inserted into trigonometric or other analytic functions? [a]

Response. Yes, but the power series terminates immediately. For one odd generator,

$$\sin\epsilon = \epsilon, \qquad \cos\epsilon = 1, \qquad e^{\epsilon} = 1 + \epsilon.$$

The familiar names survive, but there are too few powers for the functions to develop their usual analytic complexity.

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[a] Lecture timestamp: 00:11:06.

5.3 Differentiation Is Sign Bookkeeping

We use a left Grassmann derivative. Its defining local rules are

$$\partial_\theta a = 0, \qquad \partial_\theta(\theta B) = B, \tag{5.12}$$

when a and B are independent of θ. Before differentiating, move the relevant θ to the left. Every passage through another odd factor changes the sign.

For example, if b is odd,

$$b\theta = -\theta b, \qquad \partial_\theta(b\theta) = -b. \tag{5.13}$$

Nothing mysterious has happened to the derivative. The minus sign arose before differentiation, when the odd factors were reordered.

5.3.1 The two-variable calculation

Return to

$$f(\theta, \bar{\theta}) = a + \bar{b}\,\theta + \bar{\theta}\,b + c\,\bar{\theta}\theta. \tag{5.14}$$

Suppose first that f is even. Then $\bar{b}$ is odd and c is even. Reordering the terms that contain θ,

$$\bar{b}\,\theta = -\theta\bar{b}, \tag{5.15}$$

$$c\,\bar{\theta}\theta = -\theta c\,\bar{\theta}. \tag{5.16}$$

Therefore

$$\partial_\theta f = -\bar{b} - c\bar{\theta}. \tag{5.17}$$

If f is odd, then $\bar{b}$ is even and c is odd. The first term crosses no odd coefficient, while the last θ crosses two odd factors, c and $\bar{\theta}$. The two signs cancel:

$$\partial_\theta f = \bar{b} + c\bar{\theta}. \tag{5.18}$$

Question from the audience. Why do both surviving terms carry minus signs in the even example? [a]

Response. The first sign comes from moving θ past the odd coefficient $\bar{b}$. In the last term, c is even, but θ must pass the odd variable $\bar{\theta}$. Each reordering crosses one odd factor, so each produces one minus sign.

[a] Lecture timestamp: 00:16:49.

The answer depends on the ordering convention, but the mathematics does not. Had we begun with $c\theta\bar{\theta}$ rather than $c\bar{\theta}\theta$, the displayed coefficient would differ by a sign and the derivative formula would change correspondingly.

Question from the audience. Could the last term have been written with θ and $\bar{\theta}$ reversed, and does the resulting sign have a deeper significance? [a]

Response. It can be written in either order, but exchanging the two odd variables changes the sign. One may absorb that sign into the coefficient c. At this stage the effect is bookkeeping: order matters, parity does not change, and a sign change never turns an odd object into an even one. The physical significance comes when the same bookkeeping enforces fermionic statistics and supersymmetry.

[a] Lecture timestamp: 00:18:50.

Question from the audience. Are the coefficients a, b, c simply drawn from the complex-number field? [a]

Response. They may be ordinary complex numbers when their required parity is even, but they may also contain generators other than the selected $\theta, \bar{\theta}$. The algebra of interest normally contains all complex scalars together with a chosen set of odd generators and their products.

[a] Lecture timestamp: 00:20:47.

Question from the audience. The derivatives of the even and odd functions look like opposites. Is that a general symmetry? [a]

Response. Not by itself. The signs follow mechanically from the parity of each coefficient and the number of odd factors crossed. Sometimes two answers happen to be opposite, but the reliable rule is simpler: every exchange of two odd objects contributes a minus sign.

[a] Lecture timestamp: 00:22:31.

5.3.2 Coordinate and derivative as operators

There is an instructive ordinary analogue. Let multiplication by x and differentiation by x act on an arbitrary function $g(x)$. Then

$$\partial_x(xg) = g + xg', \tag{5.19}$$

$$x\partial_x g = xg', \tag{5.20}$$

so

$$[\partial_x, x]g = g. \tag{5.21}$$

Because this holds for every g,

$$[\partial_x, x] = 1. \tag{5.22}$$

With $p = -i\partial_x$, this is the familiar position-momentum algebra, in units with $\hbar = 1$.

The Grassmann analogue uses an anticommutator. Take

$$f(\theta) = a + \theta b, \tag{5.23}$$

with a even and b odd, so that f is even. First,

$$\theta \partial_\theta f = \theta b. \tag{5.24}$$

In the opposite order,

$$\partial_\theta(\theta f) = \partial_\theta(\theta a + \theta^2 b) \tag{5.25}$$

$$= a. \tag{5.26}$$

Adding the two orders gives the original function:

$$\left(\partial_\theta \theta + \theta \partial_\theta\right) f = a + \theta b = f. \tag{5.27}$$

Hence

$$\boxed{\{\partial_\theta, \theta\} = 1.} \tag{5.28}$$

The derivative ∂_θ is itself odd. That is why its canonical relation with the odd coordinate is an anticommutator.

Question from the audience. May one use a function containing both even and odd parts rather than assigning a definite parity? [a]

Response. Such inhomogeneous elements can be formed, but calculations are normally organized into homogeneous even and odd objects. A definite parity makes every sign in a graded product unambiguous; a mixed object is handled by splitting it into its even and odd parts.

[a] Lecture timestamp: 00:31:05.

Question from the audience. Was the test function $g(x)$ in the ordinary commutator calculation supposed to be odd? [a]

Response. No. That was an ordinary function of an ordinary commuting coordinate. It was introduced only to show how the operator identity $[\partial_x, x] = 1$ is proved before repeating the argument with the Grassmann pair ∂_θ, θ.

[a] Lecture timestamp: 00:31:45.

Question from the audience. Is there a limit to the number of independent Grassmann generators? [a]

Response. There is no algebraic upper limit. For n generators the basis has 2^n monomials, so the bookkeeping grows rapidly, but the construction itself works for any finite n.

[a] Lecture timestamp: 00:32:16.

5.3.3 Integration

The integral over one Grassmann variable is fixed by

$$\int d\theta\, 1 = 0, \qquad \int d\theta\, \theta = 1. \tag{5.29}$$

Consequently, for $f(\theta) = a + \theta b$,

$$\int d\theta\, f(\theta) = b = \partial_\theta f(\theta). \tag{5.30}$$

Thus integration and differentiation perform the same component extraction. This is the Berezin integral. Its normalization is the unique simple choice compatible with translation invariance and Grassmann integration by parts. [2]

5.4 Ordinary Symmetry as the Baseline

Consider a continuous unitary group with Hermitian generators G_i. Near the identity,

$$U(\alpha) = \mathbf{1} + i\alpha^i G_i + O(\alpha^2), \tag{5.31}$$

where the α^i are ordinary commuting parameters. The infinitesimal multiplication law is encoded by the Lie algebra

$$[G_i, G_j] = i f_{ij}{}^k G_k. \tag{5.32}$$

The constants $f_{ij}{}^k$ are the structure constants. Closure means that commuting any two generators returns a linear combination of generators already in the algebra.

For rotations,

$$[L_i, L_j] = i\varepsilon_{ijk} L_k. \tag{5.33}$$

The three Hermitian components may be combined into the non-Hermitian ladder operators

$$L_\pm = L_x \pm i L_y, \tag{5.34}$$

which raise or lower the magnetic quantum number m by one. Complex combinations are useful without changing the Hermiticity of the underlying physical generators.

Only the first two indices of $f_{ij}{}^k$ must be antisymmetric in a general basis. For a compact algebra with an invariant orthonormal metric, one can lower the final index and choose conventions in which f_{ijk} is totally antisymmetric, as it is for rotations. [3]

For G_i to generate an actual symmetry of a time-independent system,

$$[H, G_i] = 0. \tag{5.35}$$

Equivalently, G_i maps every energy eigenspace into itself. With no explicit time dependence, the Heisenberg equation gives

$$\frac{dG_i}{dt} = i[H, G_i] = 0, \tag{5.36}$$

[2] Editorial clarification: The component rules make precise the lecture's statement that Grassmann integration and differentiation are the same operation.

[3] Editorial clarification: This qualification separates the universally antisymmetric commutator from the stronger property available in the familiar rotation basis.

so the generator is a conserved charge.

Ordinary observables are even. A fermion field is odd, but an isolated odd field is not itself a measurable observable; physical operators contain an even total number of fermionic factors. Supersymmetry changes the parity of the generators, not the parity required of observable results.

The clean notation for all cases is the graded commutator

$$[X, Y]_{\mathrm{gr}} = XY - (-1)^{|X||Y|} YX. \tag{5.37}$$

It is an ordinary commutator if either operator is even and an anticommutator if both are odd.

5.5 The Smallest Supersymmetric Oscillator

5.5.1 One bosonic mode and one fermionic mode

Take a single bosonic mode with creation and annihilation operators a_+ and a_-:

$$[a_-, a_+] = 1, \qquad [a_+, a_+] = [a_-, a_-] = 0. \tag{5.38}$$

Its number operator is

$$N_B = a_+ a_-, \tag{5.39}$$

and its occupation number can be $0, 1, 2, \ldots$.

For a single fermionic mode, introduce c_+ and c_-:

$$\{c_-, c_+\} = 1, \qquad \{c_+, c_+\} = \{c_-, c_-\} = 0. \tag{5.40}$$

Therefore

$$c_+^2 = c_-^2 = 0, \qquad N_F = c_+ c_-, \tag{5.41}$$

and the occupation number is only 0 or 1. The bosonic operators are even, the fermionic operators are odd, and every $a_\pm$ commutes with every $c_\pm$.

[4]

Assume the boson and fermion have the same rest energy m, and define

$$Q = \sqrt{m}\, a_+ c_-, \qquad \bar{Q} = Q^\dagger = \sqrt{m}\, a_- c_+. \tag{5.42}$$

The operator Q removes one fermion and creates one boson. Its conjugate removes one boson and creates one fermion. Each contains one fermionic operator and is therefore odd.

On number states, their action is explicit:

$$Q\,|n, 1\rangle = \sqrt{m(n+1)}\;|n+1, 0\rangle\,, \qquad Q\,|n, 0\rangle = 0, \tag{5.43}$$

$$\bar{Q}\,|n, 0\rangle = \sqrt{mn}\;|n-1, 1\rangle\,, \qquad \bar{Q}\,|n, 1\rangle = 0. \tag{5.44}$$

[5]

[4] Editorial clarification: The nilpotence restricts the fermionic mode to one occupied state. The bosonic mode has unrestricted occupation; a brief contrary phrase in the classroom discussion is resolved by the oscillator algebra itself.

[5] Editorial clarification: The number-state formulas are the direct normalized form of the particle-replacement action described in the lecture.

5.5.2 The anticommutator closes on energy

The graded analogue of finding a Lie algebra is to compute

$$\{Q,\bar{Q}\} = Q\bar{Q} + \bar{Q}Q \tag{5.45}$$
$$= m\left(a_+c_-a_-c_+ + a_-c_+a_+c_-\right). \tag{5.46}$$

Since the a's are even, they pass through the c's without a sign:

$$\{Q,\bar{Q}\} = m\left(a_+a_-c_-c_+ + a_-a_+c_+c_-\right). \tag{5.47}$$

Now use

$$c_-c_+ = 1 - c_+c_-, \qquad a_-a_+ = a_+a_- + 1. \tag{5.48}$$

Then

$$\{Q,\bar{Q}\} = m\left[N_B(1-N_F) + (N_B+1)N_F\right] \tag{5.49}$$
$$= m(N_B + N_F). \tag{5.50}$$

The temporary extra factor of two that arose during the live rearrangement disappears: one contribution supplies N_B, the other supplies N_F.

For particles at rest with equal masses, the Hamiltonian is

$$H = m(N_B + N_F). \tag{5.51}$$

Thus

$$\boxed{\{Q,\bar{Q}\} = H.} \tag{5.52}$$

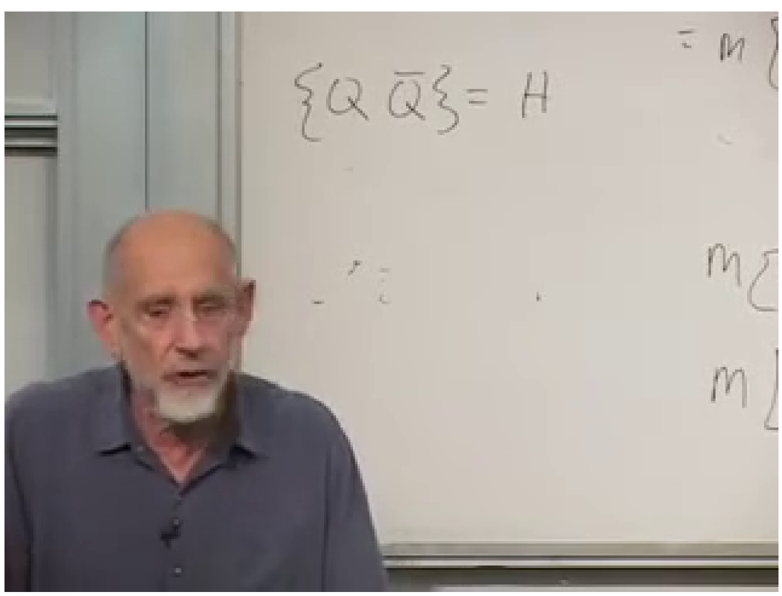

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Figure 5.1: The central closure relation isolated on the board. The braces and $Q, \bar{Q}, H$ are visible; the comma is supplied by standard anticommutator notation.

The additive zero of energy has been chosen so that the empty state has $H = 0$. In the supersymmetric oscillator this is the natural convention: the bosonic and fermionic zero-point contributions cancel. [6]

The Hamiltonian is even. It is also a symmetry generator: it generates translations in time. The odd generators therefore fail to close among themselves unless the even time-translation generator is included.

Question from the audience. Which member of the relation is even? Is it the Hamiltonian? [a]

Response. Yes. Energy is an ordinary observable, so H is even. The supercharge Q is odd, and the product of two odd quantities is even; accordingly, $\{Q, \bar{Q}\}$ can equal H. An odd and an even generator are related by a commutator, not an anticommutator.

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[a] Lecture timestamp: 00:58:24.

5.5.3 Completing the algebra

The same-parity relations are immediate:

$$\{Q, Q\} = 2Q^2 = 0, \qquad \{\bar{Q}, \bar{Q}\} = 2\bar{Q}^2 = 0, \tag{5.53}$$

because Q^2 contains c_-^2 and $\bar{Q}^2$ contains c_+^2.

What remains is the relation with H. Let

$$H\,|E\rangle = E\,|E\rangle\,. \tag{5.54}$$

If the boson and fermion masses are equal, Q replaces one by the other without changing the energy. Hence $Q\,|E\rangle$ is either zero or another eigenstate with the same eigenvalue:

$$HQ\,|E\rangle = EQ\,|E\rangle\,. \tag{5.55}$$

On the other hand,

$$QH\,|E\rangle = Q(E\,|E\rangle) = EQ\,|E\rangle\,. \tag{5.56}$$

Subtracting,

$$[H, Q]\,|E\rangle = 0. \tag{5.57}$$

If the energy eigenstates form a complete basis, this holds as an operator identity:

$$[H, Q] = 0, \qquad [H, \bar{Q}] = 0. \tag{5.58}$$

[6] Editorial clarification: The lecture works directly with total rest energy and does not display separate zero-point terms; the stated convention makes that choice explicit.

Question from the audience. What happens if the boson and fermion do not have the same mass? [a]

Response. Then replacing one by the other changes the energy, so the proposed supercharge does not commute with the Hamiltonian and the simple supersymmetry is lost. For

$$H = m_B N_B + m_F N_F,$$

the explicit result is

$$[H, Q] = (m_B - m_F)Q.$$

It vanishes precisely when $m_B = m_F$. [b]

[a]Lecture timestamp: 01:01:16.

[b]Editorial clarification: The unequal-mass commutator is the direct operator form of the energy-change argument and makes the equal-mass condition explicit.

Question from the audience. Could the energy-eigenstate argument be repeated more slowly? [a]

Response. Begin with $H|E\rangle = E|E\rangle$. Equal masses imply that $Q|E\rangle$ has the same definite energy E, so $HQ|E\rangle = EQ|E\rangle$. The other order gives $QH|E\rangle = QE|E\rangle = EQ|E\rangle$, because E is an ordinary scalar. Their difference vanishes on every energy eigenstate. Completeness then promotes the statement from those states to the operator equation $[H, Q] = 0$.

[a]Lecture timestamp: 01:01:37.

The complete toy algebra is therefore

$$\{Q, \bar{Q}\} = H, \tag{5.59}$$

$$\{Q, Q\} = \{\bar{Q}, \bar{Q}\} = 0, \tag{5.60}$$

$$[H, Q] = [H, \bar{Q}] = 0. \tag{5.61}$$

The supercharges are conserved odd charges. More surprisingly, their algebra mixes an operation that changes particle type with the generator of spacetime translation.

5.6 Why the Hamiltonian Means Time Translation

In this stripped-down model there is no spatial coordinate, but there is time. The Schrödinger equation reads

$$i\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle. \tag{5.62}$$

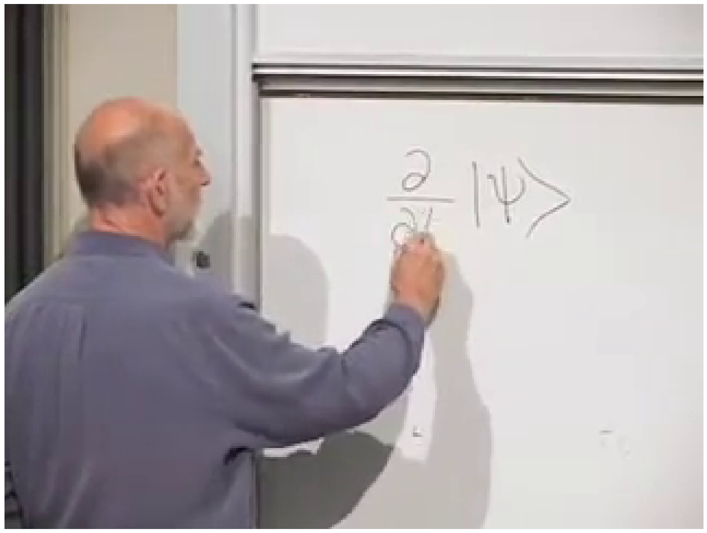

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Figure 5.2: The time derivative being written beside $|\psi\rangle$. The hand obscures part of the denominator; the complete Schrödinger equation is given above.

For a finite shift δ,

$$|\psi(t+\delta)\rangle = e^{-iH\delta}\, |\psi(t)\rangle\,. \tag{5.63}$$

Expanding to first order,

$$|\psi(t+\delta)\rangle = (\mathbf{1} - i\delta H)\, |\psi(t)\rangle + O(\delta^2). \tag{5.64}$$

Thus H is the infinitesimal generator that updates a state in time.

Internal symmetries such as isospin and color also change particle labels. Isospin can mix proton and neutron states; color rotations mix red, green, and blue quarks. Yet no composition of those internal transformations translates a system in space or time. Supersymmetry is different:

$$\begin{gathered}\text{odd particle-changing transformation}\\ \xRightarrow{\{\,,\,\}}\\ \text{time translation}\end{gathered} \tag{5.65}$$

That is the clue that the odd generators may themselves be translations, but along coordinates unlike the usual ones.

5.7 Superspace

5.7.1 A function of time and odd coordinates

Introduce a function

$$\Psi = \Psi(\theta, \bar{\theta}, t). \tag{5.66}$$

It may be regarded as a wavefunction in the toy model or, more generally, as a superfield. Because the odd expansion terminates,

$$\Psi(\theta, \bar{\theta}, t) = A(t) + \bar{\chi}(t)\theta + \bar{\theta}\chi(t) + F(t)\bar{\theta}\theta. \tag{5.67}$$

The single object packages even and odd component functions. [7]

An ordinary time translation leaves $\theta, \bar{\theta}$ fixed and sends

$$t \longmapsto t + \delta. \tag{5.68}$$

To first order,

$$\Psi(\theta, \bar{\theta}, t + \delta) = \Psi(\theta, \bar{\theta}, t) + \delta\, \partial_t \Psi(\theta, \bar{\theta}, t). \tag{5.69}$$

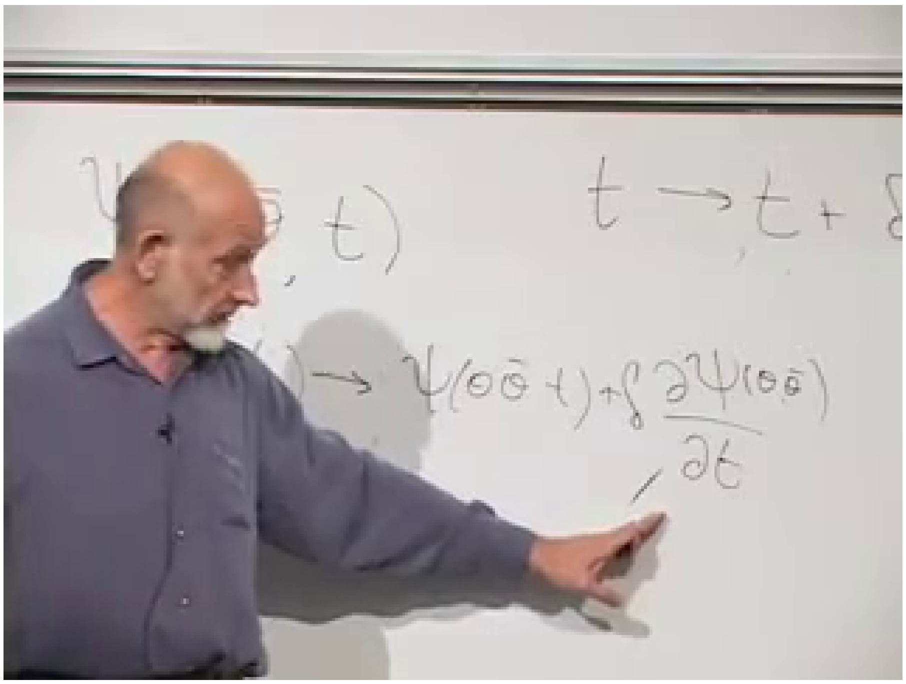

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Figure 5.3: The board places $t \to t + \delta$ above the first-order change in Ψ, with the derivative term indicated directly.

5.7.2 Translation in an odd direction

Now invent a transformation that shifts an odd coordinate:

$$\theta \longmapsto \theta + \epsilon, \tag{5.70}$$

where ϵ is another odd Grassmann parameter. At the same time, let time acquire an even nilpotent displacement,

$$t \longmapsto t + i\bar{\theta}\epsilon. \tag{5.71}$$

Both $\bar{\theta}$ and ϵ are odd, so their product is even and can consistently be added to the even coordinate t.

With the parameter moved to the left,

$$i\bar{\theta}\epsilon = -i\epsilon\bar{\theta}. \tag{5.72}$$

[7] Editorial clarification: This component expansion applies the Grassmann polynomial developed earlier; it is the standard explicit content of a function on the toy superspace.

The induced variation is therefore

$$\delta_\epsilon \Psi = \epsilon\, \partial_\theta \Psi - i\epsilon\bar{\theta}\, \partial_t \Psi \tag{5.73}$$
$$= \epsilon q_0 \Psi, \tag{5.74}$$

where

$$q_0 = \partial_\theta - i\bar{\theta}\partial_t. \tag{5.75}$$

This odd differential operator is the generator of the combined shift. A compatible conjugate generator, using left derivatives, is

$$\bar{q}_0 = -\partial_{\bar{\theta}} + i\theta\partial_t. \tag{5.76}$$

The exact signs depend on whether one uses left or right Grassmann derivatives and on how conjugation reverses odd factors. What must not depend on convention is the closure on time translation. With the definitions above,

$$\{q_0, \bar{q}_0\} = i\{\partial_\theta, \theta\}\partial_t + i\{\bar{\theta}, \partial_{\bar{\theta}}\}\partial_t \tag{5.77}$$
$$= 2i\partial_t. \tag{5.78}$$

The derivative-derivative term vanishes, as does the term quadratic in the odd coordinates. The two mixed terms each contribute $i\partial_t$.

Since $H = i\partial_t$ on Schrödinger-picture states,

$$\{q_0, \bar{q}_0\} = 2H. \tag{5.79}$$

If we normalize

$$q = \frac{q_0}{\sqrt{2}}, \qquad \bar{q} = \frac{\bar{q}_0}{\sqrt{2}}, \tag{5.80}$$

then

$$\boxed{\{q, \bar{q}\} = H.} \tag{5.81}$$

[8]

Question from the audience. Where does the final factor of two in the superspace anticommutator come from? [a]

Response. There are two equal mixed contributions. One comes from $\{\partial_\theta, \theta\} = 1$, and the other from $\{\partial_{\bar{\theta}}, \bar{\theta}\} = 1$. Each multiplies the time derivative. The unnormalized generators therefore give $2i\partial_t$; a factor $1/\sqrt{2}$ in each supercharge restores the earlier normalization.

[a] Lecture timestamp: 01:24:21.

Question from the audience. Did the construction use the assumption that ϵ is small? [a]

Response. It used nilpotence rather than an ordinary numerical magnitude. Since $\epsilon^2 = 0$, every expansion in a single ϵ stops after first order. Calling it infinitesimal or small is a useful analogy, but the truncation is exact.

[a] Lecture timestamp: 01:24:51.

[9]

[8] Editorial clarification: The lecture deliberately works through uncertain signs and then notices a factor of two. The displayed left-derivative convention preserves that classroom calculation while fixing one self-consistent normalization. Other standard conventions move signs between the coordinate shifts and the generators.

[9] Editorial clarification: Grassmann parameters have no ordinary positive magnitude. Their role as infinitesimal parameters comes from nilpotence and the exact termination of the expansion.

5.7.3 The geometric meaning

The algebra now has two equivalent realizations. In the oscillator picture, Q and $\bar{Q}$ replace fermions by bosons and bosons by fermions. In the geometric picture, q and $\bar{q}$ translate a function through odd directions while inducing an ordinary time shift. In both pictures,

$$\begin{aligned}&\{\text{supercharge}, \text{conjugate supercharge}\}\\ &= \text{time-translation generator}.\end{aligned} \tag{5.82}$$

The generalized coordinate space containing $t, \theta, \bar{\theta}$ is superspace. In a relativistic field theory it also contains the ordinary spatial coordinates, and the supercharges close on the full spacetime momentum P_μ, not only on $H = P_0$. Curving this structure leads toward supergravity.

In one sense Grassmann variables are an efficient bookkeeping device for bosons and fermions. In another, the closure relation says something much deeper: the distinction between particle type and spacetime geometry is no longer absolute. Moving twice in odd directions produces motion in an ordinary direction. Whether nature realizes that structure experimentally is separate from the mathematical fact that the structure is coherent, economical, and remarkably rigid.

Chapter 6

Why Supersymmetry Might Matter

Overview. Vacuum fluctuations connect the subjects of this chapter. Boundary conditions produce the Casimir force, loop effects threaten the Higgs mass, new virtual particles alter running gauge couplings, and vacuum energy governs cosmic expansion. These problems motivate low-energy supersymmetry as a way to stabilize the Higgs scale, supply a dark-matter candidate, and improve gauge-coupling unification. We then build superfields using Grassmann coordinates and use Berezin integration to extract a supersymmetric action.

6.1 Vacuum Fluctuations Between Reflecting Plates

We return to supersymmetry by first settling a question about the Casimir effect. Place two parallel reflecting plates a distance d apart. Even when no real particles are present, a quantum field has vacuum fluctuations. We may picture them as virtual processes, or equivalently as the zero-point motion of the field's oscillator modes. The plates impose boundary conditions, so the allowed fluctuations between them differ from those in unrestricted space.

The important quantity is not an absolute zero of energy but the change produced when the separation changes. Denote the vacuum energy per unit area by $\mathcal{E}(d) = E(d)/A$. Once $\mathcal{E}$ depends on d, it is a potential energy, and the force per unit area is

$$\frac{F}{A} = -\frac{d\mathcal{E}}{dd}. \tag{6.1}$$

Thus a rearrangement of virtual processes becomes an observable force. In the familiar electromagnetic example with ideal conducting plates, the energy and pressure are

$$\mathcal{E}(d) = -\frac{\pi^2}{720\,d^3}, \tag{6.2}$$

$$\frac{F}{A} = -\frac{\pi^2}{240\,d^4}, \tag{6.3}$$

in units $\hbar = c = 1$.[1] The minus sign denotes attraction.

[1]Editorial clarification: These coefficients and the distinction between the d^{-3} energy per area and the d^{-4} pressure resolve the dimensional question raised at the board. They are the standard ideal-plate result, not a derivation carried out here.

Question from the audience. Does the fourth power describe the force or the energy? [a]

Response. For parallel plates in four spacetime dimensions, the energy per unit area falls as d^{-3}, while differentiating with respect to d makes the force per unit area fall as d^{-4}. Dimensional analysis already fixes these powers once $\hbar$ and c are restored.

[a]Lecture timestamp: 00:04:39.

One sometimes summarizes the calculation by saying that the plates interrupt virtual photon loops. That picture is useful provided it is not taken too literally. The precise calculation is a change in the spectrum of allowed field modes or, equivalently, a change in the quantum effective action. Either language leads to the same separation-dependent energy.

Question from the audience. Can the Casimir force be repulsive rather than attractive? [a]

Response. It can. The sign is not universal; it depends on the fields, geometry, materials, and boundary conditions. Fermionic loops often bring the opposite loop sign from bosonic loops, which motivates the suggestion made at the board, but fermionic statistics by itself is not a general theorem that every Casimir force must be repulsive.[b]

[a]Lecture timestamp: 00:07:48.

[b]Editorial clarification: This qualification replaces a deliberately tentative classroom answer with the general statement needed to avoid a false universal rule.

6.1.1 What would count as evidence for vacuum energy?

The Casimir force is measured, and atomic observables are sensitive to vacuum loops as well. Yet neither fact by itself demonstrates that an absolute vacuum energy gravitates. A Casimir experiment measures an energy difference caused by boundaries; the same force can be described without assigning operational meaning to an unbounded absolute zero-point sum.

The sharp test comes from gravity. Energy is gravitational charge. If empty space carries a uniform energy density, it must enter Einstein's equations and influence cosmic expansion. The accelerated expansion of the universe is therefore the direct gravitational evidence for a vacuum-like component.

Question from the audience. Why do popular accounts call vacuum energy mysterious, as if physicists do not know what it is? [a]

Response. Quantum field theory gives every field a zero-point contribution, so the possibility of vacuum energy is not mysterious. The mystery is quantitative: why is the gravitating value so extraordinarily small compared with estimates based on known quantum fields and a high-energy cutoff? The surprising fact is its near absence, not its bare presence.

[a]Lecture timestamp: 00:08:19.

For many years the observational upper bound was so small that it was psychologically tempting to set the cosmological constant exactly to zero and search for a principle that enforced the cancellation. Improved cosmological observations instead found a small nonzero value. That discovery did not make vacuum energy conceptually alien; it made the unexplained degree of suppression impossible to ignore. The relevant comparison depends on the assumed ultraviolet scale, but every natural high-energy estimate demands an enormous cancellation.

6.2 What One Supercharge Does to a Many-Particle State

Before listing reasons to believe in supersymmetry, we should remove a possible paradox. Bosons may occupy one mode in arbitrarily large numbers, while a fermionic mode has occupation zero or one. If a symmetry exchanges bosons and fermions, does it turn a thousand bosons in one mode into a thousand forbidden fermions?

No. A fermionic generator changes one quantum at a time. In the oscillator model, let a annihilate a boson and $c^\dagger$ create its fermionic partner. One supercharge may be represented as

$$Q = c^\dagger a. \tag{6.4}$$

On a state with n bosons and no fermion,

$$Q\,|n,0\rangle = \sqrt{n}\,|n-1,1\rangle\,. \tag{6.5}$$

One boson has been removed and one fermion inserted. Acting again with the same charge gives zero:

$$Q^2 = (c^\dagger)^2 a^2 = 0. \tag{6.6}$$

This is the nilpotence inherited from fermionic creation operators.

Question from the audience. If many bosons occupy the same state, why does a supersymmetry transformation not produce many fermions in that same state and violate the exclusion principle? [a]

Response. The generator does not replace the entire population. It changes the state by one unit of fermion number. The same fermionic charge cannot be compounded like an ordinary rotation generator because its square vanishes. A complementary charge can undo the exchange, and the mixed anticommutator closes on time translation.

[a]Lecture timestamp: 00:12:05.

The last sentence is essential. The shorthand $Q^2 = 0$ applies to an individual chiral component or to the simple charge just written; it does not mean that every product of supercharges vanishes. In four-dimensional supersymmetry,

$$\{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2\sigma^\mu_{\alpha\dot\beta} P_\mu, \tag{6.7}$$

while

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\} = 0 \tag{6.8}$$

for the minimal algebra without central charges.[2]

6.3 Three Reasons to Look for Low-Energy Supersymmetry

Question from the audience. Is there motivation for supersymmetry besides cancelling vacuum energy? [a]

Response. There are three closely related motivations: control of the Higgs self-energy, a possible stable dark-matter particle, and a much sharper meeting of the gauge couplings at high energy. None proves that supersymmetry is realized in nature, but together they make it a serious theory to test.

[a]Lecture timestamp: 00:15:06.

[2]Editorial clarification: This component form makes precise the classroom shorthand that repeated action of the same Q vanishes while a supercharge and its conjugate generate translation.

The phrase *low-energy supersymmetry* is relative. It does not mean that superpartners are light on an everyday scale; it means that their masses are not vastly above the electroweak and accelerator scales. All three arguments weaken if the superpartners are pushed arbitrarily high.

6.3.1 Protecting the Higgs scale

A scalar mass is unusually sensitive to high-energy physics. Schematically, a bosonic loop produces a correction of the form

$$\delta m_H^2\big|_{\mathrm{B}} \sim +\frac{\lambda}{16\pi^2}\Lambda^2, \tag{6.9}$$

where Λ denotes the scale up to which the effective theory is being used. A fermionic loop carries the opposite sign,

$$\delta m_H^2\big|_{\mathrm{F}} \sim -\frac{|y|^2}{8\pi^2}\Lambda^2. \tag{6.10}$$

The coefficients depend on the interaction and field multiplicities, but supersymmetry relates them so that the leading ultraviolet sensitivity cancels between partners.[3]

This matters because the Higgs expectation value sets the electroweak scale. Gauge-boson and fermion masses obey relations such as

$$m_W = \frac{gv}{2}, \qquad m_f = \frac{y_f v}{\sqrt{2}}. \tag{6.11}$$

If the Higgs sector were naturally driven to an enormous scale, the familiar low-energy particle spectrum would be lost unless bare parameters were adjusted against radiative corrections with extreme precision.

Exact supersymmetry would make the cancellation exact, but exact supersymmetry is already excluded by the unequal masses of observed bosons and fermions. With softly broken supersymmetry, the quadratic sensitivity is replaced by dependence on superpartner splittings, schematically

$$\delta m_H^2 \sim \frac{g^2}{16\pi^2}\left(m_{\mathrm{soft}}^2 - m_{\mathrm{ordinary}}^2\right)\ln\frac{\Lambda^2}{m_{\mathrm{soft}}^2}. \tag{6.12}$$

The cancellation remains useful only while the splitting is not too large.

6.3.2 A stable endpoint for dark-matter cascades

Separate the spectrum into ordinary particles and their superpartners. A fermion has a bosonic partner; a boson has a fermionic partner. If the interactions preserve a parity that distinguishes these two classes, a superpartner decay must leave another superpartner:

$$\widetilde{X} \longrightarrow X + \widetilde{Y} \longrightarrow X + Y + \widetilde{Z} \longrightarrow \cdots . \tag{6.13}$$

The cascade cannot discard its last superpartner. It therefore ends at the lightest supersymmetric particle, the LSP.

The commonly used symmetry is R-parity,

$$R_p = (-1)^{3(B-L)+2s}. \tag{6.14}$$

[3] Editorial clarification: The schematic loop formulas expose the mathematics behind the qualitative self-energy argument. They are not intended as a complete Standard Model Higgs-mass calculation.

Standard Model particles have $R_p = +1$, while their superpartners have $R_p = -1$. If R_p is exact, every interaction contains an even number of superpartners and the LSP is stable.[4]

Dark matter must survive for at least cosmological times. It must also interact weakly enough to pass repeatedly through the visible galactic disk without losing substantial energy. An electrically charged or strongly interacting population would scatter, dissipate energy, and tend to collapse toward the same thin structure as ordinary matter. The observed extended halo instead points toward a neutral, weakly interacting component.

This does not identify the LSP uniquely. Depending on the model and symmetry-breaking pattern, the candidate might be a neutral mixture of electroweak gauginos and Higgsinos, or it might be a gravitino. Charged or colored candidates are strongly constrained. What supersymmetry supplies is not a guaranteed answer but a natural mechanism for producing a neutral, long-lived endpoint.

6.4 Running Couplings and the Hint of Unification

The third motivation starts from charge screening. A measured coupling depends on the energy scale because virtual particles dress the interaction. For electromagnetism, an electron-positron loop inserted in a photon propagator screens the charge at long distance. At shorter distance, or higher momentum transfer, less of that polarization cloud lies between the probes and the effective charge is larger.

Write

$$\alpha_i = \frac{g_i^2}{4\pi}. \tag{6.15}$$

At one loop the inverse coupling runs linearly with the logarithm of energy:

$$\frac{d\alpha_i^{-1}}{d\ln\mu} = -\frac{b_i}{2\pi}. \tag{6.16}$$

Thus

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi}\ln\frac{\mu}{\mu_0}. \tag{6.17}$$

[5]

Above the electroweak scale the three fundamental gauge couplings are those of

$$U(1)_Y, \qquad SU(2)_L, \qquad SU(3)_c. \tag{6.18}$$

The electromagnetic coupling used for intuition at low energy is a combination of the first two after electroweak symmetry breaking. For QCD the non-Abelian gauge fields produce antiscreening: the strong coupling decreases at short distance, the phenomenon of asymptotic freedom. Consequently α_3^{-1} rises with $\ln\mu$, while the other inverse couplings have different slopes.

The Standard Model curves point suggestively toward one another but do not meet at a single point. Superpartners modify the virtual loops and therefore change the coefficients b_i. In the conventional

[4] Editorial clarification: Supersymmetry alone does not guarantee a stable LSP. Stability requires conserved R-parity or another symmetry with the same consequence; without it, the LSP may decay.

[5] Editorial clarification: This one-loop equation is the precise version of the nearly straight lines drawn as functions of logarithmic energy. Thresholds and higher-loop corrections bend the lines slightly.

grand-unified normalization,

$$(b_1, b_2, b_3)_{\mathrm{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \tag{6.19}$$

$$(b_1, b_2, b_3)_{\mathrm{MSSM}} = \left(\frac{33}{5}, 1, -3\right). \tag{6.20}$$

With superpartner thresholds near the TeV scale, the extrapolated supersymmetric curves meet much more accurately near

$$M_{\mathrm{GUT}} \sim 10^{16}\ \mathrm{GeV}. \tag{6.21}$$

No experiment directly reaches that scale. Low-energy couplings are measured, the field content determines the running, and the curves are extrapolated over many decades. Two lines always cross somewhere; three independent lines meeting near one point is the nontrivial observation.

Question from the audience. Does the improved extrapolation assume exact supersymmetry and equal partner masses? [a]

Response. No. Supersymmetry is broken, so the masses are split. What matters is when the superpartner loops enter the running. If the partners are not too heavy, the slopes change early enough for the three couplings to converge. If they are exceedingly heavy, the supersymmetric thresholds arrive too late and the improved meeting is lost.

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[a] Lecture timestamp: 00:38:09.

Threshold corrections and the unknown high-energy spectrum prevent a literal claim that the three lines meet with mathematical exactness. The result is nevertheless striking: the same approximate superpartner scale that helps the Higgs and provides a thermal dark-matter candidate also improves unification.

6.4.1 Gravity and the common high-energy scale

Gravity behaves differently because its charge is energy itself. A useful dimensionless measure at center-of-mass energy E is

$$\alpha_G(E) \sim GE^2 = \left(\frac{E}{M_{\mathrm{Pl}}}\right)^2. \tag{6.22}$$

At ordinary energies this is fantastically small, but it grows quadratically with E. Its inverse therefore falls much faster than the logarithmically running gauge inverse couplings and becomes order unity near the Planck scale.

This comparison suggests that all interactions may be governed by neighboring high scales, but it is less clean than gauge unification. The gauge couplings are dimensionless parameters in renormalizable field theories; Newton's constant is dimensionful, so translating gravity onto the same plot requires a convention.

Question from the audience. Do superpartners participate in the same three gauge interactions, or do they define new forces? [a]

Response. They carry the same gauge quantum numbers as their partners. A selectron has the electron's electric charge, and a squark has the corresponding quark's color and electroweak charges, though their spins and masses differ. Their loops therefore modify the same gauge interactions rather than introducing three new ones.

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[a] Lecture timestamp: 00:42:35.

Question from the audience. Would discovery of a superpartner at the LHC be definitive evidence for supersymmetry? [a]

Response. A single unfamiliar particle would not be enough by its name alone, but a superpartner spectrum obeying supersymmetric relations among spins, charges, masses, and couplings would be highly distinctive. Supersymmetry imposes many correlated restrictions, so the detailed interaction pattern would provide the decisive test.

[a] Lecture timestamp: 00:43:27.

6.4.2 Why the scale cannot be moved away without cost

If superpartners lie only a little beyond experimental reach, all three motivations can still operate. If their masses are raised by orders of magnitude, each argument deteriorates:

1. Higgs-mass cancellations leave a large residual sensitivity to the splitting.
2. A conventional weak-scale thermal relic no longer emerges at the expected scale.
3. Gauge-running thresholds move upward and the three couplings meet less accurately.

Supersymmetry could still exist at a much higher scale, but it would no longer solve these particular low-energy problems in the same economical way. Longevity of the idea is not evidence that nature must use it; it is evidence that the proposal is coherent enough to deserve a decisive test.

6.5 What Cosmology Can and Cannot Tell Us About Dark Matter

Question from the audience. Can standard cosmology determine the amount of dark matter and the mass of each dark-matter particle, much as nucleosynthesis constrains hydrogen and helium? [a]

Response. Cosmology measures the total dark-matter mass density very well, but density alone does not separate mass from number. The same density could be carried by a few heavy particles or an enormous number of very light ones. Additional theory or nongravitational observations are needed to identify the microscopic particle.

[a] Lecture timestamp: 00:46:32.

Two broad possibilities illustrate the ambiguity. A weakly interacting particle near the electroweak or TeV scale can freeze out from a hot thermal bath. At the other extreme, an extremely light boson such as an axion can occupy a coherent field configuration or condensate. Intermediate possibilities are not forbidden merely because the two best-developed mechanisms sit near opposite ends.

The halo structure itself supplies an important constraint. Dark matter crosses ordinary matter with little friction, so it remains spatially extended instead of collapsing into the luminous disk. Rotation curves and other gravitational phenomena tell us that the mass is present, but they do not by themselves count the particles.

Question from the audience. Could the early universe initially have contained only superparticles, which later decayed into the particles we see? [a]

Response. At temperatures much larger than all relevant masses, rapid reactions drive the species toward thermal equilibrium rather than an arbitrary one-sided population. Ordinary collisions create superpartners when they are underabundant, while superpartner collisions and decays replenish ordinary particles. The precise abundance depends on the interactions and degrees of freedom, but the hot plasma continually tends toward a balance.

[a] Lecture timestamp: 00:49:11.

As the universe cools, massive species become Boltzmann suppressed and annihilate or decay. A stable LSP ceases to track equilibrium when its interaction rate becomes slower than cosmic expansion:

$$\Gamma_{\text{ann}} \sim n\langle\sigma v\rangle \lesssim H. \tag{6.23}$$

The remaining particles form a relic abundance.[6]

Question from the audience. Which ordinary particle is the LSP the partner of, and must it be noninteracting? [a]

Response. There is no model-independent answer. A viable dark-matter LSP is expected to be electrically neutral and colorless, but it need not be completely noninteracting. It may be a mixture related to the photon, weak gauge bosons, and Higgs fields, or it may be the gravitino. Its precise identity follows from the supersymmetry-breaking spectrum.

[a] Lecture timestamp: 00:51:05.

6.6 Vacuum Diagrams, Self-Energy Diagrams, and Expansion

Question from the audience. What is the difference between the Higgs divergence and the vacuum-energy divergence? [a]

Response. A vacuum diagram has no external particle lines; it changes the effective action of empty space and contributes to the vacuum energy. A Higgs self-energy diagram has two external Higgs legs; it changes the Higgs propagator and therefore its mass. Both involve virtual fluctuations, but they answer different questions.

[a] Lecture timestamp: 00:52:09.

One may picture a real particle as an impurity inserted into the vacuum. Vacuum fluctuations that attach to this impurity shift its propagation and mass. The Casimir plates also modify vacuum fluctuations, but through macroscopic boundary conditions. The common ingredient is a response of quantum fluctuations to what has been inserted; the observable being corrected differs in each case.

6.6.1 Matter, radiation, and vacuum dilute differently

Let $a(t)$ be the cosmological scale factor and

$$H = \frac{\dot{a}}{a} \tag{6.24}$$

[6] Editorial clarification: This freeze-out criterion is the standard quantitative completion of the thermal-balance argument. Other LSP candidates, especially very weakly coupled gravitinos, can have nonthermal histories.

the Hubble parameter. Local stress-energy conservation in a homogeneous universe gives

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{6.25}$$

For an equation of state $p = w\rho$, this integrates to

$$\rho(a) \propto a^{-3(1+w)}. \tag{6.26}$$

[7]

For cold, nonrelativistic matter, $p \approx 0$, so

$$\rho_m \propto a^{-3}. \tag{6.27}$$

A fixed comoving volume grows as a^3, while the rest energy carried by its particles remains approximately fixed.

For radiation, $p = \rho/3$, and

$$\rho_r \propto a^{-4}. \tag{6.28}$$

Three powers come from dilution of the number density. The fourth comes from redshift:

$$\lambda \propto a, \qquad E_\gamma = \frac{2\pi\hbar c}{\lambda} \propto a^{-1}. \tag{6.29}$$

Question from the audience. If a photon has a wavefunction, why may we speak of it as a particle rather than a wave when deriving the dilution law? [a]

Response. The distinction does not affect the scaling. In the particle description the number in a comoving volume is fixed and each quantum redshifts as a^{-1}. In the classical-wave description the wavelength stretches and the field energy falls by the same amount. We can use whichever language makes the expansion easiest to see.

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[a] Lecture timestamp: 00:58:21.

Vacuum energy has $p_\Lambda = -\rho_\Lambda$, so

$$\rho_\Lambda = \text{constant}. \tag{6.30}$$

It does not dilute as space expands. Indeed, a comoving region contains

$$E_\Lambda = \rho_\Lambda a^3 V_0, \tag{6.31}$$

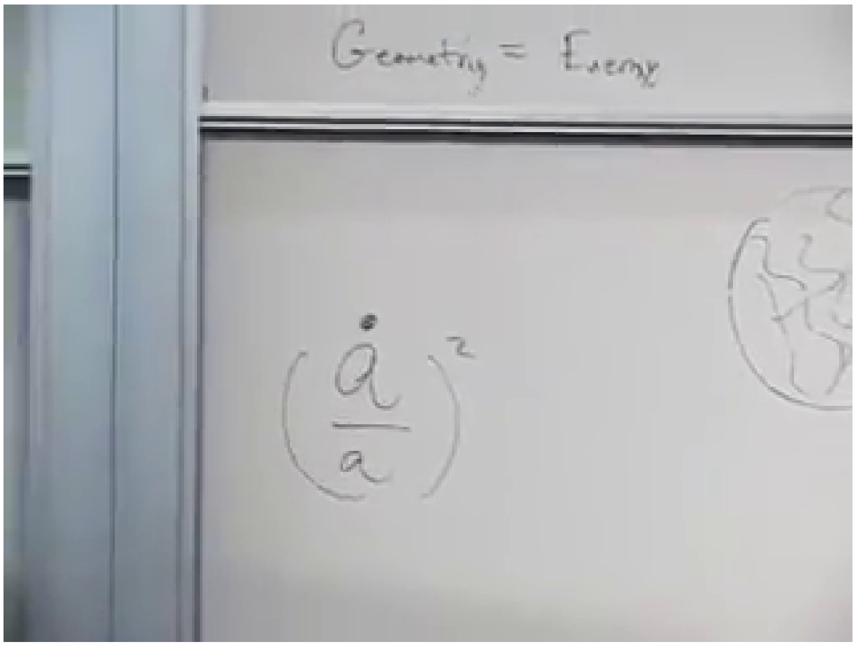
which grows with its physical volume. That fact creates the energy-conservation question we shall address shortly.

The different powers organize cosmic history. Radiation falls fastest, matter falls more slowly, and vacuum energy does not fall at all. A tiny vacuum term can therefore be negligible in the early universe and dominant at late times. Using the rough values discussed here, the present energy budget is about 70% vacuum energy and 30% matter, with radiation much smaller.

[7] Editorial clarification: The continuity equation supplies the compact derivation underlying the particle-counting and wavelength-stretching explanations.

6.7 Geometry Equals Energy

The time-time component of Einstein's equations turns the preceding scaling laws into dynamics. The board begins with a slogan and a single geometric term.



Lecture frame: 01:06:37.088.

Figure 6.1: The organizing slogan *geometry equals energy*, with the expansion term $H^2 = (\dot{a}/a)^2$ written beneath it.

For a homogeneous and isotropic universe, the first Friedmann equation is

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho, \tag{6.32}$$

in units $c = 1$, where $k = +1, 0, -1$ denotes closed, flat, or open spatial geometry and

$$\rho = \rho_{\text{ordinary}} + \rho_{\text{dark}} + \rho_{\text{radiation}} + \rho_{\text{vacuum}}. \tag{6.33}$$

The sign seen beside $1/a^2$ depends on whether the curvature term is kept on the left or moved to the right, as well as on the value of k.

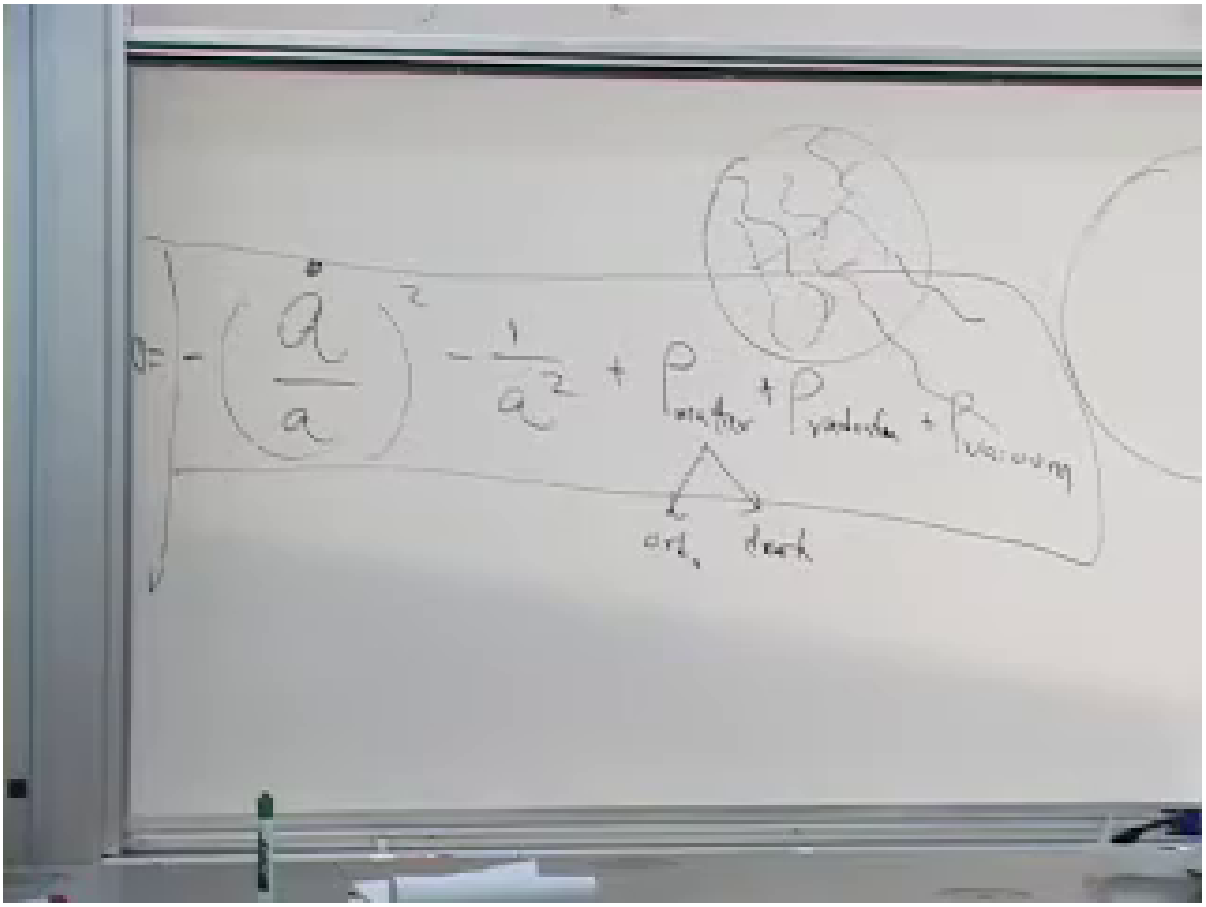


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Figure 6.2: The full board balance: expansion and spatial curvature on the geometric side, with matter, radiation, and vacuum energy on the source side. Matter is further separated into ordinary and dark components.

6.7.1 The cosmological Hamiltonian constraint

Question from the audience. If the vacuum energy in a comoving region grows as the universe expands, is there a constant net energy density or a conserved total energy? [a]

Response. General relativity does not supply a universal global energy for an arbitrary expanding spacetime. In the homogeneous cosmological reduction, however, the time-time Einstein equation is a Hamiltonian constraint. When geometry and matter are placed on one side, the constrained Hamiltonian vanishes. The changing matter terms are balanced by the gravitational degrees of freedom rather than by an ordinary fixed energy in a box.

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[a] Lecture timestamp: 01:01:51.

Schematically, the canonical statement is

$$\mathcal{H}_{\text{gravity}} + \mathcal{H}_{\text{matter}} = 0. \tag{6.34}$$

This is the controlled meaning of saying that the total energy is zero. It is an equation of motion enforced by time-reparameterization invariance, not a claim that every local energy density vanishes.

In a comoving volume, ordinary matter energy is roughly fixed, radiation energy decreases, and vacuum energy grows; meanwhile the gravitational contribution and expansion rate adjust so that Einstein's constraint remains satisfied.

Local conservation remains exact:

$$\nabla_\mu T^{\mu\nu} = 0. \tag{6.35}$$

For the homogeneous fluid this is precisely the continuity equation $\dot{\rho} + 3H(\rho + p) = 0$. What generally fails is the flat-spacetime intuition that there must also be one coordinate-independent global sum of matter energy conserved in an expanding universe.[8]

Question from the audience. Is the zero constraint the condition that decides whether the universe is open, closed, or flat? [a]

Response. No. The Hamiltonian constraint applies in every case. Open, flat, and closed universes differ through the curvature label k and therefore through the k/a^2 term. Curvature changes one contribution to the balance; it does not turn the constraint on or off.

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[a] Lecture timestamp: 01:04:42.

6.8 Grassmann Coordinates and Superfields

Question from the audience. Do Grassmann numbers actually appear in the Lagrangian of the theory? [a]

Response. They appear first because fermion fields are Grassmann odd. They appear a second time as coordinates of superspace. Superfields depend on those odd coordinates, and supersymmetric actions are built by differentiating and integrating with respect to them.

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[a] Lecture timestamp: 01:09:37.

An ordinary field is a function of spacetime,

$$\phi = \phi(x^\mu). \tag{6.36}$$

Minimal supersymmetry in four dimensions augments x^μ by a two-component odd spinor θ^α and its conjugate $\bar{\theta}^{\dot{\alpha}}$:

$$z^M = (x^\mu, \theta^\alpha, \bar{\theta}^{\dot{\alpha}}). \tag{6.37}$$

There are two complex odd components, or four real odd directions. A superfield is a function on this superspace:

$$\Phi = \Phi(x, \theta, \bar{\theta}). \tag{6.38}$$

These are not tiny spatial dimensions in the metric sense. A Grassmann coordinate has no positive numerical magnitude. Calling it small is a mnemonic for nilpotence:

$$(\theta^\alpha)^2 = 0, \qquad \theta^\alpha\theta^\beta = -\theta^\beta\theta^\alpha. \tag{6.39}$$

[9]

[8] Editorial clarification: This distinction makes the classroom statement that energy is always zero precise: local covariant conservation and the cosmological Hamiltonian constraint are exact, while a unique global matter-plus-gravity energy is not generally defined.

[9] Editorial clarification: Nilpotence, not numerical smallness, is the reason every expansion in finitely many Grassmann coordinates terminates.

With four independent odd coordinates, a completely general superfield has at most $2^4 = 16$ Grassmann monomials. Schematically,

$$\begin{aligned}\Phi(x,\Theta) &= \phi(x) + \Theta_A\psi_A(x)\\ &\quad + \frac{1}{2}\Theta_A\Theta_B Z_{AB}(x) + \cdots\\ &\quad + \Theta_1\Theta_2\Theta_3\Theta_4 D(x),\end{aligned} \tag{6.40}$$

where Θ_A collectively denotes the four real odd coordinates. Since the whole superfield may be chosen even, coefficients of even powers are bosonic and coefficients of odd powers are fermionic. One superfield therefore packages a finite multiplet of ordinary component fields.

This packaging is more than notation. Supersymmetry acts geometrically as translation in the odd directions mixed with spacetime translation. A Lagrangian constructed from superfields according to superspace rules inherits the symmetry in the same way that a scalar Lagrangian constructed from tensors inherits rotational or Lorentz invariance.

The action now includes Grassmann measures. A typical full-superspace term has the form

$$S_D = \int d^4x\, d^2\theta\, d^2\bar{\theta}\, \mathcal{K}(\Phi,\Phi^\dagger). \tag{6.41}$$

Chiral terms instead use

$$S_F = \int d^4x\, d^2\theta\, W(\Phi) + \text{h.c.} \tag{6.42}$$

[10]

We have therefore reached a question that ordinary calculus does not answer: what does integration over an anticommuting coordinate mean?

6.9 Berezin Integration

6.9.1 The ordinary model: a total derivative

Begin with an ordinary function $F(x)$ that vanishes at large positive and negative x. The fundamental theorem of calculus gives

$$\int_{-\infty}^{\infty} dx\, \frac{dF}{dx} = F(\infty) - F(-\infty) = 0. \tag{6.43}$$

[10]Editorial clarification: The distinction between full-superspace and chiral measures is the standard refinement of the schematic $d^4x\, d^4\theta$ action. Concrete examples will determine which measure is appropriate.

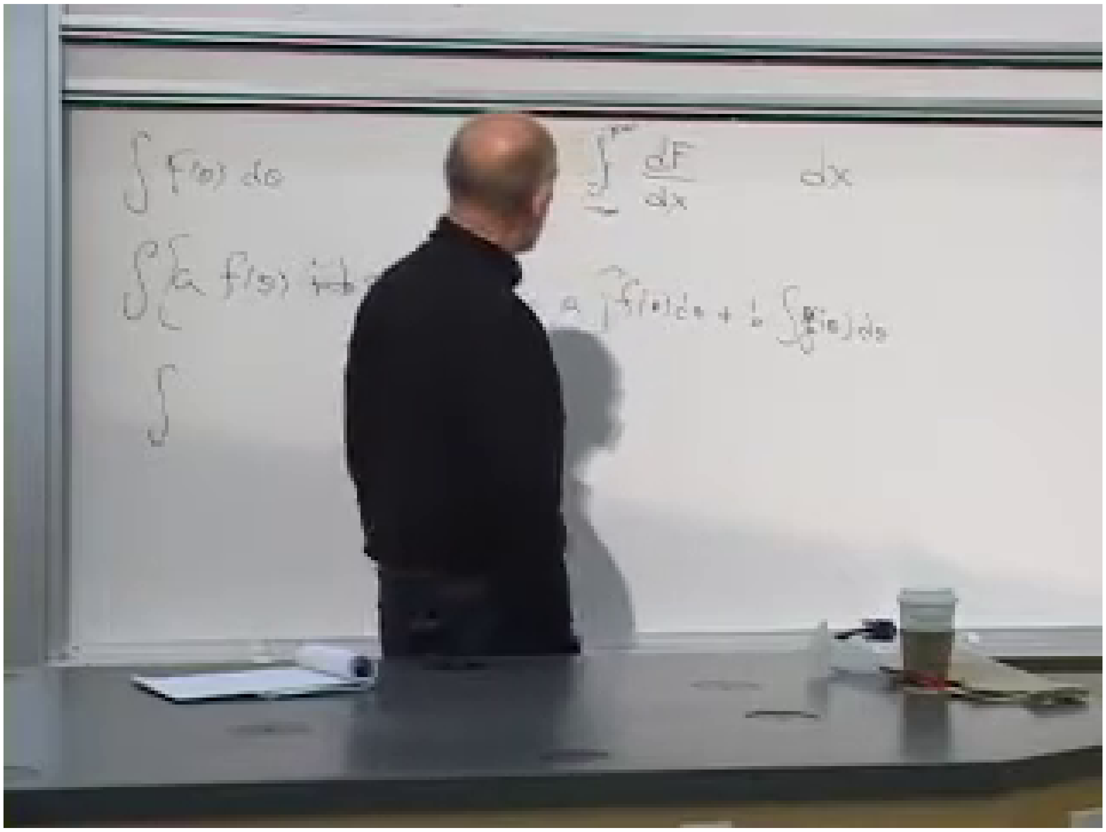

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Figure 6.3: The ordinary prototype for Berezin integration: an integral of a total derivative reduces to endpoint values and vanishes when the function dies off. The lower board line is partly obscured, so the clean equation is supplied in the text.

We want the Grassmann integral to be linear and to share the useful property that a total derivative integrates to zero. For one odd coordinate, require

$$\int d\theta\,[af(\theta) + bg(\theta)] = a\int d\theta\, f(\theta) + b\int d\theta\, g(\theta), \tag{6.44}$$

$$\int d\theta\, \partial_\theta f(\theta) = 0. \tag{6.45}$$

The second condition is the algebraic analogue of translation invariance of the measure. There are no endpoints of an odd number line and no sum over tiny intervals.

Question from the audience. If there is no indefinite Grassmann integral, are we abandoning the idea that integration is the inverse of differentiation? [a]

Response. Yes. Berezin integration is a definite linear functional, not an antiderivative operation. For one Grassmann variable it is, in fact, represented by the same algebraic operation as differentiation.

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[a] Lecture timestamp: 01:21:19.

Question from the audience. Is the d^4x integration over all spacetime also a definite integral? [a]

Response. Yes. Boundary conditions are understood when one integrates the action over spacetime. The difference is that ordinary spacetime integration can be understood through limits of sums, whereas Grassmann integration is defined algebraically by its action on the finite basis $1, \theta$.

[a]Lecture timestamp: 01:21:34.

6.9.2 One variable

Every function of one odd variable can be written with its coefficient in the fixed right-hand order

$$f(\theta) = a + \theta b. \tag{6.46}$$

Since $\partial_\theta \theta = 1$, the total-derivative rule implies

$$\int d\theta \, 1 = 0. \tag{6.47}$$

The remaining normalization is chosen to be

$$\int d\theta \, \theta = 1. \tag{6.48}$$

Consequently,

$$\int d\theta \, f(\theta) = b. \tag{6.49}$$

The integral selects the coefficient of the highest available power. With the left-derivative convention used here,

$$\partial_\theta f = b = \int d\theta \, f. \tag{6.50}$$

Question from the audience. Is there a function $F(\theta)$ satisfying $\partial_\theta F = \theta$? [a]

Response. No. Such a function would need a term proportional to θ^2, but $\theta^2 = 0$. This is another way to see why there is no Grassmann analogue of an indefinite antiderivative.

[a]Lecture timestamp: 01:30:05.

6.9.3 Two variables and the order of the measure

For a complex pair $\theta, \bar\theta$, write a general even function in the ordered form

$$f(\theta, \bar\theta) = a + \bar\theta\psi + \bar\psi\theta + b\,\bar\theta\theta. \tag{6.51}$$

No higher term exists because

$$\theta^2 = \bar\theta^2 = 0, \qquad \bar\theta\theta = -\theta\bar\theta. \tag{6.52}$$

We now fix the convention

$$\int d\theta \, d\bar\theta \, \bar\theta\theta = 1. \tag{6.53}$$

With iterated integrals, the adjacent $d\bar{\theta}$ operation acts first. The table is then

$$\int d\theta\, d\bar{\theta}\, 1 = 0, \tag{6.54}$$

$$\int d\theta\, d\bar{\theta}\, \theta = 0, \tag{6.55}$$

$$\int d\theta\, d\bar{\theta}\, \bar{\theta} = 0, \tag{6.56}$$

$$\int d\theta\, d\bar{\theta}\, \bar{\theta}\theta = 1. \tag{6.57}$$

Thus

$$\int d\theta\, d\bar{\theta}\, f(\theta, \bar{\theta}) = b. \tag{6.58}$$

Question from the audience. For a term linear in θ, is it really the θ integration that makes the double integral vanish? [a]

Response. No; the live explanation had the roles reversed. Integrating θ with respect to θ gives one. The result then contains no $\bar{\theta}$, and the remaining $\bar{\theta}$ integral of one vanishes. Either way, a double integral survives only when both variables are present.

[a] Lecture timestamp: 01:34:59.

Question from the audience. If θ and $\bar{\theta}$ are interchanged, does the integral change sign? [a]

Response. Yes. Odd variables and odd measures anticommute. With the convention above,

$$\int d\bar{\theta}\, d\theta\, \bar{\theta}\theta = -1.$$

One must therefore choose an ordering and keep it consistently. The physics is convention-independent, but intermediate signs are not.

[a] Lecture timestamp: 01:36:33.

For n independent Grassmann variables, the pattern is immediate:

$$\int d\theta_n \cdots d\theta_1\, f(\theta_1, \dots, \theta_n) \tag{6.59}$$

extracts the coefficient of the ordered top monomial $\theta_1 \cdots \theta_n$, up to the sign fixed by the chosen order of variables and measures. This simple component-extraction rule is the engine behind superspace actions. A supersymmetry variation becomes an odd total derivative, and the superspace integral discards it, just as an ordinary action discards a spacetime total derivative under suitable boundary conditions.

6.10 Why Such a Small Calculus Has So Much Power

The rules look almost trivial:

$$\int d\theta\, 1 = 0, \qquad \int d\theta\, \theta = 1. \tag{6.60}$$

Yet they let us build interacting field theories in which relations between bosonic and fermionic masses and couplings are preserved automatically. The purpose of superspace calculus is the same as the purpose of tensor calculus in relativity: arrange the notation so that the desired symmetry survives every allowed manipulation.

Sign bookkeeping is the price. Every exchange of two odd objects contributes a minus sign, and a calculation can be conceptually correct while failing by one ordering choice. That difficulty is not decorative; it is the algebraic shadow of fermionic statistics.

Question from the audience. Were Grassmann numbers invented together with supersymmetry, or did they exist earlier? [a]

Response. They are much older. Hermann Grassmann developed the underlying exterior algebra in the nineteenth century. Differential forms already use anticommuting basis elements; twentieth-century quantum field theory and supersymmetry found a new physical role for the same mathematics.

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[a] Lecture timestamp: 01:41:19.

We are now ready for concrete constructions. Superfields collect complete boson-fermion multiplets, superspace derivatives preserve the algebra, and Berezin integration extracts the component action. The next task is to use this machinery in the simplest one-dimensional model and then carry it into field theory.

Chapter 7

Superspace and the Construction of Supersymmetric Dynamics

Overview. We rebuild supersymmetry from the smallest system in which it can act: one bosonic oscillator and one fermionic oscillator of equal energy. The contrast between their second- and first-order equations leads to the supercharges and their algebra. We then reinterpret that algebra as geometry on a space with the ordinary coordinate (t) and the Grassmann coordinates $\theta, \bar{\theta}$. Superfields package bosons, fermions, and an auxiliary field into a single object; Berezin integration extracts an invariant ordinary Lagrangian; covariant derivatives make symmetry-compatible constraints possible; and the chiral constraint provides the first important example. The lecture deliberately exposes the sign and ordering difficulties of this machinery, then closes by explaining why the symmetry itself is more fundamental than a catalogue of Feynman rules.

7.1 Return to the One-Mode System

Let us begin again with fermions and bosons at rest. We suppress every spatial coordinate and every momentum label. A quantum is either present or absent in a given mode, and creation and annihilation operators are enough to describe it. Write $a^\dagger, a$ for a bosonic mode and $c^\dagger, c$ for a fermionic mode. They obey

$$[a, a^\dagger] = 1, \qquad \{c, c^\dagger\} = 1, \qquad c^2 = (c^\dagger)^2 = 0. \tag{7.1}$$

If the two quanta have the same energy m, then, apart from an irrelevant vacuum constant,

$$H = m\left(a^\dagger a + c^\dagger c\right). \tag{7.2}$$

The number operator $a^\dagger a$ may take any nonnegative integer value, while $c^\dagger c$ can only be zero or one. This is the entire kinematics of the toy model.

The simplification is strategic. We want to understand the operation that removes a boson and inserts a fermion, or does the reverse, without changing the energy. Before constructing that operation, however, we translate the same system from operator language into Lagrangian language.

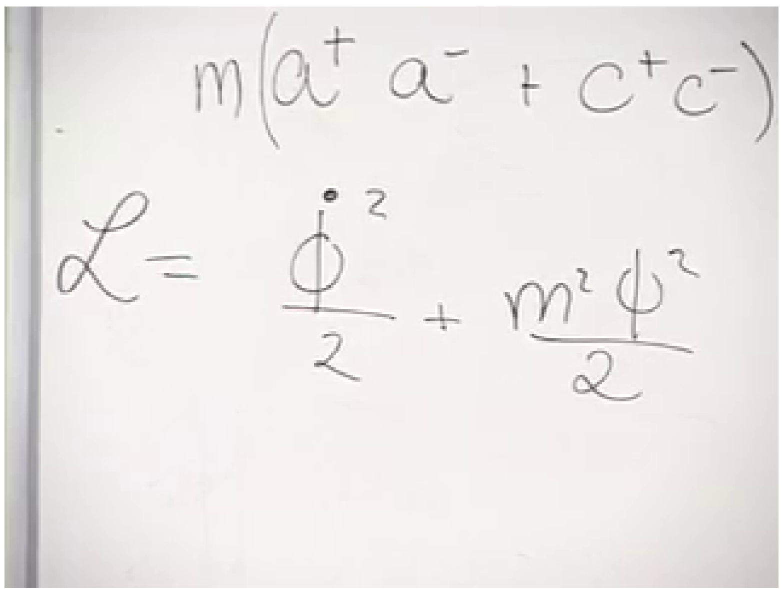
7.2 The Bosonic Oscillator

For the boson, introduce a real coordinate $\phi(t)$. With no spatial variation, a free scalar field has become an ordinary harmonic oscillator. The first expression written on the board was

$$\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m^2\phi^2.$$

That expression is kinetic plus potential energy, not the Lagrangian. The sign is corrected immediately in the lecture:

$$L_B = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}m^2\phi^2. \tag{7.3}$$



Lecture frame: 00:03:25.431.

Figure 7.1: The one-mode Hamiltonian above the bosonic Lagrangian. The frame preserves the initial plus sign before the spoken correction to $T - V$.

Question from the audience. What is wrong with the sign in the displayed oscillator Lagrangian? [a]

Response. The energy is $T + V$, but the Lagrangian is $T - V$. The potential term must therefore carry a minus sign. The correction changes the equation from exponential behavior to the intended harmonic oscillation.

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[a]Lecture timestamp: 00:04:00.

Indeed, the Euler–Lagrange equation gives

$$\frac{d}{dt}\frac{\partial L_B}{\partial \dot{\phi}} = \frac{\partial L_B}{\partial \phi}, \tag{7.4}$$

$$\ddot{\phi} = -m^2\phi, \tag{7.5}$$

or

$$\ddot{\phi} + m^2\phi = 0. \tag{7.6}$$

Its frequency is m, not m^2. Quantizing this oscillator produces the same ladder of states described by $a^\dagger$ and a in Eq. (7.2).

7.3 Why a Fermionic Action Is First Order

The fermionic Lagrangian looks simpler because it contains only one time derivative. The reason is already visible in a deliberately naive trial. For an ordinary commuting variable,

$$L_{\text{trial}} = \phi\dot{\phi} = \frac{1}{2}\frac{d}{dt}\phi^2. \tag{7.7}$$

It is a total derivative. Its Euler–Lagrange equation says only $\dot{\phi} = \dot{\phi}$, so every differentiable path is allowed. Removing a total derivative from the action changes endpoint data but not the bulk equation of motion.

Now rename the variable ψ and declare it Grassmann odd. Both ψ and $\dot{\psi}$ anticommute. With a fixed choice of left variation, reordering the odd factors introduces a minus sign, so the same formal trial gives

$$\frac{d}{dt}\frac{\partial L_{\text{trial}}}{\partial\dot{\psi}} = \dot{\psi}, \qquad \frac{\partial L_{\text{trial}}}{\partial\psi} = -\dot{\psi}. \tag{7.8}$$

The equation is therefore

$$\dot{\psi} = -\dot{\psi} \quad\Longrightarrow\quad \dot{\psi} = 0. \tag{7.9}$$

The expression that was a tautology for a commuting coordinate has acquired dynamical content for an anticommuting one. It has too much content for an oscillator, since it permits only a constant, but this is precisely the clue we need.

Question from the audience. Is the solution simply that ψ is constant? [a]

Response. Yes. Equation (7.9) is a genuine first-order differential equation, and its solution is an arbitrary Grassmann-valued constant. A mass term will turn that static solution into an oscillating one.

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[a] Lecture timestamp: 00:13:57.

The obvious attempt, $m\psi^2$, fails because $\psi^2 = 0$. We need two real odd variables ψ_1, ψ_2, or equivalently one complex variable and its conjugate. The lecture works through the two-real-variable form on the board, repeatedly checking signs and factors of two. A conventionally normalized complex form is

$$L_F = \frac{i}{2}\left(\bar{\psi}\dot{\psi} - \dot{\bar{\psi}}\psi\right) - m\bar{\psi}\psi. \tag{7.10}$$

Up to a total derivative, this is $i\bar{\psi}\dot{\psi} - m\bar{\psi}\psi$. Varying $\bar{\psi}$ gives

$$i\dot{\psi} = m\psi, \qquad \psi(t) = \psi(0)e^{-imt}. \tag{7.11}$$

The conjugate equation has the opposite phase. Thus the fermionic mode oscillates with energy m, just as the bosonic mode does, although its equation is first order rather than second order.

[1]

[1] Editorial clarification: Equation (7.10) fixes one standard ordering and normalization for the exploratory two-real-variable calculation. The lecture itself inserts the factor of i, adjusts a factor of two, and leaves one real-component sign unsettled. These changes do not alter its central conclusion: fermion kinetics are first order, the mass appears linearly, and a complex fermion supplies a nonvanishing bilinear.

This is the zero-dimensional shadow of a familiar relativistic contrast. The Klein–Gordon equation is second order in time, while the Dirac equation is first order. With space suppressed, the boson is one harmonic oscillator and the fermion is one complex Grassmann oscillator.

Question from the audience. Does the order in which ψ, $\dot{\psi}$, and their conjugates are written matter? [a]

Response. The intermediate formulas do depend on ordering because odd factors anticommute. The physical theory does not depend on an arbitrary writing preference, provided one uses a single convention for left or right variation, conjugation, and integration throughout. Reversing an odd pair changes a sign that must be changed everywhere else consistently.

[a] Lecture timestamp: 00:18:58.

Combining Eqs. (7.3) and (7.10) gives a Lagrangian description of the same matched pair represented by Eq. (7.2). We are now ready to identify the symmetry between them.

7.4 Supercharges and Their Algebra

An operator that removes a fermion and inserts a boson is

$$Q^\dagger = \sqrt{2m}\, a^\dagger c, \tag{7.12}$$

while its conjugate removes a boson and inserts a fermion,

$$Q = \sqrt{2m}\, ac^\dagger. \tag{7.13}$$

The normalization is chosen so that the algebra takes its standard form. Using $cc^\dagger = 1 - c^\dagger c$, $aa^\dagger = a^\dagger a + 1$, and the fact that bosonic and fermionic operators commute with one another,

$$\begin{aligned}
\{Q, Q^\dagger\} &= 2m\left(ac^\dagger a^\dagger c + a^\dagger c a c^\dagger\right) \\
&= 2m\left[(a^\dagger a + 1)c^\dagger c + a^\dagger a(1 - c^\dagger c)\right] \\
&= 2m\left(a^\dagger a + c^\dagger c\right) = 2H.
\end{aligned} \tag{7.14}$$

The bosonic operators may be moved through the fermionic ones. The same nilpotence that limits fermion occupation gives

$$Q^2 = (Q^\dagger)^2 = 0. \tag{7.15}$$

Finally,

$$[H, Q] = [H, Q^\dagger] = 0. \tag{7.16}$$

Each supercharge changes the boson number and fermion number in opposite directions, so the two changes in energy cancel when the frequencies are equal.

Question from the audience. What is $[H, Q]$? Is the answer proportional to Q? [a]

Response. No. The answer is zero. A lone creation or annihilation operator would acquire its energy under commutation with H, but the bosonic and fermionic contributions in Q cancel. That cancellation is exactly the equal-energy condition.

[a] Lecture timestamp: 00:25:10.

Question from the audience. Was that commutator not already written on the board? [a]

Response. It was. The rediscovery is useful because it forces us to interpret rather than merely copy the relation: commuting with the Hamiltonian means that the transformation is a symmetry and that its generator is conserved.

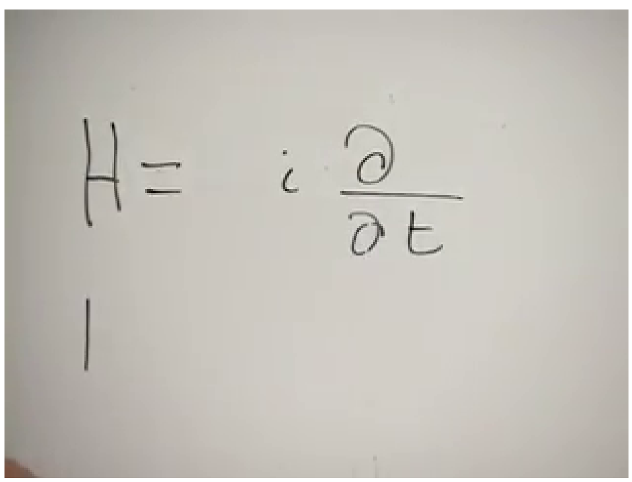
[a]Lecture timestamp: 00:25:45.

The adjective *charge* reflects this conservation law. We should not, however, treat Q as an ordinary simultaneously measurable number attached to a state. It is Grassmann odd, nilpotent, and maps a bosonic state into a fermionic one. The useful statement is that it is a conserved generator acting within degenerate supermultiplets. [2]

7.5 Generators as Derivatives

The algebra is compact, but by itself it does not tell us how to build interacting theories. The next step is geometric. Familiar symmetry generators already act as differential operators. In the convention of the lecture,

$$H = i\frac{\partial}{\partial t}, \qquad P = -i\frac{\partial}{\partial x}. \tag{7.17}$$



Lecture frame: 00:28:12.720.

Figure 7.2: The Hamiltonian represented as the generator of time translations.

Angular momentum is similarly a coordinate times a derivative. If supercharges are odd generators, it is natural to ask whether they can be represented by derivatives with respect to odd coordinates. We therefore enlarge the time axis to the superspace

$$(t, \theta, \bar{\theta}), \qquad \theta^2 = \bar{\theta}^2 = 0, \qquad \theta\bar{\theta} = -\bar{\theta}\theta. \tag{7.18}$$

Fields on this space depend on an ordinary time and a conjugate pair of Grassmann coordinates.

[2]Editorial clarification: This wording refines the classroom analogy with labeling states by energy or electric charge. Individual non-Hermitian odd supercharges are not ordinary observables with numerical quantum labels. Their conserved action and the representation of the full superalgebra provide the correct classification.

The lecture deliberately exposes how treacherous the signs are. A sign from complex conjugation, a sign from moving an odd parameter, and a sign from choosing left rather than right derivatives are different operations. Rather than preserve mutually incompatible board stages, we now fix one convention: all odd derivatives are left derivatives, and

$$q = \partial_\theta + i\bar{\theta}\partial_t, \qquad \bar{q} = \partial_{\bar{\theta}} + i\theta\partial_t. \tag{7.19}$$

Then

$$q^2 = \bar{q}^2 = 0, \qquad \{q, \bar{q}\} = 2i\partial_t = 2H. \tag{7.20}$$

The factor of two comes from two cross terms: $\{\partial_\theta, \theta\} = 1$ and $\{\partial_{\bar{\theta}}, \bar{\theta}\} = 1$. If the generators contained only ∂_θ and $\partial_{\bar{\theta}}$, their anticommutator would vanish and could not reproduce the Hamiltonian.

Question from the audience. Should the two time-derivative terms in q and $\bar{q}$ carry the same sign? [a]

Response. They must be chosen consistently so that the mixed anticommutator is $2H$. Either an overall plus convention or its consistently conjugated alternative can be used. The calculation, not visual intuition about the symbol i, fixes the relative signs.

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[a] Lecture timestamp: 00:31:01.

The room repeatedly checks the signs, and the difficulty becomes a joke: the signs appear to change during the commute to work. The joke is apt. Anticommuting a Grassmann parameter through a coordinate can change a sign even when no physical convention has changed. Equation (7.19) is the fixed reference for everything that follows.

7.5.1 Supersymmetry as a coordinate shift

Let $\xi, \bar{\xi}$ be infinitesimal odd parameters. Acting on a superfield with

$$\delta_\xi = \xi q + \bar{\xi}\bar{q} \tag{7.21}$$

is equivalent, in the convention above, to the infinitesimal coordinate change

$$\theta \longrightarrow \theta + \xi, \tag{7.22}$$

$$\bar{\theta} \longrightarrow \bar{\theta} + \bar{\xi}, \tag{7.23}$$

$$t \longrightarrow t + i\left(\xi\bar{\theta} + \bar{\xi}\theta\right). \tag{7.24}$$

The change in a scalar superfield is therefore

$$\delta_\xi \Phi = \left(\xi\partial_\theta + \bar{\xi}\partial_{\bar{\theta}} + i\xi\bar{\theta}\partial_t + i\bar{\xi}\theta\partial_t\right)\Phi = \left(\xi q + \bar{\xi}\bar{q}\right)\Phi. \tag{7.25}$$

This is the geometric content of supersymmetry. It is a translation in odd directions tied to a translation in ordinary time. Two odd shifts close into a time translation because of Eq. (7.20).

Question from the audience. Could the disputed minus sign represent a motion backward in time? [a]

Response. A different sign convention can reverse the displayed infinitesimal time shift, but no separate physical backward-time process is being introduced. What matters is that the coordinate transformation and the differential generators obey the same algebra.

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[a] Lecture timestamp: 00:58:13.

Question from the audience. Does this extended time have a physical realization, or is it only formal bookkeeping? [a]

Response. It is bookkeeping in one useful sense: it packages the algebra into geometry. But the packaging becomes physically valuable when interactions are introduced. Superspace gives a systematic calculus for constructing Lagrangians that preserve the boson–fermion symmetry at every step, just as tensor calculus helps us construct Lorentz scalars.

[a]Lecture timestamp: 00:58:23.

7.6 Superfields and the Auxiliary Component

Because there are only two odd coordinates, the expansion of an even superfield terminates:

$$\Phi(t,\theta,\bar\theta) = \phi(t) + \bar\theta\psi(t) + \bar\psi(t)\theta + D(t)\bar\theta\theta. \tag{7.26}$$

The coefficients of even monomials, ϕ and D, are bosonic; the coefficients of odd monomials, ψ and $\bar\psi$, are fermionic. No θ^2, $\bar\theta^2$, or higher term survives. We leave the time dependence unexpanded because an ordinary Taylor series in t would not terminate and would serve no purpose here.

Question from the audience. What is the component $D(t)$ in the top term? [a]

Response. At this stage it is simply the final bosonic coefficient in the finite expansion. In the component action it normally has no kinetic term and is eliminated by an algebraic equation of motion, so it is called an auxiliary field. The lecture playfully suggests “D for dummy,” which captures its nondynamical role without being the historical definition.

[a]Lecture timestamp: 00:41:46.

The superfield is one object, while $\phi, \psi, \bar\psi, D$ are its components. A supersymmetry transformation mixes them. Thus a boson and a fermion are related in much the same structural sense that the x- and y-components of a vector are related by a rotation, even though Grassmann directions cannot be pictured as ordinary axes.

7.7 The Superaction and the Top Component

To obtain equations of motion we need an action. In ordinary field theory one forms a scalar Lagrangian and integrates it over spacetime. Here the analogous expression is

$$S = \int dt\, d\theta\, d\bar\theta\; \Lambda\big(\Phi,\ \text{superspace derivatives of } \Phi\big), \tag{7.27}$$

where Λ is an even superfield. Berezin integration is algebraic. We choose

$$\int d\theta\, d\bar\theta\; \bar\theta\theta = 1. \tag{7.28}$$

Therefore, if

$$\Lambda = A + \bar\theta B + \bar B\theta + C\bar\theta\theta, \tag{7.29}$$

then

$$\int d\theta\, d\bar\theta\; \Lambda = C. \tag{7.30}$$

The odd integral extracts the top component, leaving an ordinary Lagrangian to be integrated over time.

Why is the result invariant? A supersymmetry variation translates the integration variables. The variation of the top component is an ordinary total derivative in time, while odd total derivatives vanish under Berezin integration. With suitable endpoint conditions, the action is unchanged. This is the superspace analogue of building a Lorentz scalar before integrating it over spacetime.

7.8 Multiplying Superfields

The simplest way to make a new superfield from an old one is to multiply it by itself. Starting from Eq. (7.26), every term containing θ^2 or $\bar{\theta}^2$ vanishes. Keeping the order $\bar{\theta}\theta$ and writing barred component fields to the left gives

$$\begin{aligned}\Phi^2 = \phi^2 + 2\phi\bar{\theta}\psi + 2\phi\bar{\psi}\theta \\ + \left(2\phi D - 2\bar{\psi}\psi\right)\bar{\theta}\theta.\end{aligned} \tag{7.31}$$

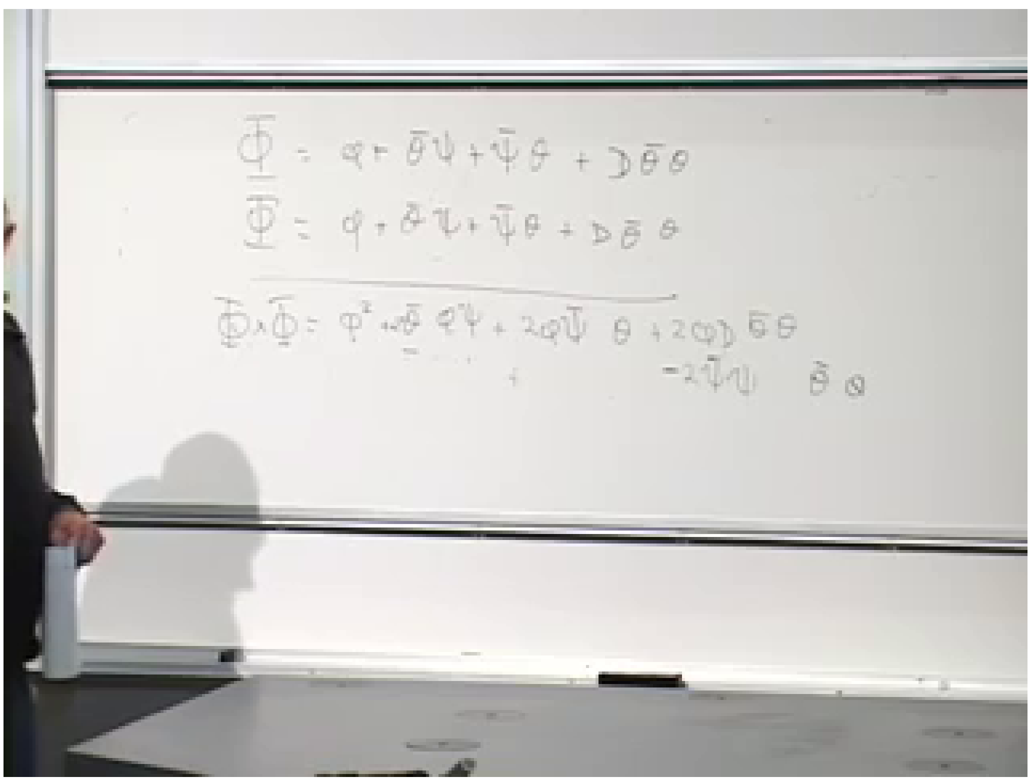

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Figure 7.3: The component-by-component superfield product and the collection of its $\bar{\theta}\theta$ terms.

The two terms ϕD and $D\phi$ add because both fields are even. The two fermion cross terms also add after each is reordered, but their common sign is negative:

$$(\bar{\theta}\psi)(\bar{\psi}\theta) + (\bar{\psi}\theta)(\bar{\theta}\psi) = -2\bar{\psi}\psi\,\bar{\theta}\theta. \tag{7.32}$$

Integrating the square over the odd coordinates yields

$$\int d\theta\, d\bar{\theta}\ \Phi^2 = 2\phi D - 2\bar{\psi}\psi. \tag{7.33}$$

This term is supersymmetric by construction, but it is not yet a kinetic Lagrangian. It contains no time derivatives.

The board then proposes Φ^3, Φ^4, and Φ^n as exercises. There is a compact rule. For any ordinary function W,

$$W(\Phi)|_{\bar{\theta}\theta} = W'(\phi)D - W''(\phi)\bar{\psi}\psi. \tag{7.34}$$

For $W(\phi) = \phi^2$, this reproduces Eq. (7.31); for $W(\phi) = \phi^n$, it gives

$$\Phi^n|_{\bar{\theta}\theta} = n\phi^{n-1}D - n(n-1)\phi^{n-2}\bar{\psi}\psi. \tag{7.35}$$

[3]

At this point the lecturer asks whether the room is exhausted. The audience says no; he answers that he is. That exchange is more than comic relief. Powers of superfields are easy because the expansion terminates, but derivatives and constraints are where the sign bookkeeping becomes genuinely demanding.

7.9 Which Derivatives Respect Supersymmetry?

There is one correction to make before following the next board argument. The ordinary time derivative is harmless:

$$[\partial_t, q] = [\partial_t, \bar{q}] = 0. \tag{7.36}$$

Consequently $\partial_t\Phi$ transforms as a superfield whenever Φ does. The subtle objects are the bare odd derivatives $\partial_\theta\Phi$ and $\partial_{\bar{\theta}}\Phi$, because they do not anticommute with both supercharges. We need modified odd derivatives for constraints and for manifestly supersymmetric component constructions. [4]

7.9.1 The "magic field" analogy

The lecture motivates the criterion with ordinary rotations. Let $f(x, y)$ satisfy

$$\partial_y f = 0. \tag{7.37}$$

Call such a function a "magic field." It depends on x but not on y. Is the class of magic fields preserved by rotations? For

$$L_z = -i\left(x\partial_y - y\partial_x\right), \tag{7.38}$$

we find

$$[\partial_y, L_z] = i\partial_x \neq 0. \tag{7.39}$$

[3]Editorial clarification: The lecture assigns the n-th power as an exercise and states the expected kinds of terms. Equation (7.34) supplies the completed rule in the fixed ordering convention.

[4]Editorial clarification: The spoken claim that time differentiation fails to produce another superfield is not correct in the convention used here. Since ∂_t commutes with both supercharges, it is covariant. The subsequent construction of modified odd derivatives remains necessary and is the actual mathematical point of the lecture.

Therefore $L_z f$ generally acquires y-dependence even when f initially obeys Eq. (7.37). The constraint is not rotationally invariant.

The general lesson is simple. If a field obeys $C\Phi = 0$ and a symmetry is generated by G, the transformed field remains constrained when the graded commutator of C and G vanishes. For two odd operators, the graded commutator is an anticommutator.

Question from the audience. When is a constraint on a field consistent with a symmetry? [a]

Response. When the constraint operator graded-commutes with the symmetry generators. Then $C(G\Phi)$ can be moved past G and reduced to $G(C\Phi) = 0$. The rotational example fails because ∂_y does not commute with the rotation generator.

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[a]Lecture timestamp: 01:17:56.

7.9.2 Covariant odd derivatives

With the supercharges fixed by Eq. (7.19), define

$$\mathcal{D} = \partial_\theta - i\bar{\theta}\partial_t, \qquad \bar{\mathcal{D}} = \partial_{\bar{\theta}} - i\theta\partial_t. \tag{7.40}$$

The only change is the sign of the time-derivative term. Direct calculation gives

$$\{\mathcal{D}, q\} = \{\mathcal{D}, \bar{q}\} = \{\bar{\mathcal{D}}, q\} = \{\bar{\mathcal{D}}, \bar{q}\} = 0, \tag{7.41}$$

$$\{\mathcal{D}, \bar{\mathcal{D}}\} = -2i\partial_t. \tag{7.42}$$

The two potentially nonzero cross terms cancel. This is the odd analogue of a covariant derivative: it differentiates while preserving the transformation law.

The lecture uses D both for the auxiliary component and for this derivative. We reserve $D(t)$ for the component and use calligraphic $\mathcal{D}$ for the operator.

7.10 The Chiral Constraint

We can now impose a nontrivial constraint that survives supersymmetry:

$$\bar{\mathcal{D}}\Phi = 0. \tag{7.43}$$

Such a field is called chiral. If it is chiral before a supersymmetry transformation, then

$$\bar{\mathcal{D}}(q\Phi) = -q(\bar{\mathcal{D}}\Phi) = 0, \qquad \bar{\mathcal{D}}(\bar{q}\Phi) = -\bar{q}(\bar{\mathcal{D}}\Phi) = 0. \tag{7.44}$$

Thus the entire transformed superfield remains inside the constrained class. Unlike the magic-field condition, chirality is compatible with the symmetry.

To solve the constraint, define a shifted time coordinate

$$\tau = t + i\bar{\theta}\theta. \tag{7.45}$$

Because we use left derivatives,

$$\partial_{\bar{\theta}}\tau = i\theta, \tag{7.46}$$

and hence

$$\bar{\mathcal{D}}F(\tau,\theta) = \left(\partial_{\bar{\theta}} - i\theta\partial_t\right) F(\tau,\theta) = i\theta\partial_\tau F - i\theta\partial_\tau F = 0. \tag{7.47}$$

The general local chiral superfield therefore depends on $\bar{\theta}$ only through τ:

$$\Phi_{\text{ch}}(\tau,\theta) = \phi(\tau) + \bar{\psi}(\tau)\theta. \tag{7.48}$$

Question from the audience. Is dependence on τ and θ the only form that satisfies the chiral constraint? [a]

Response. Locally, yes. The differential equation $\bar{\mathcal{D}}\Phi = 0$ says precisely that the independent $\bar{\theta}$-dependence can be absorbed into the shifted coordinate $\tau = t + i\bar{\theta}\theta$. The remaining finite expansion is a function of τ plus one term proportional to θ.

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[a]Lecture timestamp: 01:32:35.

Re-expanding in the original time coordinate,

$$\phi(\tau) = \phi(t) + i\bar{\theta}\theta\dot{\phi}(t), \tag{7.49}$$

$$\bar{\psi}(\tau)\theta = \bar{\psi}(t)\theta, \tag{7.50}$$

because the correction to the second line contains θ^2. Thus

$$\Phi_{\text{ch}}(t,\theta,\bar{\theta}) = \phi(t) + \bar{\psi}(t)\theta + i\bar{\theta}\theta\dot{\phi}(t). \tag{7.51}$$

Question from the audience. Should the Taylor expansion of $\phi(t + i\bar{\theta}\theta)$ contain a second-order term? [a]

Response. No. The first correction is $i\bar{\theta}\theta\dot{\phi}$, and every higher correction contains $(\bar{\theta}\theta)^2 = 0$. The audience catches this immediately; nilpotence has already supplied the first two terms and terminated the series.

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[a]Lecture timestamp: 01:33:51.

This is the economy of chirality. The unconstrained superfield in Eq. (7.26) has two bosonic and two fermionic component slots. The chiral condition removes the independent auxiliary slot in this minimal one-dimensional example and ties the top component to $\dot{\phi}$.

7.11 The Calculation That Properly Stops

The natural next move is to combine the chiral field with its conjugate and integrate over superspace. The conjugate expansion is started, and an audience member asks whether the order $\theta\bar{\theta}$ must reverse under conjugation. At that point the signs no longer look trustworthy. Rather than bluff, the calculation stops.

Question from the audience. Can we have the punchline even if the conjugate-field multiplication is deferred? [a]

Response. Yes. With the signs and conjugations handled consistently, a superspace expression built from a chiral field and its conjugate produces the desired ordinary bosonic and fermionic kinetic terms. The important result for this lecture is the construction principle, not an unreliable final line copied from a tired calculation.

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[a]Lecture timestamp: 01:37:30.

The stopping point should remain visible. The formalism is unforgiving: one misplaced odd factor changes a sign, and the same convention must survive conjugation, differentiation, multiplication, and integration. The correct response to losing that chain is to repair it before proceeding. In a standard convention the familiar kinetic action is built from a conjugate product, but its detailed component reduction belongs to the next clean calculation, not to a reconstruction of an unfinished board line.

7.12 Why the Symmetry Is the Punchline

A supersymmetric Lagrangian does more than place a boson and a fermion side by side. It constrains their masses and interactions so that they form a representation of the superalgebra. In an exactly supersymmetric vacuum, members of a supermultiplet have equal masses. Their loop contributions are correlated, and bosonic and fermionic ultraviolet effects cancel in precisely the combinations enforced by the symmetry.

This is why the construction machinery matters. One could list Feynman vertices and verify cancellations diagram by diagram, but the cancellations would then look accidental. Supersymmetry explains them in the same way Lorentz symmetry explains why scattering amplitudes transform consistently in every inertial frame.

[5]

Question from the audience. How is "chiral" spelled? [a]

Response. C-H-I-R-A-L. The word labels the constraint $\bar{\mathcal{D}}\Phi = 0$; it is not the ordinary geometric handedness of an object in three-dimensional space, although the terminology descends from the left- and right-handed structure of relativistic spinors.

[a]Lecture timestamp: 01:39:48.

Question from the audience. Could one bypass the Lagrangian and go directly to supersymmetric Feynman rules? [a]

Response. One can certainly state the rules, but then the relations among couplings and the cancellations they produce look like unexplained coincidences. The deeper content is the generalized spacetime symmetry. The Lagrangian or an equivalent invariant formulation makes that reason visible and generates the rules systematically.

[a]Lecture timestamp: 01:40:00.

Suppose every perturbative order happened to give Lorentz-invariant amplitudes although no Lorentz transformations had been mentioned. After enough calculations, we would demand an explanation. The answer would be the spacetime symmetry that organizes all orders at once. The role of superspace is analogous: shifts in $\theta, \bar{\theta}$ mix bosonic and fermionic components, and their algebra closes on translations in spacetime.

[5]Editorial clarification: Exact supersymmetry does not imply that every individual self-energy correction vanishes. It relates renormalizations and removes particular ultraviolet sensitivities; nonrenormalization statements depend on the quantity, dimension, and amount of supersymmetry. The lecture's broad cancellation claim is retained in this qualified form.

Question from the audience. Should ordinary spatial coordinates also be expanded in a finite power series, as the Grassmann coordinates are? [a]

Response. No. A power series in an ordinary coordinate does not terminate and is not usually the natural decomposition. A Fourier transform is more useful because it resolves a field into waves and momenta. Expansion in θ is special precisely because nilpotence makes it finite.

[a]Lecture timestamp: 01:44:55.

Question from the audience. If a new particle appeared at the LHC, how could we tell that it was a superpartner rather than merely an unknown particle? [a]

Response. Supersymmetry predicts correlated quantum numbers and interactions, not just one extra state. Spins differ by one half, gauge charges are related, and many cubic and quartic couplings are governed by fewer independent parameters than in a generic theory. Cross sections, branching fractions, and the rest of the spectrum would have to satisfy those relations.

[a]Lecture timestamp: 01:45:42.

One unfamiliar resonance would not by itself prove supersymmetry. A network of coupling and spectrum relations would. That is the experimental counterpart of the superspace construction: apparently independent component interactions descend from one invariant object.

Question from the audience. Observed particles and their hypothetical superpartners do not have equal masses. How does that change the theory? [a]

Response. It changes it enormously. Supersymmetry must be broken, usually through spontaneous breaking communicated to the observed sector. The breaking splits multiplet masses while retaining enough of the structure to control interactions and, if the scale is low enough, ultraviolet corrections. Working out that mechanism is a substantial theory of its own, especially when gravity is included.

[a]Lecture timestamp: 01:47:01.

7.13 Simple Principle, Difficult Calculus

The closing assessment is unusually candid. The starting idea sounds simple: bosons and fermions are related by a symmetry. Yet implementation rapidly becomes a discipline of indices, conjugations, Grassmann orderings, and signs. Even after constructing the symmetry, realistic particle physics requires breaking it; coupling it to gravity leads toward supergravity and adds another layer of structure.

This difficulty is not evidence that the idea is empty. Nature is under no obligation to make a simple principle easy to calculate with. Specialists learn the subject by doing the details repeatedly until the sign rules become second nature. The frustration comes from the distance between the compact algebra

$$\{Q, Q^\dagger\} = 2H \tag{7.52}$$

and the amount of machinery required to realize it in an interacting theory.

Question from the audience. Does another layer of abstraction eventually simplify the formalism, or does the complication keep accumulating? [a]

Response. No general simplification has removed the detailed work. The underlying rules may be short, but using them reliably requires repeated control of the conventions. That is why a full treatment normally occupies an entire technical course rather than a fraction of a broad particle-physics course.

[a] Lecture timestamp: 01:49:33.

Question from the audience. Do the transformations tell us what fermions and bosons really are, or why their ordinary behavior is different? [a]

Response. They tell us that the two are components of one superfield and images of one another under the supercharges. Acting with Q maps a bosonic component to a fermionic one, while the next transformation produces derivatives and auxiliary components. This is analogous to a rotation mixing vector components, but it cannot be translated into an equally literal picture: the appropriate language is the Grassmann geometry itself.

[a] Lecture timestamp: 01:51:42.

That answer returns us to the central idea. A supersymmetry transformation is a small motion in superspace. It mixes $\phi, \psi, \bar{\psi}, D$ just as an ordinary rotation mixes components, and it does so while closing on a translation in time. The picture is abstract, but it is not arbitrary: every term in the superfield, every sign in the covariant derivatives, and every relation among component interactions exists to realize the algebra.

7.14 What We Have Built

We began with the matched oscillator Hamiltonian and recovered its two different Lagrangian descriptions. The boson obeys a second-order oscillator equation; the fermion requires a first-order Grassmann action and a complex pair of odd components. The supercharges exchange the two modes, square to zero individually, anticommute to the Hamiltonian, and commute with it.

Representing those charges as differential operators led to superspace. A finite superfield expansion assembled bosonic, fermionic, and auxiliary components; Berezin integration selected the top component of a super-Lagrangian; and multiplication generated invariant interaction candidates. The demand for symmetry-compatible odd differentiation then led to $\mathcal{D}, \bar{\mathcal{D}}$, the chiral constraint, and the shifted coordinate $\tau = t + i\bar{\theta}\theta$.

The unfinished conjugate-product calculation is part of the lesson rather than a defect to hide. Supersymmetry begins with a simple physical demand, but its calculus is sensitive to every ordering choice. Once those choices are controlled, the reward is powerful: masses and couplings are organized by symmetry, quantum corrections satisfy nonaccidental relations, and bosons and fermions become components of one geometric object.

Chapter 8

Four-Dimensional Supersymmetry and the Wess–Zumino Model

Overview. The one-dimensional supersymmetry algebra of the preceding lecture involved only the Hamiltonian. Lorentz invariance now forces energy to rejoin the spatial momentum, and the supercharges must become two-component spinors. We reach that algebra by returning first to a massless Weyl fermion and its Hamiltonian $H = \boldsymbol{\sigma} \cdot \mathbf{p}$. We then represent the supercharges as differential operators on four-dimensional superspace, isolate chiral superfields with the constraint $\bar{D}_{\dot{\alpha}}\Phi = 0$, and use superspace integration to construct a concrete field theory. The successive choices $\Phi^\dagger\Phi$, Φ^2, and Φ^3 produce propagation, equal boson–fermion masses, Yukawa interactions, and a related scalar interaction. The auxiliary field F makes these relations transparent, while the closing loop diagrams show why supersymmetry softens ultraviolet sensitivity.

8.1 Why the Hamiltonian Cannot Stand Alone

In the toy model, space was suppressed and the basic relation was

$$\{Q, Q^\dagger\} = 2H. \tag{8.1}$$

That form is adequate when time is the only coordinate. In a relativistic theory, however, $H = P_0$ is one component of the four-momentum P_μ. A Lorentz boost mixes it with P_i. An algebra whose right-hand side singled out P_0 could not transform covariantly.

The repair must therefore do two things at once. It must replace H by all four components P_μ, and it must provide indices on the left that can be matched to a Lorentz vector on the right. The massless Dirac equation supplies the needed clue: a relativistic fermion may be described by a two-component spinor, and the matrices σ^μ convert a vector index into a pair of spinor indices.

8.2 Massless Fermions, Helicity, and Chirality

Begin with the physical distinction between massive and massless particles. Take a massive spin-$\frac{1}{2}$ particle moving in the direction of its spin. Because its speed is less than the speed of light, an observer may overtake it. In the new frame the momentum reverses while the spin does not, so the helicity reverses. A Lorentz-invariant massive theory must consequently contain both helicities.

No observer can overtake a massless particle. Proper Lorentz transformations cannot reverse its momentum relative to its spin, so one irreducible massless representation may carry a single helicity. A spatial rotation can turn the flight direction around, but it does not turn a right hand into a left hand.

Question from the audience. Is the overtaking argument really about helicity rather than chirality? [a]

Response. Yes. Helicity is the projection of spin along momentum, whereas chirality is the eigenvalue of the chiral matrix, conventionally γ^5. For a massless mode the two labels are tied together, which is why the language can sound interchangeable in this example. For a massive particle they are distinct.

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[a] Lecture timestamp: 00:05:42.

The state count requires one qualification.[1]

Write the Weyl field used in this opening argument as

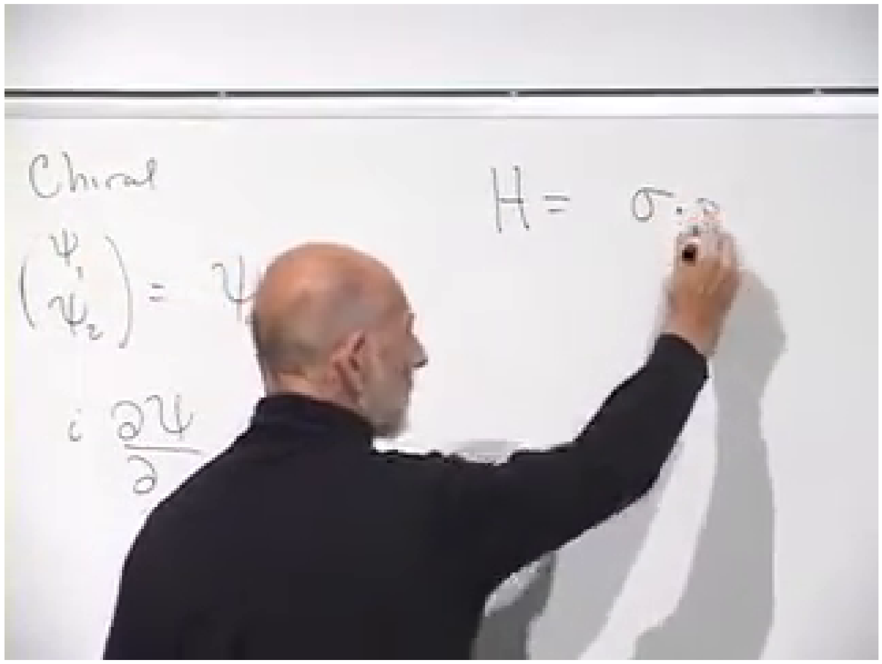
$$\chi = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix}, \tag{8.2}$$

The lecture denotes this field by ψ_α; we call it χ here so that it is not confused with the conjugate Weyl representation conventionally placed in a chiral superfield below.

$$H\chi = \boldsymbol{\sigma} \cdot \mathbf{p}\, \chi. \tag{8.3}$$

For momentum $\mathbf{p}$, the matrix $\boldsymbol{\sigma} \cdot \mathbf{p}$ has eigenvalues $\pm|\mathbf{p}|$. The positive-frequency branch supplies the particle mode; the negative-frequency branch is reinterpreted through the antiparticle construction rather than retained as a spectrum unbounded below.

[1] Editorial clarification: A single massless Weyl field has one particle helicity and the opposite antiparticle helicity. Thus "half as many states" refers to the reduction from a four-component Dirac field to one two-component Weyl field; it does not mean that antiparticle states have disappeared.



Lecture frame: 00:07:44.271.

Figure 8.1: The chiral two-component spinor and the Hamiltonian $H = \boldsymbol{\sigma} \cdot \mathbf{p}$. The final momentum symbol is partly covered, but the spoken equation fixes it.

The handedness label also depends on convention.[2]

Replacing H by $i\partial_t$ and $\mathbf{p}$ by $-i\nabla$ gives

$$i\partial_t \chi = -i\boldsymbol{\sigma} \cdot \nabla \chi. \tag{8.4}$$

Introduce

$$\begin{aligned} \sigma^\mu &= (\mathbf{1}, \boldsymbol{\sigma}), \\ \bar{\sigma}^\mu &= (\mathbf{1}, -\boldsymbol{\sigma}), \\ \eta_{\mu\nu} &= \mathrm{diag}(1, -1, -1, -1). \end{aligned} \tag{8.5}$$

Then the equation just obtained is

$$i\sigma^\mu \partial_\mu \chi = 0. \tag{8.6}$$

The conjugate Weyl representation uses $\bar{\sigma}^\mu$. It is this conjugate representation, named ψ_α below, whose free equation is $i\bar{\sigma}^\mu \partial_\mu \psi = 0$. Proving covariance requires the spinor representation of the Lorentz group, but the important fact here is structural: the four matrices σ^μ join one spacetime index to two spinor indices.

8.3 The Four-Dimensional Superalgebra

The odd coordinates must now transform like Weyl spinors:

$$\theta^\alpha = (\theta^1, \theta^2), \qquad \bar{\theta}^{\dot{\alpha}} = (\bar{\theta}^{\dot{1}}, \bar{\theta}^{\dot{2}}). \tag{8.7}$$

[2]Editorial clarification: The board associates a handedness directly with the positive eigenvalue of $\boldsymbol{\sigma} \cdot \mathbf{p}$. Which chirality receives that sign depends on the definitions of σ^μ, $\bar{\sigma}^\mu$, and the Weyl field. We retain the eigenvalue statement but do not attach a convention-independent left/right label to it.

The bar denotes complex conjugation, and the dot distinguishes the conjugate Lorentz representation. There are two complex, or four real, Grassmann components. A general even superfield

$$\mathcal{S} = \mathcal{S}(x^\mu, \theta^\alpha, \bar{\theta}^{\dot{\alpha}}) \tag{8.8}$$

therefore has a finite Grassmann expansion. Its highest monomial is

$$\theta^2\bar{\theta}^2 \equiv (\theta^1\theta^2)(\bar{\theta}^{\dot{1}}\bar{\theta}^{\dot{2}}), \tag{8.9}$$

the product of all four independent odd coordinates. The coefficient of this monomial is commonly called the D-component.

The supercharges also carry spinor indices. In four-dimensional $N = 1$ supersymmetry their algebra is

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma^\mu_{\alpha\dot{\beta}} P_\mu, \tag{8.10}$$

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0, \tag{8.11}$$

$$[P_\mu, Q_\alpha] = 0, \qquad [P_\mu, \bar{Q}_{\dot{\alpha}}] = 0, \qquad [P_\mu, P_\nu] = 0. \tag{8.12}$$

The factor of two is conventional. The matrix $\sigma^\mu_{\alpha\dot{\beta}}$ contracts the vector index on P_μ and leaves precisely the two spinor indices appearing on the left.

Question from the audience. Are α and β on the sigma matrix its matrix indices? [a]

Response. Yes. More precisely, one is an undotted Weyl index and the other is a dotted conjugate index; μ is the spacetime-vector index. The sigma matrix is the bridge between those descriptions.

[a] Lecture timestamp: 00:22:01.

The cleaned notation makes that bridge explicit.[3]

The last line says that the supercharges are conserved and compatible with translations. Momentum components commute because their derivative representatives commute. In the differential convention used below,

$$P_\mu = i\partial_\mu, \tag{8.13}$$

so Eq. (8.10) may equally be written

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu. \tag{8.14}$$

8.4 Supercharges as Superspace Derivatives

A momentum translates an ordinary coordinate; a supercharge translates an odd coordinate while also shifting spacetime. Fix left Grassmann derivatives and define

$$Q_\alpha = \partial_\alpha - i\sigma^\mu_{\alpha\dot{\alpha}}\bar{\theta}^{\dot{\alpha}}\partial_\mu, \tag{8.15}$$

$$\bar{Q}_{\dot{\alpha}} = -\bar{\partial}_{\dot{\alpha}} + i\theta^\alpha\sigma^\mu_{\alpha\dot{\alpha}}\partial_\mu. \tag{8.16}$$

[3] Editorial clarification: The lecture deliberately suppresses dotted indices and debates the order $\alpha\beta$ versus $\beta\alpha$ at the board. Equations (8.10) and (8.12) restore the standard dotted/undotted notation without changing the argument being made.

Acting with $\epsilon^\alpha Q_\alpha + \bar{\epsilon}_{\dot{\alpha}}\bar{Q}^{\dot{\alpha}}$ gives the infinitesimal supersymmetry variation of a superfield. Its components are simply the coefficients of the independent Grassmann monomials, and the differential operators mix those coefficients.

Question from the audience. Which x-derivative is meant in the shorthand $\bar{\theta}\sigma\partial$? [a]

Response. All four. The repeated μ in $\bar{\theta}^{\dot{\alpha}}\sigma^\mu_{\alpha\dot{\alpha}}\partial_\mu$ is summed, so each component of σ^μ is paired with the matching derivative $\partial/\partial x^\mu$.

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[a]Lecture timestamp: 00:25:40.

Question from the audience. Should the conjugate charge contain θ, not another $\bar{\theta}$? [a]

Response. Yes. Q differentiates with respect to θ and contains $\bar{\theta}$; $\bar{Q}$ differentiates with respect to $\bar{\theta}$ and contains θ. They are conjugate operators.

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[a]Lecture timestamp: 00:26:16.

Question from the audience. Is σ^μ symmetric? [a]

Response. The Pauli matrices are Hermitian. Three of the four matrices in $(\mathbf{1}, \boldsymbol{\sigma})$ are symmetric, while σ^2 is antisymmetric under ordinary transpose. Hermiticity, together with dotted and undotted index placement, is the relevant conjugation property.

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[a]Lecture timestamp: 00:27:04.

Question from the audience. Does the sigma matrix in the conjugate equation carry the indices in the opposite order? [a]

Response. Conjugation exchanges dotted and undotted representations, so the safe notation is $\sigma^\mu_{\alpha\dot{\alpha}}$ and its conjugate $\bar{\sigma}^{\mu\,\dot{\alpha}\alpha}$, rather than treating the two labels as interchangeable matrix slots.

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[a]Lecture timestamp: 00:27:13.

The formulas in Eqs. (8.15) and (8.16) differ in superficial signs from the shorthand used at the board. Those signs are linked to whether P_μ is $+i\partial_\mu$ or $-i\partial_\mu$, whether derivatives act from the left or right, and how conjugate odd variables are ordered. What matters is that one convention obeys the entire algebra at once.

8.5 A Constraint That Survives Supersymmetry

A general superfield contains many components. We seek a smaller class without destroying the symmetry. The lecture first gives an ordinary analogy. For a vector $\mathbf{V}$, the constraint

$$V_x = 0 \tag{8.17}$$

is not rotationally invariant: after a rotation, V_y contributes to the new x-component. By contrast,

$$\mathbf{V}^2 = 1 \tag{8.18}$$

is invariant and removes one independent degree of freedom. A useful constraint must map constrained fields into constrained fields under the symmetry.

Suppose a linear odd operator $\bar{D}_{\dot{\alpha}}$ defines

$$\bar{D}_{\dot{\alpha}}\Phi = 0. \tag{8.19}$$

After a supersymmetry variation $\delta\Phi = Q\Phi$, the same condition will hold if $\bar{D}$ anticommutes with the supercharges:

$$\bar{D}(Q\Phi) = -Q(\bar{D}\Phi) = 0. \tag{8.20}$$

This is the abstract consistency test that motivates the construction, rather than a definition appearing without explanation.

Question from the audience. Why may Q and $\bar{D}$ be switched in the last expression? [a]

Response. They are odd operators chosen to anticommute. Switching their order therefore introduces a minus sign; once $\bar{D}$ reaches Φ, the original constraint makes the result zero.

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[a] Lecture timestamp: 00:36:56.

The required covariant derivatives are obtained by reversing the spacetime-derivative signs relative to the supercharges:

$$D_\alpha = \partial_\alpha + i\sigma^\mu_{\alpha\dot{\alpha}}\bar{\theta}^{\dot{\alpha}}\partial_\mu, \tag{8.21}$$

$$\bar{D}_{\dot{\alpha}} = -\bar{\partial}_{\dot{\alpha}} - i\theta^\alpha\sigma^\mu_{\alpha\dot{\alpha}}\partial_\mu. \tag{8.22}$$

They satisfy

$$\{D_\alpha, \bar{D}_{\dot{\beta}}\} = -2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu, \tag{8.23}$$

$$\{D_\alpha, Q_\beta\} = \{D_\alpha, \bar{Q}_{\dot{\beta}}\} = 0, \tag{8.24}$$

$$\{\bar{D}_{\dot{\alpha}}, Q_\beta\} = \{\bar{D}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0. \tag{8.25}$$

Question from the audience. Is the only change from $\bar{Q}$ to $\bar{D}$ the sign of the spacetime-derivative term? [a]

Response. Exactly. That sign reversal makes the cross terms cancel when D is anticommuted with either supercharge.

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[a] Lecture timestamp: 00:37:58.

We impose only Eq. (8.19). Imposing both $D_\alpha\Phi = 0$ and $\bar{D}_{\dot{\alpha}}\Phi = 0$ would, through their anticommutator, force $\partial_\mu\Phi = 0$, leaving at most a spacetime-independent remnant. [4]

8.6 Solving the Chiral Constraint

Define the shifted coordinate

$$y^\mu = x^\mu + i\theta\sigma^\mu\bar{\theta}. \tag{8.26}$$

With the left-derivative convention above, a function

$$\Phi = \Phi(y, \theta) \tag{8.27}$$

[4] Editorial clarification: The board says that imposing both constraints would force $\Phi = 0$. The algebra strictly forces spacetime independence; additional boundary or component conditions may then reduce the field further.

obeys $\bar{D}_{\dot\alpha}\Phi = 0$. The chain rule makes the cancellation explicit. Differentiating through y^μ gives the same $i\theta\sigma^\mu\partial_\mu\Phi$ term that appears with the opposite sign in $\bar{D}_{\dot\alpha}$.

Thus a chiral superfield has no independent $\bar\theta$-dependence. It may still contain $\bar\theta$, but only through the combination y^μ. Its finite expansion is

$$\Phi(y,\theta) = \phi(y) + \sqrt{2}\,\theta\psi(y) + \theta^2 F(y). \tag{8.28}$$

The three component types are a complex scalar ϕ, a Weyl fermion ψ_α, and a complex scalar F. The last is auxiliary rather than a second propagating boson.

Question from the audience. Has x been replaced by a new coordinate y orthogonal to it? [a]

Response. No. y^μ is shorthand for the particular combination in Eq. (8.26); it is not an additional spacetime direction. Working in y makes chirality simple, and we translate back to x when constructing the ordinary field theory.

[a]Lecture timestamp: 00:49:08.

Expanding Eq. (8.28) at fixed x gives

$$\begin{aligned}\Phi(x,\theta,\bar\theta) = \phi &+ \sqrt{2}\,\theta\psi + \theta^2 F + i\theta\sigma^\mu\bar\theta\,\partial_\mu\phi \\ &- \frac{i}{\sqrt{2}}\theta^2\partial_\mu\psi\,\sigma^\mu\bar\theta + \frac{1}{4}\theta^2\bar\theta^2\Box\phi.\end{aligned} \tag{8.29}$$

All component fields on the right are evaluated at x. The conjugate antichiral field depends on $\bar{y}^\mu = x^\mu - i\theta\sigma^\mu\bar\theta$ and $\bar\theta$. A chiral superfield cannot consistently be real, because its conjugate satisfies the opposite chirality constraint.

8.7 Superspace Integrals

A full-superspace action has the form

$$S_D = \int d^4x\, d^2\theta\, d^2\bar\theta\; K, \tag{8.30}$$

where K is a real superfield. Berezin integration selects its $\theta^2\bar\theta^2$ coefficient. After the odd integrations, only an ordinary spacetime Lagrangian remains.

There is a second invariant construction. If $W(\Phi)$ is chiral, then

$$S_F = \int d^4x\, d^2\theta\; W(\Phi) + \int d^4x\, d^2\bar\theta\; \overline{W}(\Phi^\dagger) \tag{8.31}$$

is supersymmetric. The first integral selects the θ^2 component and the second its conjugate.

Question from the audience. Why is it legitimate to integrate only over $d^2\theta$ when the expression has no $\bar\theta$-dependence? [a]

Response. Chirality is preserved by supersymmetry, and the highest θ-component of a chiral superfield changes only by a spacetime total derivative. Its spacetime integral is therefore invariant, just as the full D-component integral is invariant.

[a]Lecture timestamp: 01:16:30.

8.8 The Free Chiral Multiplet

Take the apparently structureless expression

$$S_{\text{free}} = \int d^4x\, d^2\theta\, d^2\bar{\theta}\ \Phi^\dagger\Phi. \tag{8.32}$$

The two factors depend on oppositely shifted coordinates:

$$\Phi = \Phi(x + i\theta\sigma\bar{\theta}, \theta), \qquad \Phi^\dagger = \Phi^\dagger(x - i\theta\sigma\bar{\theta}, \bar{\theta}). \tag{8.33}$$

Inside the spacetime integral we may shift the dummy variable x. One shift can remove the Grassmann displacement from one factor, but then it doubles the relative displacement in the other. Taylor expansion in that nilpotent displacement terminates after finitely many terms.

This is where the derivatives come from. The superspace expression $\Phi^\dagger\Phi$ contains no explicit spacetime derivative, yet the shifted arguments generate first and second derivatives of the component fields. Multiplying the expansions and retaining only the top component gives, in the convention of Eq. (8.5),

$$\mathcal{L}_{\text{free}} = \partial_\mu\phi^*\partial^\mu\phi + i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi + F^*F. \tag{8.34}$$

Up to integration by parts, the scalar piece may instead be written schematically as $-\phi^*\Box\phi$, which is the form approached in the board calculation.

Question from the audience. Where does the second line in the Taylor expansion of the superfield come from? [a]

Response. It comes from the component expansion $\Phi(y,\theta) = \phi(y) + \sqrt{2}\theta\psi(y) + \theta^2 F(y)$. Each component is then Taylor-expanded because y differs from x by a nilpotent Grassmann bilinear.

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[a] Lecture timestamp: 01:03:00.

Question from the audience. Should that last component carry θ^2? [a]

Response. Yes. The auxiliary component is $F\theta^2$. Since it already contains both independent θ's, any further Taylor shift carrying another θ vanishes.

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[a] Lecture timestamp: 01:04:54.

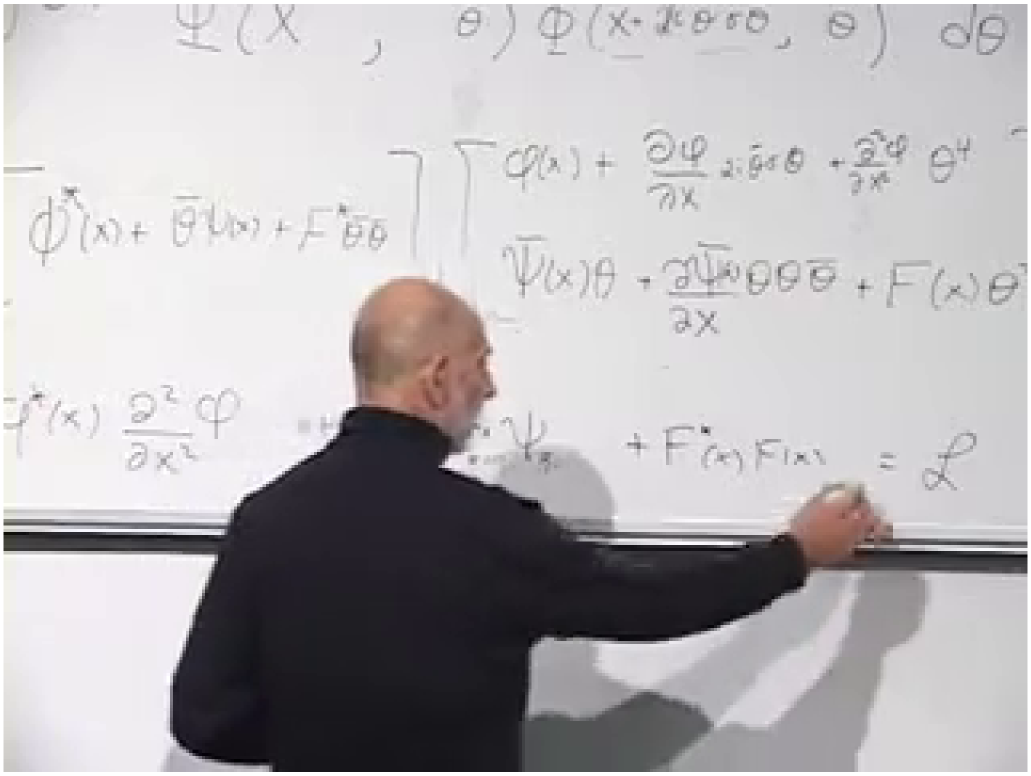

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Figure 8.2: The shifted-coordinate expansion being collected into scalar, fermion, and auxiliary terms of the ordinary Lagrangian.

The first two terms in Eq. (8.34) yield the massless Klein–Gordon and Weyl equations. The last contains no derivative:

$$\frac{\partial \mathcal{L}}{\partial F} = F^* = 0, \qquad \frac{\partial \mathcal{L}}{\partial F^*} = F = 0. \tag{8.35}$$

It introduces no initial data and no on-shell particle.

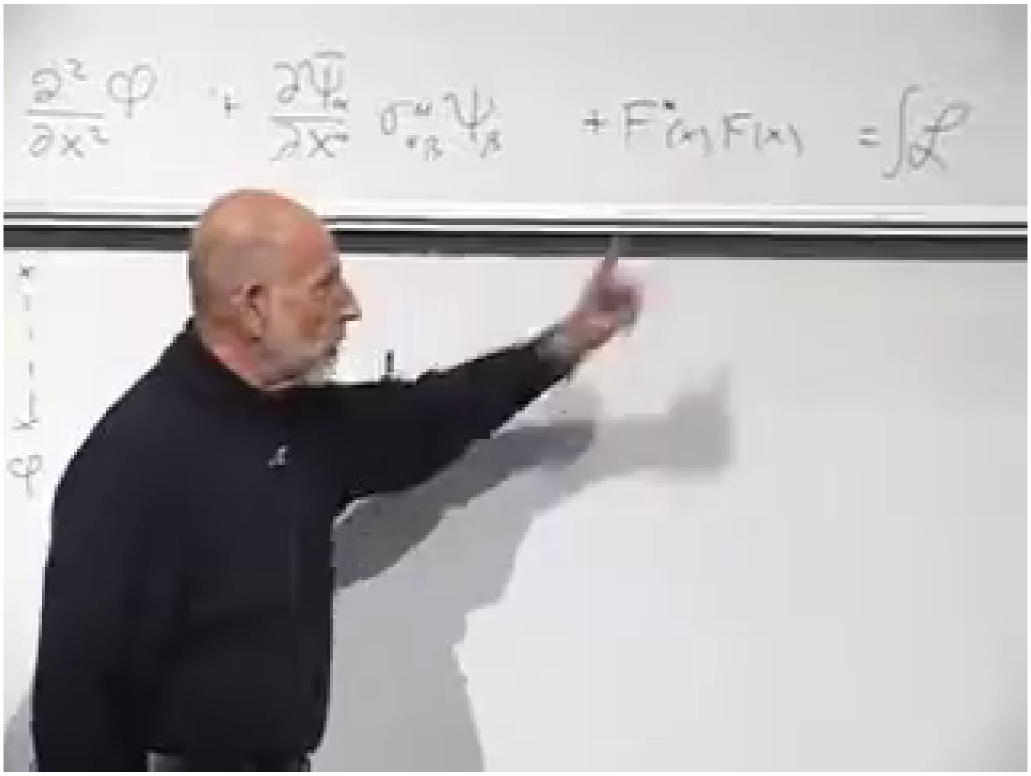

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Figure 8.3: The no-derivative auxiliary term F^*F at the end of the component Lagrangian.

The propagator language says the same thing. The inverse quadratic operator for F is momentum independent. Its Fourier transform is local:

$$\langle F(x)F^*(y)\rangle \propto \delta^{(4)}(x-y). \tag{8.36}$$

An F-line may organize a diagram, but its endpoints coincide. It is a contact contraction, not the trajectory of a physical particle.

8.9 The Quadratic Superpotential and Equal Masses

Propagation came from a D-term. Mass comes from the chiral integral. Choose the conventional normalization

$$W_2(\Phi) = \frac{1}{2}m\Phi^2. \tag{8.37}$$

The θ^2 component is

$$W_2(\Phi)|_{\theta^2} = m\phi F - \frac{1}{2}m\psi\psi. \tag{8.38}$$

Together with the conjugate term, the component Lagrangian becomes

$$\begin{aligned}\mathcal{L} = {} & \partial_\mu\phi^*\partial^\mu\phi + i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi + F^*F \\ & + m\phi F + m^*\phi^*F^* - \frac{1}{2}m\psi\psi - \frac{1}{2}m^*\bar{\psi}\bar{\psi}.\end{aligned} \tag{8.39}$$

The fermion bilinear is a Majorana mass for the Weyl field. The F-equations are algebraic:

$$F^* = -m\phi, \qquad F = -m^*\phi^*. \tag{8.40}$$

Substitution gives

$$\begin{aligned}\mathcal{L}_{\text{on shell}} &= \partial_\mu\phi^*\partial^\mu\phi + i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi \\ &\quad - |m|^2|\phi|^2 \\ &\quad - \frac{1}{2}\left(m\psi\psi + m^*\bar{\psi}\bar{\psi}\right).\end{aligned} \tag{8.41}$$

Thus the scalar mass and fermion mass are both $|m|$. In the board's diagrammatic language, a scalar turns locally into F through one $m\phi F$ insertion and immediately turns back through its conjugate. Two insertions supply $|m|^2|\phi|^2$, exactly the scalar mass term found by solving Eq. (8.40).

The normalization can now be stated precisely.[5]

8.10 The Cubic Superpotential and Interactions

Now add

$$W(\Phi) = \frac{1}{2}m\Phi^2 + \frac{1}{3}g\Phi^3. \tag{8.42}$$

For a general holomorphic W, the F-component is

$$W(\Phi)|_{\theta^2} = FW'(\phi) - \frac{1}{2}W''(\phi)\psi\psi. \tag{8.43}$$

Here

$$W'(\phi) = m\phi + g\phi^2, \qquad W''(\phi) = m + 2g\phi. \tag{8.44}$$

Before eliminating F, the cubic term contains

$$g\phi^2 F - g\phi\psi\psi + \text{h.c.} \tag{8.45}$$

The second term is a Yukawa vertex joining one scalar to two fermion lines. The first joins two scalars to an auxiliary line.

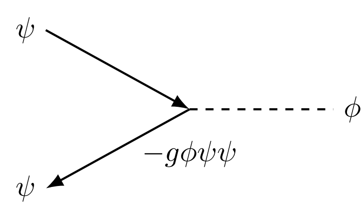


Figure 8.4: Pedagogical redraw of the Yukawa interaction: two fermionic legs and one scalar leg meet with coupling proportional to g.

The complete auxiliary-field equation is

$$F^* = -W'(\phi) = -(m\phi + g\phi^2). \tag{8.46}$$

[5] Editorial clarification: The lecture intentionally suppresses numerical coefficients while multiplying superfields. The factors 1/2, 1/3, and $\sqrt{2}$ used here are the conventional Wess–Zumino normalization; they make the component masses and couplings exact while preserving every qualitative step of the board derivation.

Eliminating it produces the scalar potential

$$V(\phi) = |W'(\phi)|^2 = |m\phi + g\phi^2|^2. \tag{8.47}$$

Besides the mass term, this contains cubic scalar terms proportional to m^*g and a quartic term

$$V(\phi) \supset |g|^2|\phi|^4. \tag{8.48}$$

In the auxiliary-line picture, two ϕ^2F vertices are connected by the local F-contraction. Because Eq. (8.36) forces their positions to coincide, the result is precisely a four-scalar contact vertex.

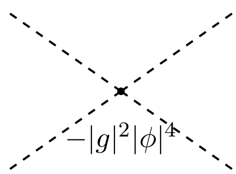


Figure 8.5: After the auxiliary field is eliminated, two local F-vertices become one four-scalar contact interaction.

Supersymmetry has done more than place a boson and a fermion in the same theory. It has forced their masses to agree, tied the Yukawa coupling to g, and fixed the quartic scalar coupling to $|g|^2$. These relations could be tested through masses, scattering amplitudes, and decays; they are not optional numerical coincidences.

8.11 Paired Loop Effects

The relations become most useful when interactions correct particle masses. The scalar self-energy receives a bosonic loop from the quartic vertex and a fermionic loop from two Yukawa vertices. Each carries two powers of the same coupling. A closed fermion loop has the relative minus sign required by Fermi statistics.

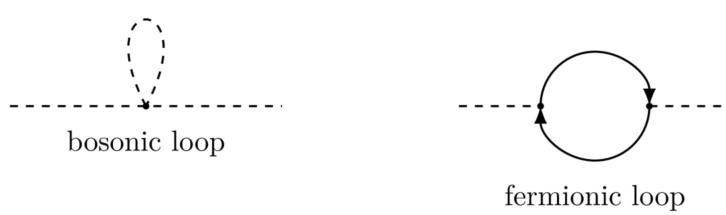


Figure 8.6: The lecture's paired scalar self-energy mechanisms. Supersymmetry fixes the relative couplings, while the closed fermion loop supplies the opposite sign.

At one loop the leading ultraviolet-sensitive pieces cancel between the complete bosonic and fermionic contributions. The same symmetry organizes higher orders through Ward identities and nonrenormalization properties. The statement concerns symmetry-related sums, not a literal one-to-one cancellation of every graph; wave-function renormalization, finite effects, and other symmetry-preserving corrections can remain. This is the practical power of building the theory in superspace: the equality of masses and the relations among couplings are enforced before any individual Feynman integral is evaluated.[6]

8.12 What Comes Next

The model constructed here is the simplest interacting chiral theory, commonly called the Wess–Zumino model. It is an example, not the most general supersymmetric field theory, but the pattern persists: represent fields by supermultiplets, write invariant superspace integrals, and let the Grassmann expansion enforce component relations.

Exact supersymmetry is not visible in the observed particle spectrum, so the next problem is how the symmetry can be broken while retaining enough of its ultraviolet control to be useful. The other promised direction is grand unification: how

$$SU(3) \times SU(2) \times U(1) \tag{8.49}$$

may fit into a larger group such as $SU(5)$, and how supersymmetric particle content changes that unification story. Those are the questions to which the remaining lectures turn.

[6]Editorial clarification: The board states the cancellation more broadly; the main text supplies the standard qualification.

Chapter 9

Spontaneous Supersymmetry Breaking and the First $SU(5)$ Multiplet

Overview. Symmetry breaking first appears here as a visible fact about spectra: a rotationally invariant atom has degenerate angular-momentum levels, while a magnetic field splits them. A ferromagnet then teaches the subtler lesson that the equations may retain a symmetry which the vacuum does not. Gapless magnons, the condition $J|0\rangle \neq 0$, and a nonzero order parameter are three descriptions of that same phenomenon. We carry this pattern into a chiral supersymmetric field theory, derive the scalar potential $V = |W'|^2$ by eliminating the auxiliary field F, and distinguish supersymmetric from F-breaking vacua. The resulting vacuum energy, boson–fermion splitting, goldstino, hidden sector, and super-Higgs mechanism complete the supersymmetry discussion. The final part changes subject and opens grand unification: $SU(3)\times SU(2)\times U(1)$ is embedded blockwise in $SU(5)$, whose first matter multiplet combines a lepton doublet with three anti-down quarks.

9.1 A Spectrum Remembers Its Symmetry

The purpose of beginning with an atom is not merely to review elementary quantum mechanics. Energy levels and particle masses are the same kind of spectral data. If a symmetry forces atomic levels to coincide, then its breaking can split those levels; in field theory the corresponding statement is that members of a particle multiplet can acquire different masses.

Let an atomic state have total angular momentum j, including both spin and orbital contributions. Relative to any chosen axis,

$$J_z|j,m\rangle = m\hbar|j,m\rangle, \qquad m = -j, -j+1, \ldots, j, \tag{9.1}$$

so there are $2j+1$ states. The familiar textbook arrows should not be read as saying that the angular momentum vector literally has only $2j+1$ possible spatial directions. What is quantized is its component along the chosen axis.

In empty, rotationally invariant space there is no physical reason to prefer that axis. A rotation mixes the states within the spin-j multiplet, and invariance of the Hamiltonian implies

$$[H, \mathbf{J}] = 0, \qquad E_{j,-j} = E_{j,-j+1} = \cdots = E_{j,j}. \tag{9.2}$$

Question from the audience. What symmetry says that rotating the whole atom cannot change its energy? [a]

Response. Rotational symmetry. Since a rotation carries one orientation of the angular momentum into another without changing the physical system, all $2j+1$ states have the same energy.

[a]Lecture timestamp: 00:05:53.

Now apply a magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$. The interaction

$$H_B = -\boldsymbol{\mu} \cdot \mathbf{B} \tag{9.3}$$

distinguishes the component parallel to the field. Schematically,

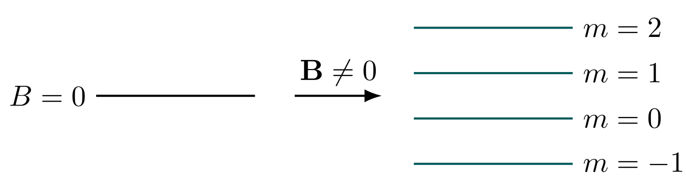
$$\Delta E_m \propto mB. \tag{9.4}$$

Levels with different m separate linearly at weak field, and transitions among them split spectral lines.

Question from the audience. What is the splitting of atomic levels in a magnetic field called? [a]

Response. The Zeeman effect. It is the prototype for the entire discussion: levels made degenerate by a symmetry separate when a background breaks that symmetry.

[a]Lecture timestamp: 00:07:13.



9.2 The Ferromagnet: Symmetric Laws, Asymmetric Vacuum

A simple microscopic picture of a ferromagnet assigns a small magnetic moment to each site. If neighboring moments lower their energy by becoming parallel, the low-temperature phase is ferromagnetic. If antiparallel alignment is preferred, the result is antiferromagnetic. Strong thermal or quantum fluctuations can instead wash out any net magnetization.

Suppose a ferromagnet fills the world and settles into a uniformly aligned state. Its internal magnetic field splits the levels of an atom just as an externally imposed field would. Yet there is a decisive difference. The underlying interactions do not prefer north, east, or any other direction. Every uniformly rotated configuration has the same energy:

$$\mathbf{M}(\mathbf{x}) = M_0 \hat{\mathbf{n}}, \qquad \hat{\mathbf{n}} \in S^2. \tag{9.5}$$

The equations are rotationally invariant, but one actual ground state chooses one $\hat{\mathbf{n}}$. This is spontaneous rather than explicit symmetry breaking.

An inhabitant made from excitations of this medium cannot grab the entire universe and rotate it. Locally, therefore, the chosen direction looks as fundamental as an external field. An atom still sees the same level pattern, with its angular momentum resolved along $\hat{\mathbf{n}}$ rather than along an externally named z-axis. The question is how an internal observer could discover that the apparent asymmetry belongs to the state, not to the laws.

9.2.1 The Goldstone Diagnostic

A uniform rotation of every moment moves from one vacuum in Eq. (9.5) to another and costs no energy. Let the orientation instead change very gradually across space. Neighboring moments then pull on one another and the distortion propagates as a spin wave, whose quanta are magnons. As the wavelength tends to infinity, the configuration approaches a uniform rotation, so its excitation energy tends to zero:

$$k = \frac{2\pi}{\lambda} \longrightarrow 0 \quad \Longrightarrow \quad E(k) \longrightarrow 0. \tag{9.6}$$

Two inhabitants can wiggle nearby moments, send such waves across the medium, and measure this arbitrarily low energy. That experiment reveals spontaneous breaking even though they cannot rotate the world.

[1]

The same idea applies to continuous internal symmetries, including the gauge and flavor groups of particle physics. When the broken symmetry is gauged, however, the Goldstone mode does not remain as an independent massless particle.

Question from the audience. What happens when spontaneous symmetry breaking is combined with gauge invariance? [a]

Response. The Goldstone boson supplies an additional polarization to the gauge field. The gauge boson becomes massive; this is the Higgs mechanism, to which we will return in the gravitational analogy.

---

[a]Lecture timestamp: 00:15:56.

9.3 Three Equivalent Signals of Breaking

The ferromagnet can now be described in three mutually reinforcing languages.

Degenerate vacua. There is one minimum-energy state for every orientation $\hat{\mathbf{n}}$. Physics takes place in one chosen member of this family even though the symmetry relates all of them.

A generator that does not annihilate the vacuum. Let J_a be a generator of rotations. A finite transformation is

$$U(\epsilon) = e^{-i\epsilon J_a}, \tag{9.7}$$

and $J_a|0\rangle$ is proportional to the infinitesimal change of the vacuum, not the entire rotated state.

Question from the audience. What are the generators of rotational symmetry? [a]

Response. The components of total angular momentum. We call them J_a because they rotate everything in the state, not merely one atom.

---

[a]Lecture timestamp: 00:17:29.

[1]Editorial clarification: A ferromagnetic magnon is gapless but generally has a nonrelativistic quadratic dispersion $E \propto k^2$, rather than the relativistic relation $E \propto |k|$. The lecture uses the word massless in the physically relevant sense of vanishing gap. Both are Goldstone modes.

An unbroken generator satisfies

$$J_a|0\rangle = 0. \tag{9.8}$$

A broken generator instead gives

$$J_a|0\rangle \neq 0. \tag{9.9}$$

Question from the audience. If $J_a|0\rangle$ is nonzero, does it simply give the same vacuum? [a]

Response. No. It gives the infinitesimal difference between the chosen vacuum and a slightly rotated one. Physically that difference is an infinitely long-wavelength Goldstone excitation.

---

[a] Lecture timestamp: 00:19:43.

Question from the audience. Does this infinitesimal rotation change the energy? [a]

Response. No. The rotated vacuum is degenerate with the original one, so the zero-momentum Goldstone state has zero energy.

---

[a] Lecture timestamp: 00:20:14.

Symbolically,

$$J_a|0\rangle \sim |\pi_a, \mathbf{k} = 0\rangle, \tag{9.10}$$

where π_a denotes the Goldstone mode associated with the broken generator.

A nonzero order parameter. The local magnetization $\mathbf{M}(\mathbf{x})$ is a field. In a warm or quantum-disordered phase,

$$\langle \mathbf{M} \rangle = 0, \tag{9.11}$$

and the vacuum does not select a direction. In the ordered phase,

$$\langle M_i \rangle \neq 0 \quad \text{for at least one component } i, \tag{9.12}$$

so a rotation mixes that nonzero component into the others. Degenerate vacua, a broken generator, and a nonzero order parameter are the three patterns to be carried into supersymmetry.

9.4 The Chiral Multiplet Recalled

The simplest four-dimensional example contains one complex scalar, one Weyl fermion, and one auxiliary complex scalar. In chiral coordinates,

$$y^\mu = x^\mu + i\bar{\theta}\bar{\sigma}^\mu\theta, \tag{9.13}$$

a chiral superfield may be expanded schematically as

$$\Phi(y, \theta) = \phi(y) + \sqrt{2}\,\theta\psi(y) + \theta^2 F(y). \tag{9.14}$$

The lecture suppresses the conventional factor $\sqrt{2}$; we include it from this point so that the component coefficients are standard. A chiral superfield is intrinsically complex: imposing reality is not preserved by its supersymmetry transformation.

The supersymmetric kinetic action is

$$S_{\text{kin}} = \int d^4x\, d^4\theta\, \Phi^\dagger \Phi. \tag{9.15}$$

Grassmann integration selects the coefficient of $\theta^2\bar{\theta}^2$. With mostly-minus metric, the component result is

$$\mathcal{L}_{\text{kin}} = \partial_\mu \phi^*\, \partial^\mu \phi + i\bar{\psi}\bar{\sigma}^\mu \partial_\mu \psi + F^* F. \tag{9.16}$$

The first two terms propagate the scalar and fermion. The last has no derivative and therefore propagates no particle.

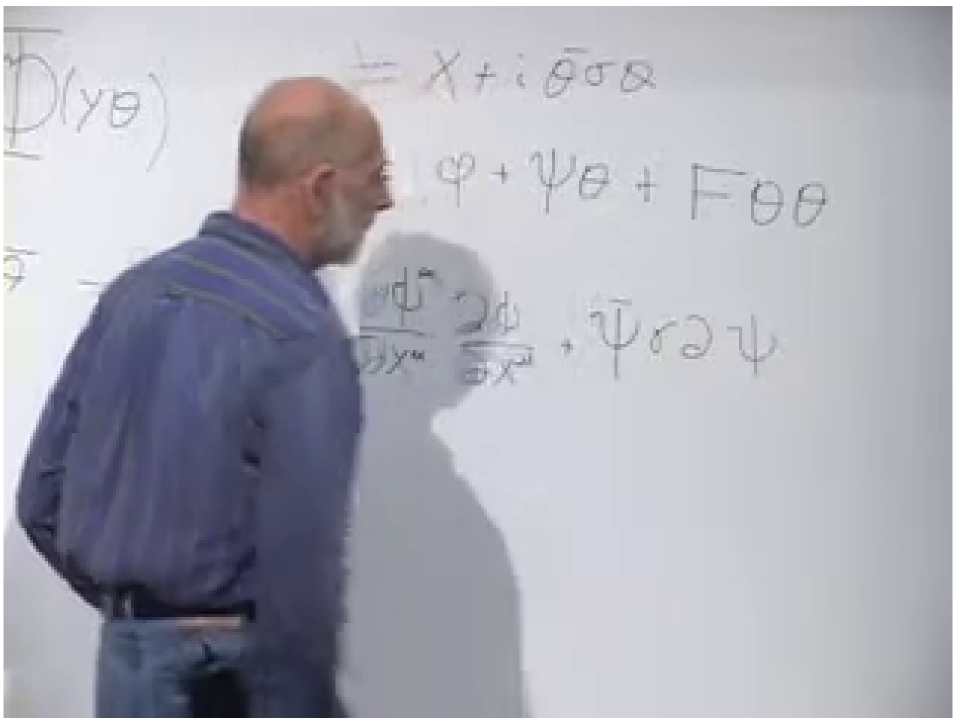


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Figure 9.1: The chiral coordinate and component expansion above the scalar, fermion, and auxiliary terms.

9.5 What the $d^2\theta$ Integral Selects

Interactions and masses arise from a holomorphic function $W(\Phi)$, the superpotential:

$$S_W = \int d^4x \left[\int d^2\theta\, W(\Phi) + \text{h.c.} \right]. \tag{9.17}$$

A chiral expression has no independent $\bar{\theta}$ dependence, so integrating it over all four odd coordinates would give zero. The $d^2\theta$ integral instead selects its θ^2 component.

To see the combinatorics before taking a general function, let $W(\Phi) = \Phi^n$. In

$$(\phi + \sqrt{2}\,\theta\psi + \theta^2 F)^n, \tag{9.18}$$

there are two ways to obtain exactly two θ's. One may choose a single $F\theta^2$, or choose two fermionic factors.

Question from the audience. How many terms contain one chosen $F\theta^2$ and otherwise only scalars: $n-1$ or n? [a]

Response. There are n. Any one of the n factors may supply $F\theta^2$; the remaining $n-1$ factors supply ϕ. The contribution is $nF\phi^{n-1}$.

---

[a] Lecture timestamp: 00:31:33.

Question from the audience. How many ways can two factors supply the two fermions? [a]

Response. There are $\binom{n}{2} = n(n-1)/2$ unordered choices. Grassmann contraction fixes the corresponding $\psi\psi$ term.

---

[a] Lecture timestamp: 00:32:26.

Thus, in the lecture's coefficient-suppressing notation,

$$\int d^2\theta\, \Phi^n + \text{h.c.} \;\rightsquigarrow\; nF\phi^{n-1} - \frac{n(n-1)}{2}\phi^{n-2}\psi\psi + \text{h.c.} \tag{9.19}$$

The minus sign shown here follows the standard normalized expansion and Grassmann convention; changing conventions can move superficial signs.

Question from the audience. What happened to the term ϕ^n, which contains no θ's at all? [a]

Response. It is present in the product, but $d^2\theta$ discards it. The arrow means that integration selects the θ^2 coefficient; it is not equality of the full expressions.

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[a] Lecture timestamp: 00:33:10.

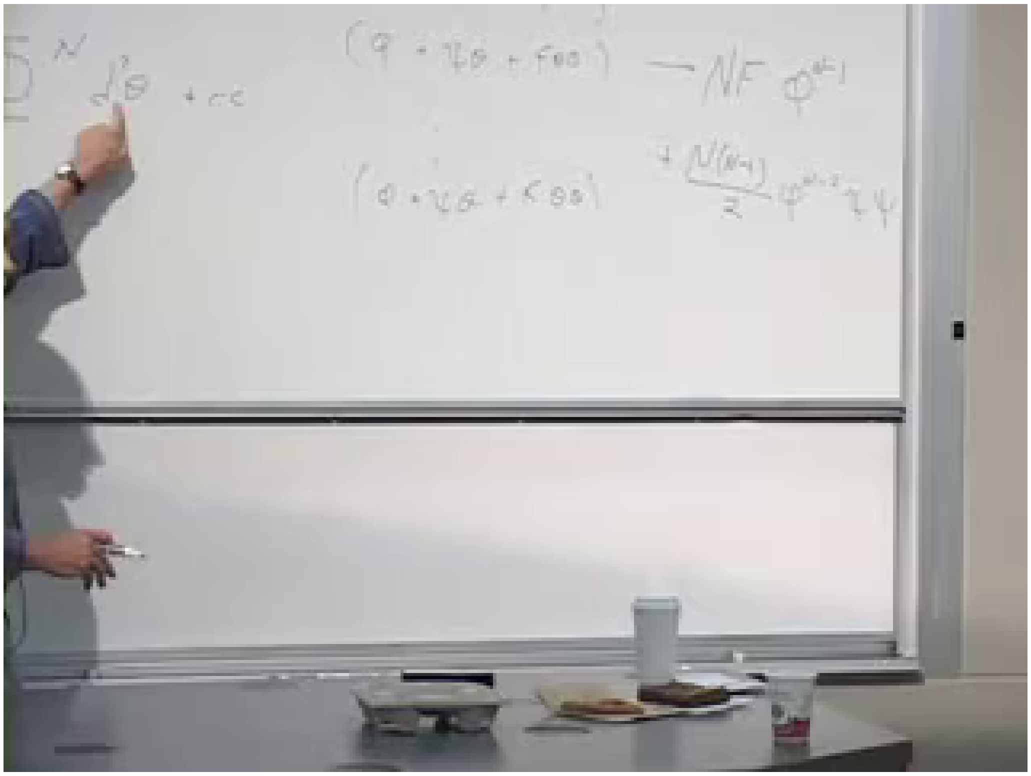

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Figure 9.2: The repeated superfield factors and the implication arrow that extracts, rather than equals, their two-θ component.

Summing powers with arbitrary coefficients converts the combinatorial factors into derivatives:

$$\int d^2\theta\, W(\Phi) + \text{h.c.} \;\rightsquigarrow\; FW'(\phi) - \frac{1}{2}W''(\phi)\psi\psi + \text{h.c.} \tag{9.20}$$

Question from the audience. Why does the fermion term contain ϕ^{n-2}, rather than ϕ^{n-1}? [a]

Response. Two factors have already supplied the fermions, leaving $n-2$ scalar factors. The coefficient is one half of the second derivative $W''(\phi)$.

---

[a] Lecture timestamp: 00:36:32.

The fermion term is important: it gives a mass when W'' is constant and a Yukawa interaction when W'' depends on ϕ. For the vacuum analysis, however, the auxiliary term FW' carries the immediate information.

9.6 Eliminating the Auxiliary Field

Collect the terms that contain F:

$$\mathcal{L}_F = F^*F + FW'(\phi) + F^*\overline{W'(\phi)}. \tag{9.21}$$

Since there is no $\partial_\mu F$, the full Euler–Lagrange equation

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu F)} - \frac{\partial \mathcal{L}}{\partial F} = 0 \tag{9.22}$$

reduces immediately to an algebraic equation. Treating F and F^* as independent variables gives

$$\frac{\partial \mathcal{L}_F}{\partial F} = F^* + W'(\phi) = 0, \qquad \frac{\partial \mathcal{L}_F}{\partial F^*} = F + \overline{W'(\phi)} = 0, \tag{9.23}$$

and hence

$$F^* = -W'(\phi), \qquad F = -\overline{W'(\phi)}. \tag{9.24}$$

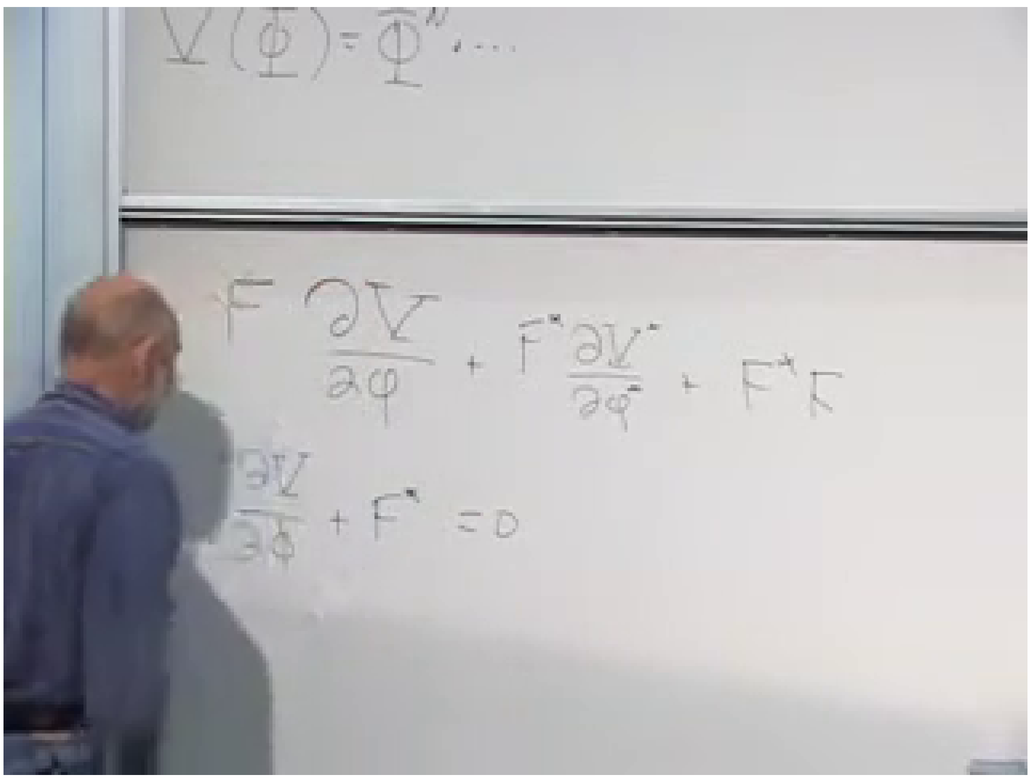

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Figure 9.3: The complete auxiliary-field expression and its algebraic equation of motion, visible together after the board is cleared of occlusion.

Question from the audience. Should that derivative be taken with respect to ϕ, rather than the conjugate field written on the board? [a]

Response. Yes. The correction restores the conjugate pair in Eq. (9.23): $F^* = -W'(\phi)$ and $F = -\overline{W'(\phi)}$.

---

[a] Lecture timestamp: 00:41:21.

Substitution into Eq. (9.21) gives

$$\mathcal{L}_F\big|_{\text{on shell}} = -|W'(\phi)|^2. \tag{9.25}$$

Since a scalar Lagrangian has the form kinetic energy minus potential energy, the ordinary scalar potential is

$$\boxed{V(\phi,\phi^*) = |W'(\phi)|^2 \geq 0.} \tag{9.26}$$

This V is not the superpotential. We reserve W for the holomorphic superspace function and V for ordinary potential energy.

9.6.1 Two Immediate Examples

With exact conventional normalizations,

$$\begin{gathered} W(\Phi) = \frac{1}{2}m\Phi^2, \\ W'(\phi) = m\phi, \qquad V = |m|^2|\phi|^2. \end{gathered} \tag{9.27}$$

The same term gives the fermion mass $-\frac{1}{2}m\psi\psi + \text{h.c.}$, so the scalar and fermion masses agree before supersymmetry is broken.

For

$$W(\Phi) = \frac{g}{3}\Phi^3, \tag{9.28}$$

one obtains

$$\begin{gathered} W'(\phi) = g\phi^2, \qquad V = |g|^2|\phi|^4, \\ \mathcal{L}_{\text{Yukawa}} = -g\phi\psi\psi + \text{h.c.} \end{gathered} \tag{9.29}$$

The quartic scalar vertex describes two-scalar scattering, while the Yukawa term is its supersymmetric companion.

9.7 Two Kinds of Vacuum

The vacuum is a constant field configuration at a minimum of V; its derivative terms vanish. Because Eq. (9.26) is nonnegative, two cases are possible.

9.7.1 A Zero-Energy Minimum

If some ϕ_0 satisfies

$$W'(\phi_0) = 0, \tag{9.30}$$

then

$$V(\phi_0) = 0, \qquad \langle F \rangle = 0. \tag{9.31}$$

The vacuum is supersymmetric. This is the analogue of vanishing magnetization: no auxiliary component has selected a superspace direction.

9.7.2 A Lifted Minimum

If the equations

$$W'(\phi) = 0 \tag{9.32}$$

have no simultaneous solution at the minimum, then

$$V_{\min} > 0, \qquad \langle F \rangle \neq 0, \tag{9.33}$$

and supersymmetry is spontaneously broken. The action remains supersymmetric; its lowest-energy state does not.

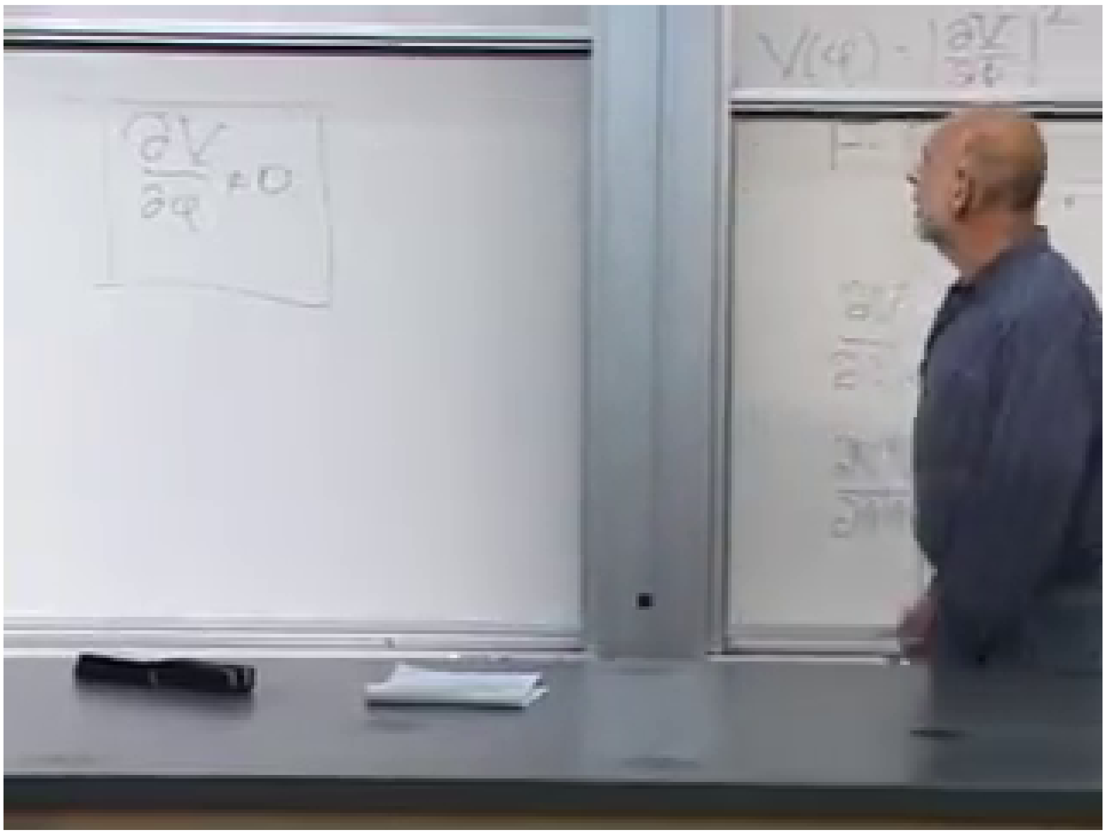

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Figure 9.4: The board keeps the positive potential, the F-equation, and the vacuum condition in view while distinguishing symmetric equations from a non-symmetric solution.

The order-parameter statement can be made precise through the fermion transformation:

$$\delta\psi_\alpha = i\sqrt{2}(\sigma^\mu \bar{\epsilon})_\alpha \partial_\mu \phi + \sqrt{2}\,\epsilon_\alpha F. \tag{9.34}$$

For a homogeneous vacuum the derivative term vanishes. If $\langle F \rangle \neq 0$, supersymmetry shifts the fermion by a nonzero constant, so the vacuum cannot be invariant.[2]

The same conclusion follows from the superalgebra. In the rest frame,

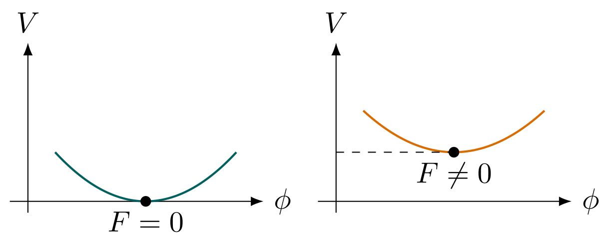
$$\{Q_\alpha, Q^\dagger_{\dot\alpha}\} = 2\sigma^0_{\alpha\dot\alpha} H. \tag{9.35}$$

Taking a vacuum expectation value expresses its energy as a sum of norms of supercharges acting on the vacuum:

$$E_0 \geq 0, \qquad E_0 = 0 \iff Q_\alpha|0\rangle = Q^\dagger_{\dot\alpha}|0\rangle = 0. \tag{9.36}$$

[2]Editorial clarification: Equation (9.34) is the standard component form that makes explicit the lecture's statement that superfield components rotate into one another.

Thus positive vacuum energy and spontaneous supersymmetry breaking are two sides of the same rigid-supersymmetry statement.



[3]

9.8 What a Broken Supersymmetry Produces

9.8.1 Mass Splitting

The ferromagnetic analogy now becomes spectral again. A weak internal magnetic field separates atomic levels without destroying the identity of the atom. Similarly, a weakly communicated supersymmetry-breaking order parameter separates bosons and fermions that would otherwise lie in one supermultiplet.

Question from the audience. Which particles are rotated into one another by supersymmetry and therefore lose their mass degeneracy when the vacuum breaks it? [a]

Response. Bosons and fermions. Their masses need not remain equal once the state no longer respects the transformation that relates them.

[a]Lecture timestamp: 00:51:40.

The splitting is not produced by $\langle F \rangle$ in isolation; it depends on couplings that transmit the breaking to the visible fields. If those couplings are weak, the visible particles remain recognizable and the splittings are small compared with grand-unification or Planck scales. If they were strong, the analogue would be an atom so distorted by an enormous magnetic field that it became a long, unrecognizable object.

9.8.2 The Goldstino

For an ordinary broken generator, $J|0\rangle$ gives a bosonic Goldstone state. A supercharge is fermionic and changes bosonic states into fermionic ones. Since the vacuum is counted as bosonic, a broken supercharge creates a fermionic zero-energy state:

$$Q_\alpha|0\rangle \sim |\widetilde{G}_\alpha, \mathbf{k} = 0\rangle. \tag{9.37}$$

Its propagating continuation is a massless Weyl fermion, the Goldstone fermion or goldstino.

[3]Editorial clarification: A minimal illustration is $W = f\Phi$, for which $W' = f$ never vanishes and $V = |f|^2$. That one-field example has a flat scalar direction; realistic stable F-term breaking is normally implemented with several chiral fields, as in O'Raifeartaigh-type hidden sectors.

Question from the audience. If a supersymmetry transformation does not leave the vacuum fixed, what new state does it make? [a]

Response. A massless fermion. The ordinary Goldstone theorem pattern remains, but a fermionic generator turns the bosonic vacuum into the goldstino.

[a] Lecture timestamp: 00:53:24.

Rigid spontaneous supersymmetry breaking therefore has two unavoidable signatures:

$$\boxed{V_{\min} > 0 \qquad \text{and} \qquad \text{a massless goldstino.}} \tag{9.38}$$

9.9 The Hidden Sector

It is useful to call the atom and the ferromagnet two sectors. The ferromagnet develops the order parameter; the atom feels only the field transmitted from it. When the coupling is weak, manipulating the atom scarcely excites a magnon and gives little direct access to the magnetic medium. The ferromagnet is then hidden, though its presence is inferred from small spectral splittings.

Supersymmetric model building uses the same language. Additional fields in a hidden sector develop nonzero F- or D-term expectation values. Interactions called the mediation mechanism communicate that breaking to the visible sector and generate soft masses for superpartners. Weak transmission explains how the visible theory can remain approximately supersymmetric at a much higher fundamental scale while showing no exact low-energy boson–fermion degeneracy.

The construction is not automatic. Extra fields, a stable breaking vacuum, and a controlled mediation mechanism are needed. This is why the lecture describes realistic supersymmetry breaking as complicated and somewhat contrived rather than as a single inevitable consequence of the minimal model.

9.10 Gravity Changes Both Signatures

9.10.1 Vacuum Energy Becomes a Source

Without gravity, shifting every energy by a constant does not change nongravitational experiments. With gravity, a vacuum energy density enters Einstein's equation as a cosmological constant. A positive $V_{\min}$ drives accelerated expansion. If its scale were comparable to the particle-physics scale needed for superpartner masses, the resulting curvature would be enormously larger than observed.

Question from the audience. What does a positive nonzero vacuum energy do to the world once gravity is included? [a]

Response. It acts as a cosmological constant and drives accelerated expansion. At ordinary particle-physics scales the effect would be intolerably large.

[a] Lecture timestamp: 01:00:40.

Local supersymmetry, or supergravity, alters the positive-square formula. For orientation, the standard $N = 1$ F-term potential is

$$\begin{aligned} V_F &= e^{K/M_P^2} \left(K^{i\bar{j}} D_i W \, D_{\bar{j}} \overline{W} - \frac{3|W|^2}{M_P^2} \right), \\ D_i W &= \partial_i W + \frac{\partial_i K}{M_P^2} W. \end{aligned} \tag{9.39}$$

Because of the negative term, positive energy from supersymmetry breaking can, in principle, be partly canceled. The remainder may be a small cosmological constant.
[4]

9.10.2 The Super-Higgs Mechanism

The second gravitational effect is easiest to understand by counting polarizations. A massive spin-j particle has $2j + 1$ spin states in its rest frame.

Question from the audience. How many spin states does a massive spin-j particle have? [a]

Response. $2j+1$. In particular a massive spin-one particle has three states, whereas a massless gauge boson has only helicities $+1$ and -1.

---

[a] Lecture timestamp: 01:05:16.

The distinction is kinematic. A massive particle can be boosted to rest and then sent in a new direction while its spin orientation is retained, so all projections must exist. A massless particle cannot be overtaken or brought to rest. It may therefore carry only the two extremal helicities.

For the electroweak gauge field,

$$\underbrace{(+1,-1)}_{\text{massless vector}} + \underbrace{(0)}_{\text{Goldstone}} \longrightarrow \underbrace{(+1,0,-1)}_{\text{massive vector}}. \tag{9.40}$$

Question from the audience. Where does the longitudinal polarization of a massive W or Z come from? [a]

Response. From the Goldstone boson. It is absorbed by the gauge field and completes the three-state massive spin-one representation.

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[a] Lecture timestamp: 01:08:50.

Gravity has a massless spin-two graviton. Its supersymmetric partner is the massless spin-$\frac{3}{2}$ gravitino.

Question from the audience. What is the spin of the graviton, and therefore of its superpartner? [a]

Response. The graviton has spin 2; the fermionic partner differs by one half and has spin 3/2.

---

[a] Lecture timestamp: 01:10:24.

[4] Editorial clarification: The lecture intentionally gives only a schematic correction and explicitly questions the placement of the Planck mass. Equation (9.39) records the conventional formula for chiral multiplets with Kähler potential K; gauge D-terms would add a further nonnegative contribution. It does not explain the observed tiny value, which remains the cosmological-constant problem.

A massless gravitino has helicities $\pm\frac{3}{2}$, while a massive gravitino must also have $\pm\frac{1}{2}$. The goldstino supplies precisely those missing states:

$$\begin{gathered}\underbrace{\left(+\frac{3}{2},-\frac{3}{2}\right)}_{\text{massless gravitino}} \oplus \underbrace{\left(+\frac{1}{2},-\frac{1}{2}\right)}_{\text{goldstino}} \\ \Downarrow \\ \underbrace{\left(+\frac{3}{2},+\frac{1}{2},-\frac{1}{2},-\frac{3}{2}\right)}_{\text{massive gravitino}}.\end{gathered} \tag{9.41}$$

This is the super-Higgs mechanism. Once supersymmetry is local, the goldstino is absorbed and no longer appears as an independent massless particle. The resulting gravitino interacts gravitationally and is correspondingly difficult to produce.

9.11 Changing Subjects: Why $SU(5)$?

The supersymmetry formalism now gives way to grand unification. The Standard Model gauge group is

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y. \tag{9.42}$$

Its three factors separately organize color, weak isospin, and weak hypercharge. The unification question is whether this product is the surviving subgroup of one larger simple group, so that apparently separate particle multiplets and gauge forces are pieces of a single structure. The minimal candidate developed here is $SU(5)$.

9.11.1 Group Elements and Generators

For $U \in SU(N)$,

$$U^\dagger U = \mathbf{1}, \qquad \det U = 1. \tag{9.43}$$

The first condition is unitary; the second is the meaning of special. Write an infinitesimal element as

$$U = \mathbf{1} + i\epsilon G + \mathcal{O}(\epsilon^2). \tag{9.44}$$

Unitarity requires

$$G^\dagger = G, \tag{9.45}$$

and

$$\det e^{i\epsilon G} = e^{i\epsilon \operatorname{tr} G} = 1 \tag{9.46}$$

requires

$$\operatorname{tr} G = 0. \tag{9.47}$$

Thus the generators are traceless Hermitian matrices.

An $N \times N$ Hermitian matrix contains N^2 real parameters: N real diagonal entries and $N(N-1)/2$ complex off-diagonal entries. Tracelessness removes one. Therefore

$$\dim SU(N) = N^2 - 1, \qquad \dim SU(5) = 24. \tag{9.48}$$

For comparison, $SU(2)$ has three Pauli generators and $SU(3)$ has eight color generators.

9.12 The Standard Model Blocks Inside $SU(5)$

Split the five-dimensional space into a weak two-dimensional block and a color three-dimensional block:

$$\mathbf{5} = \mathbf{2} \oplus \mathbf{3}. \tag{9.49}$$

The embedded generators have the forms

$$T_L^a = \begin{pmatrix} \frac{1}{2}\tau^a & 0 \\ 0 & 0_{3\times 3} \end{pmatrix}, \qquad a = 1, 2, 3, \tag{9.50}$$

$$T_c^A = \begin{pmatrix} 0_{2\times 2} & 0 \\ 0 & \frac{1}{2}\lambda^A \end{pmatrix}, \qquad A = 1, \ldots, 8. \tag{9.51}$$

The Pauli matrices τ^a generate weak $SU(2)$; the Gell-Mann matrices λ^A generate color $SU(3)$. Because the blocks do not overlap, the two subgroups commute.

One additional diagonal generator is proportional to the identity within each block and therefore commutes with both. In the fundamental representation one convenient doubled-hypercharge convention is

$$T_Y^{(\mathbf{5})} = \mathrm{diag}\left(1, 1, -\frac{2}{3}, -\frac{2}{3}, -\frac{2}{3}\right). \tag{9.52}$$

Its trace vanishes:

$$1 + 1 - 3\left(\frac{2}{3}\right) = 0. \tag{9.53}$$

The conjugate matter representation has the opposite eigenvalues,

$$Y_{\overline{\mathbf{5}}} = \mathrm{diag}\left(-1, -1, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right), \tag{9.54}$$

which are the values written for the matter column in the lecture. The overall sign and normalization are conventional; these entries use the Standard Model convention

$$Q = T_3 + \frac{Y}{2}. \tag{9.55}$$

Equations (9.50)–(9.52) display

$$SU(3)_c \times SU(2)_L \times U(1)_Y \subset SU(5). \tag{9.56}$$

The three weak generators, eight color generators, and one hypercharge generator account for 12 of the 24 $SU(5)$ generators.[5]

Question from the audience. Which gauge particles correspond to the familiar subgroup generators? [a]

Response. The eight $SU(3)$ generators give the eight gluons. The three $SU(2)$ generators give the weak gauge fields, which after electroweak mixing become $W^{\pm}$ and part of the Z-photon system; the $U(1)$ field supplies the other neutral combination.

[a] Lecture timestamp: 01:31:05.

[5] Editorial clarification: The other twelve occupy the off-diagonal 2×3 and 3×2 blocks. Their gauge bosons convert quark-type and lepton-type components into one another; this is the structural origin of the baryon- and lepton-changing interactions developed in the continuation.

9.13 One Family in Left-Handed Language

Focus on the first generation; the same pattern is repeated for the muon/charm/strange and tau/top/bottom generations. It is most economical to count every independent Weyl field as left-handed. Right-handed particles are represented by their left-handed charge-conjugate fields:

$$Q_L = (u_L, d_L), \quad L_L = (\nu_{eL}, e_L), \quad u^c_L, \quad d^c_L, \quad e^c_L. \tag{9.57}$$

Their multiplicities are

$$\underbrace{6}_{Q_L} + \underbrace{2}_{L_L} + \underbrace{3}_{u^c_L} + \underbrace{3}_{d^c_L} + \underbrace{1}_{e^c_L} = 15. \tag{9.58}$$

[6]

The five-dimensional multiplet must place colorless weak-doublet entries in the first two slots and colored weak singlets in the last three.

Question from the audience. Does the proposed column contain six fields rather than five? [a]

Response. No. The up quark written during the first pass is erased. The column is the neutrino, the electron, and three color copies of the anti-down quark: five entries in all.

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[a]Lecture timestamp: 01:36:14.

That gives

$$\overline{\mathbf{5}} = \begin{pmatrix} \nu_{eL} \\ e_L \\ d^c_{rL} \\ d^c_{gL} \\ d^c_{bL} \end{pmatrix}. \tag{9.59}$$

[7] The first two entries transform under $SU(2)_L$ but not color. The last three transform as an anti-triplet under color but are weak singlets.

Question from the audience. Which quarks carry color but no weak $SU(2)_L$ quantum number, once every field is written left-handed? [a]

Response. The charge conjugates of the right-handed quarks. For this five-component multiplet, the charge test selects the three anti-down fields $d^c_{r,g,b}$.

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[a]Lecture timestamp: 01:40:13.

The electric charges in Eq. (9.59) sum to zero:

$$0 - 1 + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 0. \tag{9.60}$$

That is required because the electric-charge operator is a traceless combination of $SU(5)$ generators.

[6]Editorial clarification: The minimal Standard Model contains no independent right-handed-neutrino Weyl field, so the count is 15. This counting alone does not assert that the neutrino is a Majorana particle. Adding a singlet ν^c_L makes 16 fields and points naturally toward $SO(10)$, beyond the present $SU(5)$ construction.

[7]Editorial clarification: The lecture initially calls this simply a five-dimensional multiplet. In standard all-left-handed $SU(5)$ notation, its color and hypercharge assignments identify it as the conjugate representation $\overline{\mathbf{5}}$.

Question from the audience. Why do anti-down quarks, rather than anti-up quarks, occupy the three color slots? [a]

Response. The neutrino and electron contribute total charge -1. Three anti-down colors contribute $+1$, making the trace zero. Three anti-up fields would have the wrong total charge.

[a]Lecture timestamp: 01:43:38.

The remaining ten left-handed fields are

$$Q_L\ (6) \oplus u_L^c\ (3) \oplus e_L^c\ (1), \tag{9.61}$$

and they fit exactly into the antisymmetric rank-two representation

$$\mathbf{10} \cong \wedge^2 \mathbf{5}, \qquad \dim \wedge^2 \mathbf{5} = \binom{5}{2} = 10. \tag{9.62}$$

Their explicit placement is the next lecture's task. The numerical fit

$$\overline{\mathbf{5}} \oplus \mathbf{10} \tag{9.63}$$

for every Standard Model family is the first striking evidence that the $SU(5)$ idea is more than an arbitrary relabeling.

Question from the audience. If the five and ten are different representations, does that mean their particles cannot interact? [a]

Response. No. A group transformation acts within each irreducible representation, so it does not directly rotate a member of the five into a member of the ten. The two sets can nevertheless interact by emitting and absorbing common gauge bosons, and Higgs couplings can connect their fields as well.

[a]Lecture timestamp: 01:46:14.

9.14 Perspective

The lecture has followed one pattern across very different scales. Rotational invariance makes atomic levels degenerate; a magnetic background splits them. A ferromagnet produces such a background while leaving the underlying laws symmetric, and its gapless magnons reveal that the asymmetry resides in the vacuum. In a chiral supersymmetric theory, the auxiliary expectation value plays the role of the order parameter:

$$\begin{aligned} \langle F \rangle = 0 \quad &\Longleftrightarrow \quad \text{unbroken supersymmetry}, \\ \langle F \rangle \neq 0 \quad &\Longrightarrow \quad \text{spontaneous breaking}. \end{aligned} \tag{9.64}$$

The broken vacuum carries positive energy in rigid supersymmetry, splits supermultiplets, and creates a goldstino. Weak mediation motivates a hidden sector; gravity turns vacuum energy into a cosmological problem and absorbs the goldstino into a massive gravitino.

The closing $SU(5)$ construction begins a new question rather than completing it. Traceless Hermitian 5×5 generators contain the Standard Model blocks, and the first five matter fields fit exactly as $(\nu_e, e^-, d_r^c, d_g^c, d_b^c)$. The remaining ten are waiting in the antisymmetric representation. That unfinished fit is where grand unification properly begins.

Chapter 10

The $SU(5)$ Multiplets, Leptoquark Gauge Bosons, and Proton Decay

Overview. Grand unification becomes concrete when the Standard Model group is embedded in $SU(5)$. To see what that statement means, we first build the necessary representation theory: the defining representation N, its conjugate $\overline{N}$, tensor products, symmetric and antisymmetric tensors, and the adjoint. One Standard Model family then falls into the unexpectedly economical combination $\overline{\mathbf{5}} \oplus \mathbf{10}$. The same symmetry that makes this arrangement beautiful also requires colored gauge bosons that convert quarks into leptons. Their exchange permits proton decay, forcing the unification scale toward 10^{15}–10^{16} GeV. The running gauge couplings point toward a similar scale, and supersymmetry returns at the end because it makes their numerical meeting substantially sharper.

10.1 Why Begin with Group Representations?

There are many proposed grand unified theories. Rather than survey them, we shall take apart one example and ask three questions: what makes it work, what new particles it requires, and where experiment puts it in danger. The group is $SU(5)$, and its defining structural feature is the embedding

$$SU(3)_c \times SU(2)_L \times U(1)_Y \subset SU(5). \tag{10.1}$$

Here $SU(3)_c$ acts on color, $SU(2)_L$ on weak isospin, and $U(1)_Y$ on hypercharge.

Supersymmetry is not a premise of this construction. Neither supersymmetry nor grand unification logically implies the other. Their connection in this lecture is numerical: when the superpartner spectrum is included in the renormalization-group calculation, the three gauge couplings approach a common value more accurately.

Particles that a symmetry mixes among themselves form a representation of the symmetry group. Classifying unified particle multiplets therefore requires a little representation theory first.

10.1.1 The Familiar $SU(2)$ Example

The $SU(2)$ associated with ordinary angular momentum is closely related to rotations of three-dimensional space. More precisely, $SU(2)$ is the double cover of $SO(3)$. Its irreducible representations are labeled by

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots, \qquad \dim(j) = 2j + 1. \tag{10.2}$$

Thus a scalar is a singlet, a spin-$\frac{1}{2}$ particle is a doublet, and a spin-one particle is a triplet. The weak group $SU(2)_L$ is a different physical symmetry, although its abstract group theory is the same.

10.2 The Defining Representation and Its Conjugate

For $SU(N)$, the defining representation is the space of complex N-component columns,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_N \end{pmatrix}, \qquad \psi_i' = U_i{}^j \psi_j, \qquad U \in SU(N). \tag{10.3}$$

It is called either the defining representation or simply $\mathbf{N}$. The matrices U are $N \times N$, and the vector space on which they act has dimension N.

Complex conjugation gives another N-dimensional representation,

$$\psi_i^* \longmapsto U_i{}^{j*} \psi_j^*, \tag{10.4}$$

denoted $\overline{\mathbf{N}}$. For $N > 2$, the defining and conjugate representations are inequivalent. One may exchange which is called the particle representation and which the antiparticle representation, but after choosing a convention one must keep it fixed.

Question from the audience. Does the N in $SU(N)$ depend on whether N is even or odd, and does it refer to the matrices? [a]

Response. There is no even–odd restriction here. N is the size of the defining matrices and also the dimension of their defining representation. Both $\mathbf{N}$ and $\overline{\mathbf{N}}$ have that dimension.

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[a] Lecture timestamp: 00:07:33.

The antiparticle of an antiparticle is the original particle. This is why interchanging the names $\mathbf{N}$ and $\overline{\mathbf{N}}$ changes convention rather than physics.

10.3 Tensor Products and Irreducibility

Suppose one object can occupy N internal states and another object can independently occupy the same number. Ignoring position, momentum, spin, and every quantum number except this $SU(N)$ label, the pair has

$$N \times N = N^2 \tag{10.5}$$

states. If the first object is in state ψ_i and the second in state ϕ_j, their product may be written

$$\Psi_{ij} = \psi_i \phi_j, \qquad \Psi'_{ij} = U_i{}^k U_j{}^\ell \Psi_{k\ell}. \tag{10.6}$$

This is the tensor product $\mathbf{N} \otimes \mathbf{N}$.

Question from the audience. Why is the answer N^2, rather than $N^2 - 1$, and does allowing both particles the same internal label violate the exclusion principle? [a]

Response. The two slots are distinguishable here: one object is fixed at one location and the other at another. The Pauli principle constrains the complete quantum state, including position. It does not remove any of these N^2 internal-label combinations. The number $N^2 - 1$ will arise later from a tracelessness condition in a different product.

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[a] Lecture timestamp: 00:10:22.

The N^2-dimensional product representation is generally reducible. A representation is reducible when it contains a proper subspace that the group maps into itself. The invariant subspaces can be separated and treated as smaller representations; if no such proper subspace exists, the representation is irreducible.

10.3.1 Symmetric and Antisymmetric Tensors

Every two-index tensor splits uniquely into

$$\Psi_{ij} = \Psi_{ij}^{(S)} + \Psi_{ij}^{(A)}, \tag{10.7}$$

$$\Psi_{ij}^{(S)} = \frac{1}{2}(\Psi_{ij} + \Psi_{ji}), \tag{10.8}$$

$$\Psi_{ij}^{(A)} = \frac{1}{2}(\Psi_{ij} - \Psi_{ji}). \tag{10.9}$$

The two pieces obey

$$\Psi_{ij}^{(S)} = \Psi_{ji}^{(S)}, \qquad \Psi_{ij}^{(A)} = -\Psi_{ji}^{(A)}. \tag{10.10}$$

Equation (10.6) preserves each condition, so group transformations never mix the symmetric and antisymmetric sectors.

[1]

For the defining doublet of $SU(2)$,

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{3}_S \oplus \mathbf{1}_A. \tag{10.11}$$

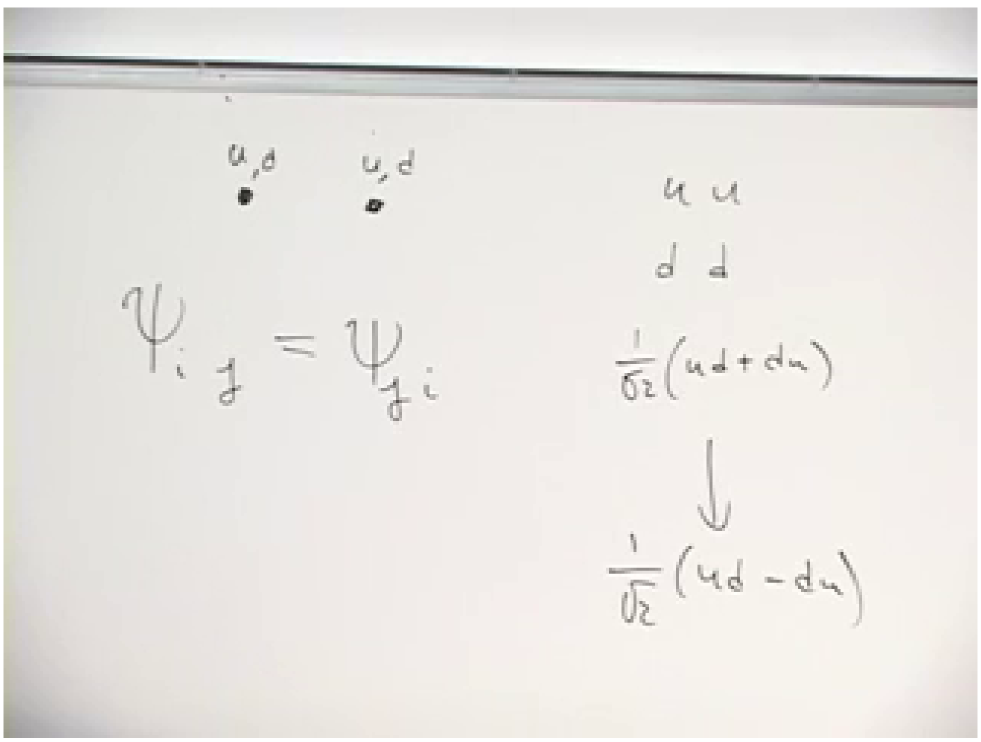
Writing the two basis states as u and d, the symmetric triplet is

$$|1, 1\rangle = |uu\rangle, \tag{10.12}$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|ud\rangle + |du\rangle), \tag{10.13}$$

$$|1, -1\rangle = |dd\rangle, \tag{10.14}$$

[1] Editorial clarification: The lecture leaves the general counting as an exercise. The symmetric and antisymmetric dimensions are $N(N+1)/2$ and $N(N-1)/2$, respectively, and their sum is N^2.



Lecture frame: 00:17:57.500.

Figure 10.1: The completed board decomposition for two spin-$\frac{1}{2}$ states: three symmetric states and one antisymmetric state.

while the antisymmetric singlet is

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|ud\rangle - |du\rangle). \tag{10.15}$$

This is the familiar addition of two spin-$\frac{1}{2}$ angular momenta: the four raw states reorganize into spin one and spin zero.

10.4 A Particle and an Antiparticle: The Adjoint

Now combine the defining representation with its conjugate. The product still has N^2 components, but it transforms as a matrix:

$$M_i{}^j \in \mathbf{N} \otimes \overline{\mathbf{N}}, \qquad M \longmapsto UMU^\dagger. \tag{10.16}$$

Unlike $\mathbf{N} \otimes \mathbf{N}$, there is no meaningful exchange symmetry between one fundamental and one antifundamental index. The natural split is instead

$$M = \frac{\operatorname{tr} M}{N}\,\mathbf{1}_N + \left(M - \frac{\operatorname{tr} M}{N}\,\mathbf{1}_N\right). \tag{10.17}$$

The first term is a singlet and the traceless part is the adjoint:

$$\mathbf{N} \otimes \overline{\mathbf{N}} = \mathbf{1} \oplus \mathbf{Adj}, \qquad \dim \mathbf{Adj} = N^2 - 1. \tag{10.18}$$

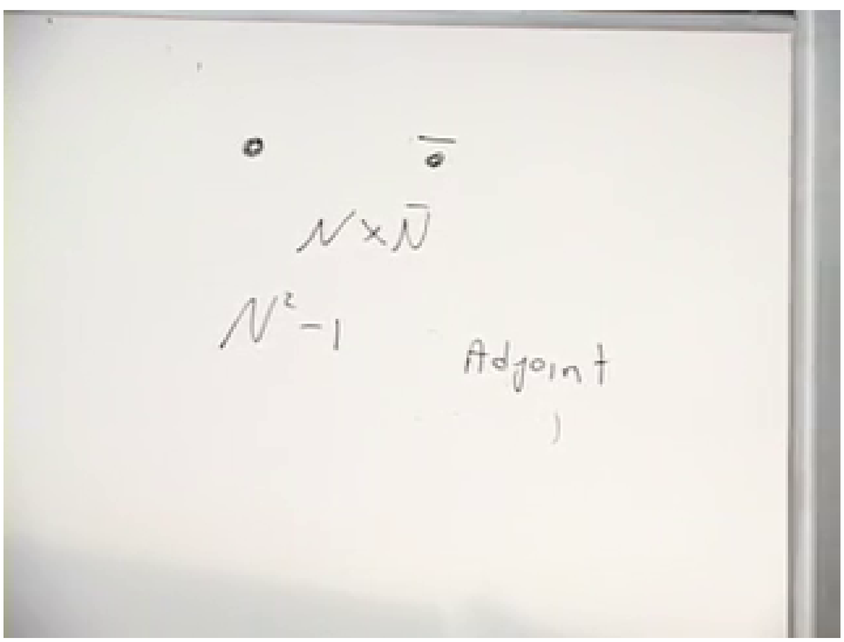


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Figure 10.2: The board conclusion for $\mathbf{N} \otimes \overline{\mathbf{N}}$: its traceless $N^2 - 1$ components form the adjoint.

For $SU(3)$, this gives the familiar octet,

$$\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \tag{10.19}$$

and the eight gluons transform in the adjoint. For $SU(2)$, the adjoint has dimension three and transforms like spin one.

10.5 One Standard Model Family

There are three observed generations, but $SU(5)$ repeats the same construction separately for each. It does not explain why there are three copies. We therefore focus on the first generation.

The fields will all be written as left-handed Weyl fields. A right-handed fermion is represented by its left-handed charge-conjugate field:

$$u_L^c, \qquad d_L^c, \qquad e_L^c. \tag{10.20}$$

The superscript c denotes charge conjugation; for example, d_L^c is a left-handed anti-down field with electric charge $+\frac{1}{3}$.

In standard $SU(5)$ notation, five of the fifteen left-handed fields fit into

$$\overline{\mathbf{5}} = \begin{pmatrix} \nu_{eL} \\ e_L \\ d_{rL}^c \\ d_{gL}^c \\ d_{bL}^c \end{pmatrix}. \tag{10.21}$$

The order has been chosen as a weak two-dimensional block followed by a color three-dimensional block. Weak transformations mix the first two entries, color transformations mix the final three, and the off-diagonal $SU(5)$ generators connect the two blocks.

[2]

The complete minimal first-generation inventory is

$$\begin{gathered} \underbrace{Q_L = (u_L, d_L)}_{6} \oplus \underbrace{L_L = (\nu_{eL}, e_L)}_{2}, \\ \underbrace{u_L^c}_{3} \oplus \underbrace{d_L^c}_{3} \oplus \underbrace{e_L^c}_{1}, \\ 6+2+3+3+1 = 15. \end{gathered} \tag{10.22}$$

The minimal Standard Model has no independent ν_L^c, so it gives no renormalizable neutrino mass. Neutrino oscillations show that this minimal statement is incomplete: nature has small nonzero neutrino masses, and an extension is required.

Question from the audience. When a fermion mass changes a left-handed state into a right-handed one, does it also turn the particle into its antiparticle? [a]

Response. No. An electron mass connects a left-handed electron to a right-handed electron of the same electric charge. In all-left-handed notation the second field is represented by e_L^c, but the four-component particle remains an electron. Only a neutral field can have a Majorana mass that identifies particle and antiparticle.

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[a] Lecture timestamp: 00:34:00.

[2] Editorial clarification: The lecture calls the column in Eq. (10.21) a $\mathbf{5}$ and later calls the antisymmetric multiplet a $\overline{\mathbf{10}}$. This is the globally conjugated convention. We use the modern standard $\overline{\mathbf{5}} \oplus \mathbf{10}$; conjugating every representation reproduces the lecture's labels without changing its physics.

10.6 Gauge Bosons Visible in the Five-Plet

Every generator of a gauge symmetry has an associated gauge field. Within Eq. (10.21), the familiar transitions are already recognizable:

- the weak gauge fields act among ν_{eL} and e_L;
- the gluons act among the three colors of d_L^c;
- neutral generators act diagonally;
- new off-diagonal generators convert a lepton entry into a colored entry or conversely.

The gluon can be pictured as carrying quark–antiquark color quantum numbers. The lecture uses the usual world-line mnemonic: follow an outgoing antiquark backward through the emission vertex and it looks like a quark running forward. The emitted field therefore carries the quantum numbers of a quark–antiquark pair. In representation language this is precisely the traceless part of $\mathbf{3} \otimes \overline{\mathbf{3}}$.

For $SU(5)$,

$$\dim \mathbf{Adj}_{SU(5)} = 5^2 - 1 = 24. \tag{10.23}$$

The decomposition of these gauge fields under the Standard Model subgroup is

$$\mathbf{24} = (\mathbf{8}, \mathbf{1})_0 \oplus (\mathbf{1}, \mathbf{3})_0 \oplus (\mathbf{1}, \mathbf{1})_0 \oplus (\mathbf{3}, \mathbf{2})_{-5/6} \oplus (\overline{\mathbf{3}}, \mathbf{2})_{5/6}, \tag{10.24}$$

where the ordered pair denotes $SU(3)_c \times SU(2)_L$ and the subscript is hypercharge.

[3]

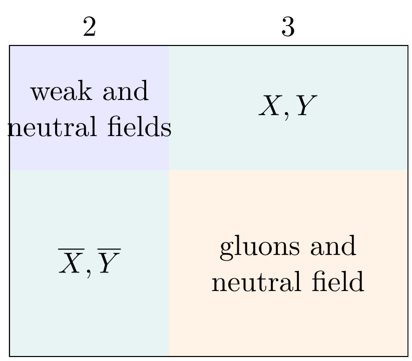

Figure 10.3: The $2 \oplus 3$ block structure of the $SU(5)$ gauge field. The off-diagonal blocks connect leptons and colored fields.

With $Q = T_3 + Y$, the colored weak doublet in Eq. (10.24) has charges $-\frac{1}{3}$ and $-\frac{4}{3}$; its conjugate has $+\frac{1}{3}$ and $+\frac{4}{3}$. The lecture calls these particles X and Y but explicitly declines to fix which name belongs to which charge. The transition and charge are the invariant content.

[3] Editorial clarification: Equation (10.24) makes explicit the block structure drawn in the lecture. The first three terms contain the familiar color and electroweak gauge fields. The final two terms are the twelve real degrees of freedom conventionally called X and Y bosons.

Question from the audience. Are there three versions of each new X or Y gauge boson? [a]

Response. Yes. A transition joins a colored quark entry to a colorless lepton entry, so the exchanged boson must carry the missing color. The X/Y fields form color triplets and can emit or absorb gluons.

[a] Lecture timestamp: 00:41:28.

Colored particles are not observed as isolated asymptotic states. The new gauge bosons must decay or hadronize into color-neutral combinations, just as quarks do.

10.7 Why the Multiplet Contains Anti-Down Quarks

The generators of $SU(N)$ are traceless Hermitian matrices. To see the connection with the word special, write an infinitesimal group element as

$$U(\epsilon) = e^{i\epsilon T}. \tag{10.25}$$

Unitarity gives $T^\dagger = T$, while

$$1 = \det U = e^{i\epsilon \operatorname{tr} T} \tag{10.26}$$

for arbitrary ϵ gives

$$\operatorname{tr} T = 0. \tag{10.27}$$

Electric charge is a linear combination of generators. On the matter column in Eq. (10.21), it acts as

$$Q_{\overline{5}} = \operatorname{diag}\left(0, -1, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \operatorname{tr} Q_{\overline{5}} = 0. \tag{10.28}$$

Thus

$$0 - 1 + 3\left(\frac{1}{3}\right) = 0. \tag{10.29}$$

Replacing the anti-down fields by down fields would instead give

$$0 - 1 + 3\left(-\frac{1}{3}\right) = -2, \tag{10.30}$$

which cannot be the trace of an $SU(5)$ generator. The apparently strange anti-down assignment is therefore forced by the group structure.

10.8 The Remaining Ten Fermions

The other ten fields arise from an antisymmetric two-index tensor. In the standard convention this is the antisymmetric part of

$$\mathbf{5} \otimes \mathbf{5} = \mathbf{15}_S \oplus \mathbf{10}_A. \tag{10.31}$$

The lecture constructs it by temporarily writing the conjugate quantum numbers

$$\chi = \begin{pmatrix} \nu^c \\ e^c \\ d_r \\ d_g \\ d_b \end{pmatrix}. \tag{10.32}$$

This is a bookkeeping device for tensoring representations, not a claim that a second physical five-plet of composite particles exists.

Let

$$\Psi^{ij} = -\Psi^{ji}, \qquad i, j = 1, \ldots, 5. \tag{10.33}$$

All diagonal entries vanish, and the lower triangle is fixed by the upper triangle. Hence

$$\dim \mathbf{10} = \binom{5}{2} = \frac{5 \cdot 4}{2} = 10. \tag{10.34}$$

The electric charge of an entry is the sum of the row and column charges in Eq. (10.32). This immediately gives e^c in the weak–weak slot, down and up quarks in the weak–color blocks, and anti-up fields in the color–color block.

Question from the audience. When the positron and down-quark labels are combined, should the resulting entry be an X boson? [a]

Response. No. At this stage we are constructing the fermion representation, not the gauge-boson adjoint. The charges add as $1 - \frac{1}{3} = \frac{2}{3}$, so the entry has the quantum numbers of an up quark.

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[a]Lecture timestamp: 00:53:08.

A consistent choice of color signs is

$$\Psi^{ij} = \begin{pmatrix} 0 & e^c & d_r & d_g & d_b \\ -e^c & 0 & u_r & u_g & u_b \\ -d_r & -u_r & 0 & u^c_b & -u^c_g \\ -d_g & -u_g & -u^c_b & 0 & u^c_r \\ -d_b & -u_b & u^c_g & -u^c_r & 0 \end{pmatrix}. \tag{10.35}$$

Changing field phases changes some displayed signs but not the representation or particle content.

Question from the audience. Must the ten-dimensional representation be written as a matrix, or can it be written as a column? [a]

Response. Either is possible. One may list the ten independent components in a column, but the antisymmetric 5×5 matrix displays how $SU(5)$ acts on its two indices and is therefore much more informative.

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[a]Lecture timestamp: 00:56:24.

Question from the audience. In the color–color block, should the entry marked u actually be an anti-up field? [a]

Response. Yes. The board label is corrected to u^c, a left-handed anti-up field. Two down-type basis charges sum to $-\frac{2}{3}$, which is the electric charge of an anti-up quark.

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[a]Lecture timestamp: 00:57:51.

Question from the audience. The colors were omitted from several matrix entries. Are they still conserved? [a]

Response. Yes. The three color slots are red, green, and blue. The labels were suppressed only to reduce clutter.

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[a]Lecture timestamp: 00:58:05.

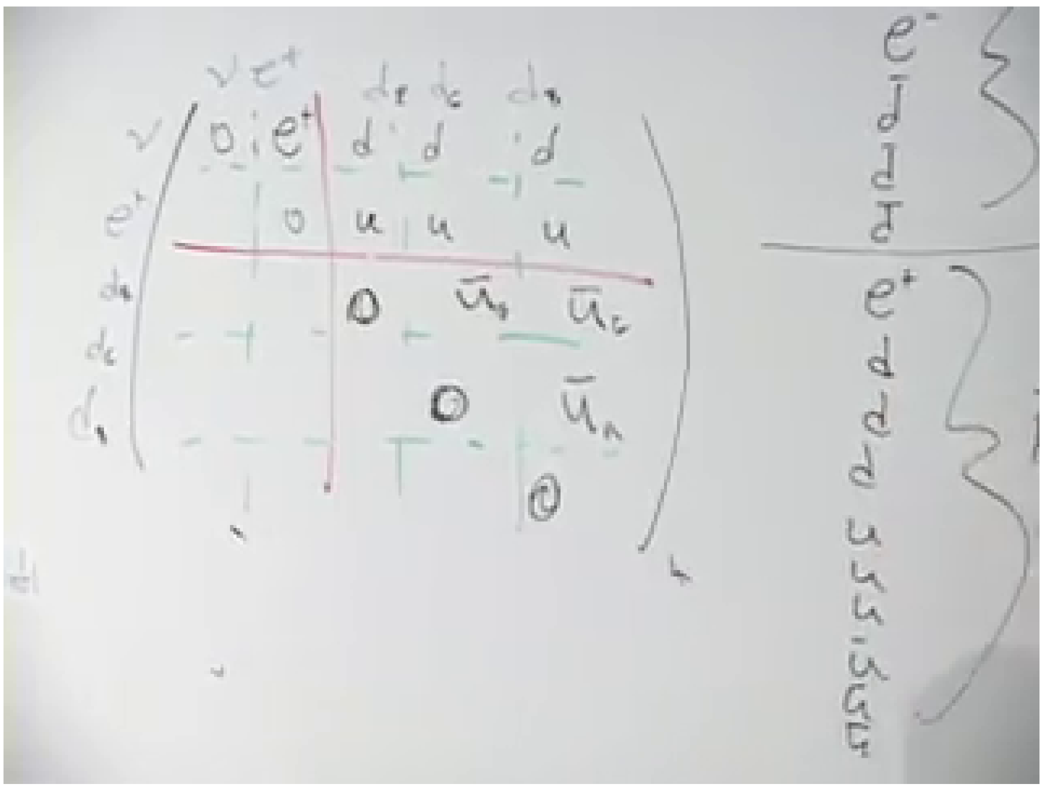

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Figure 10.4: The unobstructed $2 \oplus 3$ antisymmetric matter matrix after the entries and color labels have been checked in class.

An antisymmetric pair of color fundamentals transforms as an anti-color:

$$\wedge^2 \mathbf{3} \simeq \overline{\mathbf{3}}, \qquad (rg, rb, gb) \sim (\bar{b}, -\bar{g}, \bar{r}). \tag{10.36}$$

This is why the three independent entries in the color–color block form precisely the anti-up color triplet.

One generation has now been organized as

$$\overline{\mathbf{5}} \oplus \mathbf{10}. \tag{10.37}$$

It is striking that u_L and u_L^c occur within one irreducible multiplet, while d_L and d_L^c, or e_L and e_L^c, lie in different multiplets.

Question from the audience. If gauge transformations act only within each representation, how can a mass connect fields lying in the five-bar and ten? [a]

Response. Gauge bosons do not rotate one irreducible matter representation into the other. Higgs fields and Yukawa couplings can connect different representations while forming an overall $SU(5)$-invariant interaction; after the Higgs field acquires an expectation value, those interactions generate masses.

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[a] Lecture timestamp: 01:00:40.

10.9 Gauge Bosons Act as Shifts

After the representation has been assembled, the action of a gauge boson can be read as a shift among its entries. A neutral boson may leave a label unchanged, a weak boson moves between weak slots, a gluon moves between color slots, and an X/Y boson moves between weak and color slots. Charged gauge bosons have antiparticles, so each shift has a reverse process.

For the antisymmetric tensor, an infinitesimal transformation acts on either index:

$$\delta\Psi^{ij} = i\epsilon\left(T^i{}_k \Psi^{kj} + T^j{}_k \Psi^{ik}\right). \tag{10.38}$$

In the matrix picture, the first term moves vertically and the second moves horizontally. The constituent labels used to build the tensor are only a mnemonic; Eq. (10.38) is the exact group-theoretic statement.

Weak shifts take d into u, the quark-level process behind neutron beta decay. Color shifts change red, green, and blue without changing flavor. Off-diagonal shifts do something new:

$$\begin{aligned} d^c &\longleftrightarrow e, \quad & d^c &\longleftrightarrow \nu, \\ d &\longleftrightarrow u^c, \quad & u &\longleftrightarrow e^c, \end{aligned} \tag{10.39}$$

with the appropriate X/Y field carrying the charge and color difference.

Question from the audience. What do shifts between the fourth and fifth slots represent, and what about a shift from the first to the fourth? [a]

Response. The fourth–fifth shift changes one color into another and is a gluon transition. A first–fourth shift crosses between weak and color blocks and is one of the new X/Y transitions.

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[a] Lecture timestamp: 01:10:36.

Question from the audience. Can one entry of the lower half of the antisymmetric matrix be filled in explicitly? [a]

Response. Yes. Every lower-triangular entry is the negative of the transposed upper-triangular entry. Filling one makes the rule visible; filling all of them adds no new states.

[a] Lecture timestamp: 01:11:23.

10.10 The Dangerous Consequence: Proton Decay

An X/Y boson can couple a quark to a lepton at one vertex and two quark-type fields at another. At momenta much smaller than its mass, integrating out the heavy boson gives a dimension-six interaction of the schematic form

$$\mathcal{L}_{\text{eff}} \sim \frac{g_5^2}{M_X^2}\,(qq)(q\ell) + \text{h.c.} \tag{10.40}$$

It violates baryon and lepton number separately while preserving the combinations dictated by the unified gauge interaction.

The board follows one down quark and one up quark through the heavy exchange. The down quark becomes a positron, the up quark becomes an anti-up quark, and a second up quark is a spectator:

$$uud \longrightarrow e^+ + u\overline{u}. \tag{10.41}$$

The initial state is a proton. The $u\overline{u}$ pair can hadronize into a neutral pion, giving

$$p \longrightarrow e^+ + \pi^0. \tag{10.42}$$

The lecture also mentions $p \to e^+ + \gamma$, in which the quark–antiquark pair annihilates electromagnetically. That channel carries an additional electromagnetic suppression relative to direct hadronization.

This possibility is not an optional decoration of the model. Once a simple group contains generators that connect quarks and leptons, baryon-number-violating processes are generally expected. If they occurred on ordinary weak or electromagnetic time scales, nuclei, hydrogen, and macroscopic matter would not survive.

10.11 Why Is the Proton So Long-Lived?

Question from the audience. Is the coupling that produces an X boson the same unified coupling used by the rest of the group? [a]

Response. At the unification scale, yes. A simple $SU(5)$ gauge theory has one gauge coupling g_5. The low-energy strong, weak, and hypercharge couplings differ because they run differently after the unified symmetry is broken.

[a] Lecture timestamp: 01:18:34.

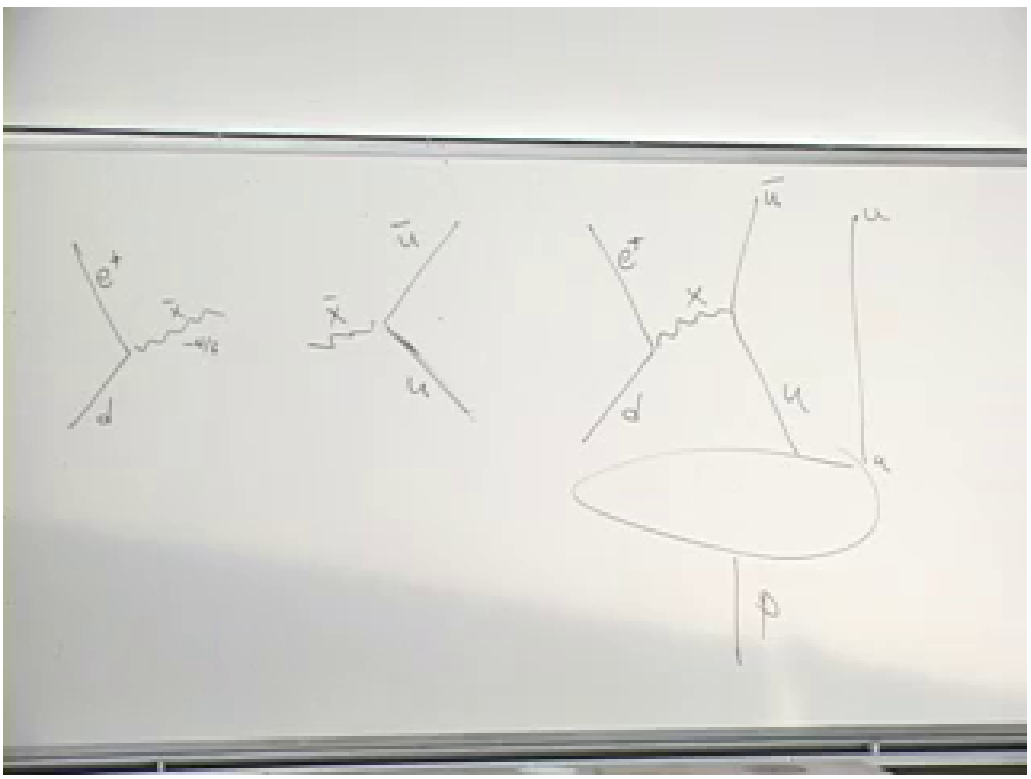

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Figure 10.5: The completed board construction of the heavy-boson exchange inside a proton, with the spectator up quark and positron channel visible.

The coupling cannot be chosen fantastically small without abandoning the unified structure. The available suppression is the heavy propagator:

$$\mathcal{A} \sim \frac{g_5^2}{M_X^2}. \tag{10.43}$$

Squaring the amplitude supplies g_5^4/M_X^4. Dimensional analysis and the proton matrix element complete the decay-rate estimate:

$$\Gamma_p \sim C_{\text{had}} \frac{\alpha_5^2\, m_p^5}{M_X^4}, \qquad \alpha_5 = \frac{g_5^2}{4\pi}, \tag{10.44}$$

where C_{had} contains group factors, renormalization effects, and the nonperturbative hadronic matrix element.

[4]

Question from the audience. Does the small factor g^4, perhaps of order 10^{-8} in the rough classroom estimate, already make the lifetime long enough? [a]

Response. No. It helps, but the gap between a microscopic strong-interaction time and the proton-lifetime bound spans far more orders of magnitude. The fourth power of the heavy mass must provide the decisive suppression.

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[a] Lecture timestamp: 01:25:13.

If all masses were of order the proton mass, the natural time unit would be approximately the light-crossing time of a proton:

$$t_p \sim \frac{1 \text{ fm}}{c} \sim 10^{-23} \text{ s}. \tag{10.45}$$

The mere existence of primordial protons already shows that their lifetime exceeds the age of the universe, but laboratory searches go vastly further. The lecture compares the microscopic estimate with the then-quoted experimental limit

$$\tau_p \gtrsim 10^{33} \text{ yr}. \tag{10.46}$$

One need not watch a single proton for that long. If the lifetime were 10^{15} years, a sample of 10^{15} protons would yield roughly one event per year. More generally, with N_p protons observed for a time T, the expected number of decays is

$$\langle n \rangle \simeq \frac{N_p T}{\tau_p}. \tag{10.47}$$

A detector containing roughly 10^{33} protons is a large tank rather than an astronomical object: the protons themselves have a mass of order 10^6 kilograms. Watching such a sample turns an enormous individual lifetime into a measurable event rate or a stronger lower bound.

Combining Eq. (10.44) with the lifetime constraint drives the leptoquark gauge-boson mass toward

$$M_X \sim 10^{15}\text{–}10^{16} \text{ GeV}, \tag{10.48}$$

with 10^{16} GeV used in the lecture as the memorable safe scale. This is several orders of magnitude below the Planck scale and therefore not automatically inconsistent with quantum field theory.

[4] Editorial clarification: The lecture isolates the essential g^4/M_X^4 dependence and notes that proton-mass factors are omitted. Equation (10.44) restores the required mass dimension without claiming a precision lifetime calculation.

10.12 Broken $SU(5)$ and the Unification Scale

The X/Y fields share an $SU(5)$ adjoint multiplet with the Standard Model gauge fields. Exact unbroken gauge symmetry would not permit some members to have masses near 10^{16} GeV while the photon and gluons remain massless. The symmetry must therefore be spontaneously broken.

Question from the audience. If the $SU(5)$ symmetry were unbroken, would all of its gauge bosons be massless? [a]

Response. Yes, in the gauge theory under discussion. Gauge invariance forbids an explicit common mass. A Higgs mechanism can give masses to the gauge bosons associated with broken generators while leaving the unbroken subgroup gauge fields massless.

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[a] Lecture timestamp: 01:29:37.

The electroweak Higgs mechanism already illustrates the pattern: it distinguishes the photon from the $W^{\pm}$ and Z. Grand unification requires a separate high-scale order parameter that distinguishes the weak 2-block from the color 3-block and makes the off-diagonal X/Y fields extremely heavy.

Compared with 10^{16} GeV, even a 100 GeV weak-boson mass is negligible. Well above the heavy threshold, all gauge-boson masses may be neglected and the full $SU(5)$ symmetry becomes a good approximation. This is the meaning of a unification scale: not that the low-energy world visibly has $SU(5)$, but that sufficiently energetic processes become insensitive to the symmetry-breaking mass splittings.

10.13 Running Couplings

Gauge couplings depend logarithmically on the energy scale. Defining

$$\alpha_i(\mu) = \frac{g_i^2(\mu)}{4\pi}, \tag{10.49}$$

the one-loop evolution has the form

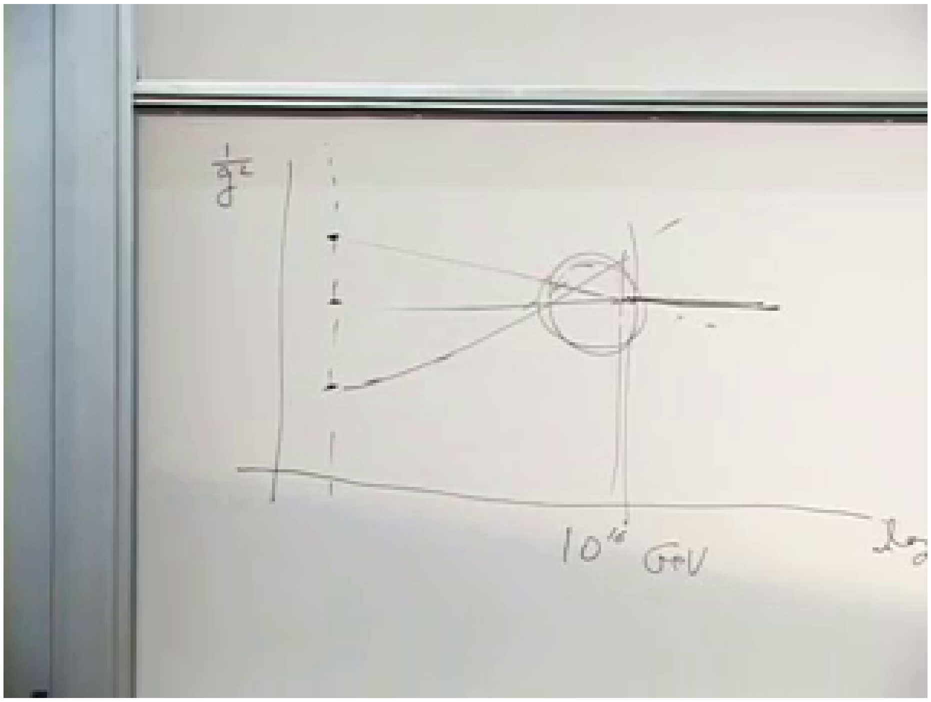
$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi} \ln\left(\frac{\mu}{\mu_0}\right). \tag{10.50}$$

The coefficients b_i depend on the gauge group and on every particle light enough to circulate in the relevant quantum loops.

[5]

At laboratory energies, the inverses of the three couplings are different. In the lecture's qualitative ordering, $1/\alpha_1$ is largest because hypercharge is weakest, $1/\alpha_2$ comes next, and $1/\alpha_3$ is smallest because the strong interaction is strongest. The curves at inaccessible energies are not directly measured. They are calculated from the observed low-energy values, the particle spectrum, and the loop diagrams that determine the beta functions. With the Standard Model particle content they become nearly straight lines when plotted against $\log \mu$.

[5] Editorial clarification: To compare hypercharge with the non-Abelian couplings inside $SU(5)$, g_1 must be given the conventional grand-unified normalization. This is the source of the numerical group-theory factors mentioned in the lecture.



Lecture frame: 01:36:32.500.

Figure 10.6: The board plot of the inverse gauge couplings approaching one another near 10^{16} GeV. The circled region marks their imperfect meeting.

Below the unification threshold the symmetry is broken, the light spectrum does not fill complete degenerate $SU(5)$ multiplets, and the three beta-function coefficients differ. Consequently the couplings run at different rates. At the high scale one asks whether they reach a common value:

$$\alpha_1(M_{\mathrm{GUT}}) \stackrel{?}{=} \alpha_2(M_{\mathrm{GUT}}) \stackrel{?}{=} \alpha_3(M_{\mathrm{GUT}}). \tag{10.51}$$

With only the nonsupersymmetric Standard Model spectrum, the lines approach one another around 10^{15}–10^{16} GeV but miss by an appreciable amount; the lecture describes the agreement as roughly fifteen percent. The important coincidence is that this is also the scale demanded by proton stability.

10.14 A Final Neutron Question

Question from the audience. If the spectator up quark in the proton-decay diagram were replaced by a down quark, would this describe neutron decay, and would the X boson matter? [a]

Response. It would define a grand-unified neutron channel such as $n \to e^+ + \pi^-$, but that is not the familiar decay of a free neutron. Ordinary neutron beta decay is $n \to p + e^- + \overline{\nu}_e$ and proceeds through the weak interaction. If the proton–neutron mass ordering were reversed, beta stability would be reversed as well, while the much slower grand-unified channel could remain.

[a] Lecture timestamp: 01:36:40.

This distinction separates kinematics from the new interaction. Which nucleon beta-decays depends on the proton–neutron mass difference; baryon-number violation from X/Y exchange is a separate, enormously suppressed mechanism.

10.15 Supersymmetry and the Larger Perspective

Adding a supersymmetric partner spectrum changes the beta-function coefficients in Eq. (10.50). In the lecture's account, the three extrapolated couplings then meet to about one-percent accuracy. This does not prove supersymmetry or grand unification, but it is the empirical reason the two ideas are often discussed together.

The matter multiplets can be unified still further. Under

$$SO(10) \supset SU(5), \tag{10.52}$$

one spinor representation decomposes as

$$\mathbf{16} \longrightarrow \mathbf{10} \oplus \overline{\mathbf{5}} \oplus \mathbf{1}. \tag{10.53}$$

The singlet is naturally identified with a left-handed antineutrino, or equivalently the charge conjugate of a right-handed neutrino. Thus all sixteen fields of an extended family fit into one irreducible $SO(10)$ multiplet.

Proton-decay experiments continue to test the idea by raising the lifetime bound when no event is seen. Supersymmetric grand-unified models do not give one universal proton lifetime: additional heavy fields and superpartner-dependent operators can change the dominant channel and rate over a broad range.

[6]

The lasting lesson is a tension rather than a verdict. The group theory fits quarks and leptons together with remarkable economy, the running couplings hint at a common high scale, and supersymmetry sharpens the numerical hint. The same construction also predicts the dangerous conversion of quarks into leptons. Any successful version must therefore preserve the elegance while keeping the proton stable for an almost inconceivably long time. In the lecture's closing spirit: may all our protons remain stable.

[6] Editorial clarification: In supersymmetric grand unification, color-triplet Higgs exchange can generate dimension-five baryon-violating operators in addition to the dimension-six gauge-boson operator discussed here. Their coefficients depend strongly on the heavy spectrum, flavor structure, and superpartner masses, which explains the lecture's warning that no single lifetime follows from supersymmetry alone.